MASTER OF SCIENCE IN INFORMATION SYSTEMS:

BUSINESS ANALYTICS

"STRESS-TESTING CONFORMAL PREDICTION INTERVALS: EMPIRICAL COVERAGE AND EFFICIENCY ACROSS CONDITIONAL VOLATILITY REGIMES IN TREE-BASED BRENT OIL FORECASTING"

Marius Bjerk Samuelsen / *107*
Truls Opsund / *104*

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Kristiania University of Applied Sciences
Kirkegata 24-26, 0153 Oslo
Norway

Abstract

This thesis investigates whether conditional volatility affects the empirical coverage and efficiency of conformal prediction intervals in one-step ahead forecasting of European Brent crude oil spot prices. The motivation is that crude oil markets are exposed to volatility clustering, regime shifts, and recurring stress. These conditions are challenging for conformal prediction as intervals are calibrated from past forecast errors, while periods of market stress might lead to intervals that perform well on average but become less reliable when uncertainty is at its greatest.

The empirical analysis uses daily observations from September 2000 to September 2025. Brent crude oil spot price is used as the forecasting target, while the feature set includes oil-market variables, macro-financial indicators, commodity prices, geopolitical risk and technical indicators. Four tree-based forecasting models are evaluated; Decision Tree, Random Forest, LightGBM, and XGBoost. For each model, three conformal prediction methods are compared; Baseline split conformal prediction, Ensemble Batch Prediction Intervals, and Adaptive Conformal Inference.

Elevated volatility is identified using two complementary measures; 30-day rolling standard deviation of Brent log returns and a GARCH(1,1) estimate of conditional volatility. Stress regimes are defined when both measures exceed their respective 90th percentile thresholds. Interval performance is evaluated using empirical coverage, mean interval width, and Winkler score, both across the full test set and separately for stress and non-stress periods.

The results show that elevated conditional volatility substantially reduces empirical coverage for Baseline CP and EnbPI. ACI provides the strongest improvement in stress-period coverage, but only by increasing mean interval width, indicating a coverage–efficiency trade-off. Overall, the findings indicate that conformal prediction intervals remain useful in Brent crude oil forecasting, but their practical reliability depends strongly on whether the calibration method can adapt to volatility-driven changes in the forecast error distribution.

Keywords: *Brent Crude Oil Forecasting, Uncertainty Quantification, Conformal Prediction, Prediction Intervals, Machine Learning*

Acknowledgements

This master's thesis marks the completion of our degree in Business Analytics at Kristiania. The work has been carried out during the spring of 2026 and focuses on uncertainty quantification in Brent crude oil price forecasting using conformal prediction.

The motivation for the thesis came from an interest in financial forecasting, machine learning, and the practical problem of making forecasts useful under unstable market conditions. Oil markets provide a particularly relevant setting for this, as they are shaped by geopolitical shocks, volatility clustering, and changing risk conditions. The process has been demanding, especially in connecting statistical theory, machine learning methods, conformal prediction frameworks, and empirical evaluation in a consistent, interpretable framework.

We would like to thank our supervisor Magnus Westerlund for guidance, intellectual sparring, and feedback throughout the process.

Any remaining errors, interpretations, and conclusions are our own.

Marius Samuelsen & Truls Opsund

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Abbreviations

AI = Artificial Intelligence

ARCH = Autoregressive Conditional Heteroskedasticity

AIC = Akaike Information Criterion

CART = Classification and Regression Trees

CP = Conformal Prediction

CPCP = Conformal Prediction Change Points

CQR = Conformalized Quantile Regression

CRPS = Continuous Ranked Probability Score

EFB = Exclusive Feature Bundling

EnbPI = Ensemble Batch Prediction Intervals

GARCH = Generalized Autoregressive Conditional Heteroskedasticity

GBDT = Gradient Boosting Decision Trees

GOSS = Gradient-based One-Side Sampling

i.i.d. = Independent and Identically Distributed

ICP = Inductive Conformal Prediction

LGBM = Light Gradient Boosting Machine

MAE = Mean Absolute Error

ML = Machine Learning

NECP/NECP-(N) = Non-Exchangeable Conformal Prediction / Non-Exchangeable Conformal Prediction with Normalization

NmtCP = Non-Exchangeable Multi-Target Conformal Prediction

RCS = Regression Coverage Score

RF = Random Forest

RMSE = Root Mean Squared Error

RMWS = Regression Mean Width Score

SDS = Switching Dynamical Systems

SHAP = SHapley Additive exPlanations

SSA = Singular Spectrum Analysis

VIX = Volatility Index

VMD = Variational Mode Decomposition

WIS = Winkler Interval Score

WTI = West Texas Intermediate

XGBoost = eXtreme Gradient Boosting

It ain't what you do not know that gets you into trouble. It is what you know for sure that just ain't so

Commonly Attributed to Mark Twain

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1 Introduction

1.1 Topic and Background

Crude oil markets exhibit volatility clustering, structural breaks, and recurrent stress episodes, implying time-varying conditional variance and distributional instability, creating a central challenge for forecasting uncertainty. A point forecast may provide an estimate of the next Brent crude oil price, but it does not quantify the uncertainty surrounding that forecast or how large the prediction error may become during volatile market periods. Most machine learning regression models are optimized for point accuracy rather than probabilistic calibration. Unless additional assumptions or post-estimation procedures are implemented, they do not return prediction intervals. This limitation is important in crude oil price forecasting as decision-makers are often concerned not only with price level, but also the range of other possible outcomes.

Conformal Prediction (CP) offers a distribution-free framework for constructing prediction intervals with finite-sample coverage guarantees under exchangeability assumptions. In the presence of heteroskedasticity and regime shifts, however, these guarantees will be strained, since time-series data generally do not satisfy exchangeability in the same way as independent and identically distributed observations. When exchangeability is violated, past forecast errors may no longer be representative of future forecast errors. This creates a direct empirical question; whether conformal prediction intervals remain valid and efficient when Brent crude oil markets enter volatility regimes.

This thesis investigates conformal prediction intervals in Brent crude oil forecasting using tree-based machine learning models, including Decision Trees, Random Forest, LightGBM and XGBoost, across three conformal frameworks; Baseline split conformal prediction, Ensemble Batch Prediction Intervals, and Adaptive Conformal Inference. The central concern is not only whether conformal intervals perform well on average, but whether they remain reliable when market conditions become unstable. This distinction is important because aggregated test-set performance can hide systematic weakness during high-volatility periods.

1.2 Purpose

The purpose of this thesis is to evaluate the empirical reliability of conformal prediction intervals in one-step ahead Brent crude oil price forecasting. Rather than assessing forecasting performance only through point prediction accuracy, this thesis examines whether the constructed intervals achieve adequate empirical coverage while remaining reasonably efficient.

Three conformal methods are applied: Baseline split conformal prediction, Ensemble Batch Prediction Intervals, and Adaptive Conformal Inference. Four tree-based forecasting models are used as the underlying point predictor: Decision Tree, Random Forest, LightGBM, and XGBoost. Performance is evaluated using MAE, RMSE, empirical coverage, mean interval width, and Winkler score. Results are reported for the full test set, non-stress periods, and stress-periods, allowing the thesis to assess whether interval performance differs systematically across volatility regimes.

The scope of the study is therefore restricted to uncertainty quantification under time-varying conditional volatility. The thesis does not seek to establish a causal effect of volatility on interval performance, but rather to empirically examine whether and to what degree interval reliability and efficiency differ across volatility regimes.

1.3 Motivation

The motivation arises from the central role oil prices continue to play in the global economy, despite an ongoing transition towards renewable energy sources. Fossil fuels still account for a substantial share of the global energy mix (Energy Institute 2025), and crude oil remains critical to industrial production, transportation, and financial stability. Volatility in crude oil prices therefore carries wide-ranging economic and political implications.

The effect of geopolitical conflicts on commodity markets is well established in the current pool of scientific literature. Huang et al. (2021), for example, document a nonlinear relationship between geopolitical risk and oil volatility. This relationship is particularly relevant for oil markets, where prices are sensitive to political uncertainty, supply disruptions, and changes in market expectations regarding both supply and demand. Recent geopolitical events have therefore further highlighted the non-stationarity and crisis-sensitive nature of crude oil markets.

Since February 2022, developments such as Russia's invasion of Ukraine (and turmoil in the Middle East with the ongoing Israel/USA – Iran conflict) have both contributed to abrupt and persistent price movements, underscoring the difficulty of forecasting oil prices in non-stationary and crisis-driven environments (Zhang et al., 2023). These events illustrate how oil prices are not only influenced by supply-demand fundamentals, but also rapidly evolving political and informational shocks. Market regimes as these generate regime-dependent error distributions and nonlinear price fundamentals, complicating both precise forecasting and assessment of uncertainty.

In these given environments, point forecasts provide limited decision value. Decision-making in energy markets requires reliable assessments of uncertainty, especially during periods of heightened volatility. This motivates the use of uncertainty-aware forecasting frameworks, such as conformal prediction, which aim to provide calibrated prediction intervals rather than a single-point estimate. Evidence from Zhou et al. (2025) further suggests that conformal prediction sets can improve human decision-making by communicating predictive uncertainty more explicitly.

1.4 Research Question

These conditions raise a narrower methodological question; whether prediction intervals calibrated on past forecast errors remain reliable when Brent crude oil markets enter high volatility regime. The thesis then seeks to answer this through the following questions;

i) "How does elevated conditional volatility affect the validity and efficiency of conformal prediction intervals in Brent crude oil price forecasting?"

ii) "Which conformal prediction method achieves the strongest coverage-efficiency trade-off during volatility-defined stress regimes?"

2 Background

2.1 Global Oil Benchmarks

Beyond the existence of multiple benchmarks, an important characteristic of the global oil market is the degree of integration between these reference prices. While crude oil is a globally traded commodity, physical constraints such as transportation costs, pipeline bottlenecks, storage capacity, and differences in crude oil quality (e.g., light to heavy) prevent prices from being perfectly equalized across regions at all times. Benchmarks instead tend to move closely together in the long run, while exhibiting short to medium-term deviations driven by regional supply-demand imbalances, infrastructure constraints and local events.

The global market is primarily anchored by three benchmarks; Brent crude, a seaborne crude oil which serves as the dominant pricing reference for international oil trade; West Texas Intermediate (WTI), a high-quality light crude benchmark centered in the United States; and Dubai/Oman, which primarily reflects crude shipped to Asian markets. Additional benchmarks such as Murban, Midland WTI and Bonny Light play important regional roles. The debate over whether the world oil market constitutes a single unified market or a set of partially segmented regional markets has been central to the energy economics literature.

Early work by Weiner (1991) analyzes the correlation between different crude oil prices across regions, using correlation analysis and a switching regression system, finds that correlations are strong within regions but weaken when compared internationally. Based on this evidence, Weiner (1991) concludes that the global oil market is characterized by a high degree of regionalization. At the same time, Weiner (1991) observes that price changes tend to appear first in certain crude oils, which serve as benchmarks towards which other crude oil prices adjust.

The existence of benchmark-led price adjustments does not imply full market integration but rather reflects the transmission of price information across partially segmented regional markets. While Weiner (1991) provides an influential early characterization of regional segmentation in oil markets, the relevance of this framework must be interpreted in light of substantial changes in production like the shale-driven expansion of U.S. oil production and increased financial integration globally among major oil consumers like China post 1991.

Subsequent research challenges the view of a strongly regionalized oil market. A growing body of empirical studies documents strong long-run relationships among major crude oil benchmarks, suggesting that oil prices across regions are linked through global arbitrage and information transmission mechanisms, even though short-term deviations may arise due to constraints within transportation, production frictions and quality differences. Fattouh (2009) emphasizes that whether the oil market constitutes one great pool or a set of regionalized markets has important implications for energy policy.

While crude oil prices may evolve independently over shorter periods, Fattouh (2009) argues that their movements are unlikely to diverge without bounds, suggesting a stationary long-run equilibrium relationship between price benchmarks. Despite their small differences in physical quality, e.g., WTI being generally lighter and sweeter than Brent, and being "landlocked" on the U.S. mainland which complicates global transport, both benchmarks are of similar quality and maintain the same liquid futures markets. This motivates a further analysis to investigate whether price movements differ across both WTI and Brent.

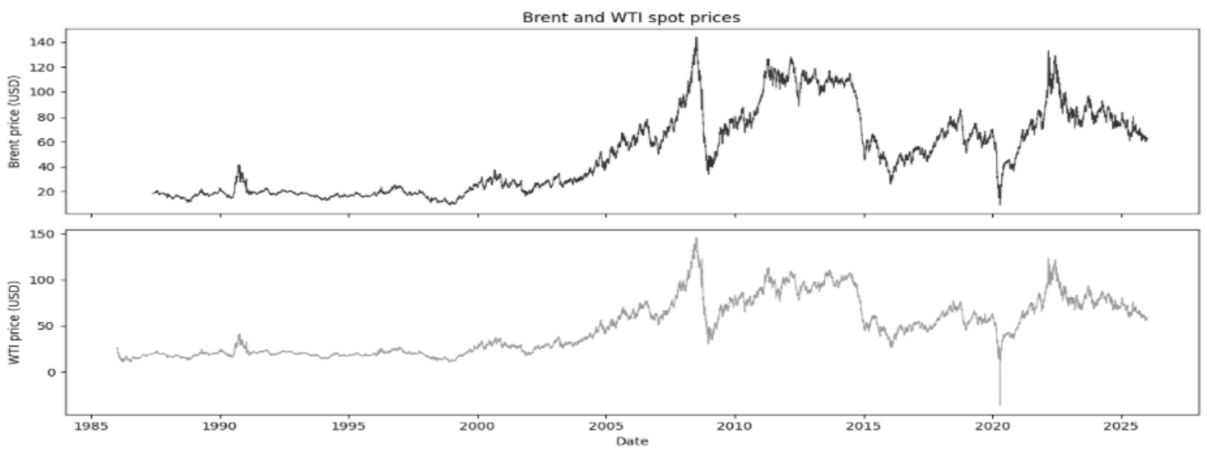

Figure 1: Brent and WTI Historical Prices

2.2 WTI versus Brent comparison

WTI and Brent crude are two of the most important global oil benchmarks, yet their prices may diverge due to differences in geography, transportation constraints, market structure and liquidity. While WTI is primarily linked to mainland U.S infrastructure and has been historically affected by local bottlenecks, Brent reflects seaborne crude traded in global markets and is therefore less exposed to regional frictions. Empirical evidence suggests that price innovations do not enter uniformly: Kaufmann and Ullman (2009) show while both WTI and Brent have highly liquid futures markets, price innovations tend to be transmitted more broadly through Brent-linked contracts, reflecting Brent's more central role in international pricing and its lower exposure to region-specific constraints.

Other empirical experiments have also been done to measure differences in price transmissions and the impact of geopolitical events on both benchmarks, as shown by Li et al (2022) and Li et al., (2024). The authors provided related empirical evidence by forecasting WTI and Brent crude oil prices using the geopolitical risk index developed by Caldara and Iacoviello (2022), including the sub-indices of geopolitical threat and geopolitical action. Their results indicate that geopolitical risk has stronger predictive performance for Brent than for WTI. They further find that geopolitical action generally provides stronger predictive accuracy than geopolitical threat, and that both sub-indices are more informative for Brent than for WTI, supporting the argument that Brent is more closely connected to global geopolitical risk than WTI. This asymmetry motivates an explicit analysis of the WTI-Brent price spread, which captures both long-run integration and short- to medium-term deviations driven by regional constraints and market-specific factors affecting prices.

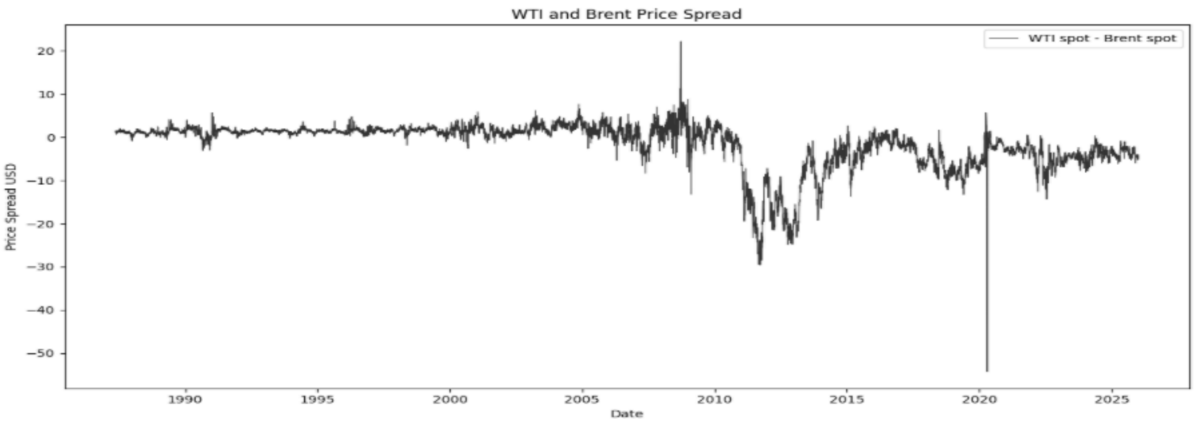

Figure 2: WTI and Brent Historical Price Spread

2.3 Empirical motivation for selecting Brent over WTI

Summary statistics indicate that WTI-Brent spread has a mean of –1.63 USD/barrel and a standard deviation of 5.33 USD/barrel over the sample period. The interquartile range is relatively narrow, with 50% of the observations lying between –3.62 and 1.62 USD/barrel, suggesting that the two benchmarks trade at similar levels during normal market conditions.

Table 1: Descriptive statistics for WTI–Brent spread (USD)

Statistic	Value
Observations	9,661
Mean	-1.63
Median	0.67
Standard deviation	5.33
Minimum	-54.34
25th percentile	-3.62
75th percentile	1.62
Maximum	22.18

The spread is measured as WTI minus Brent in USD per barrel.

The underlying distribution is strongly left-skewed due to extreme negative outliers during the COVID-19 oil market collapse when WTI briefly traded at negative prices (Zhang et al., 2023). Large negative spreads are also observed during the early 2010s that reflect regional supply increases associated with the rapid expansion of the U.S shale oil production, causing WTI to trade at discount relative to Brent, showcasing how local changes to supply and infrastructure can alter regional pricing, as proposed by Kaufmann & Ullman (2009).

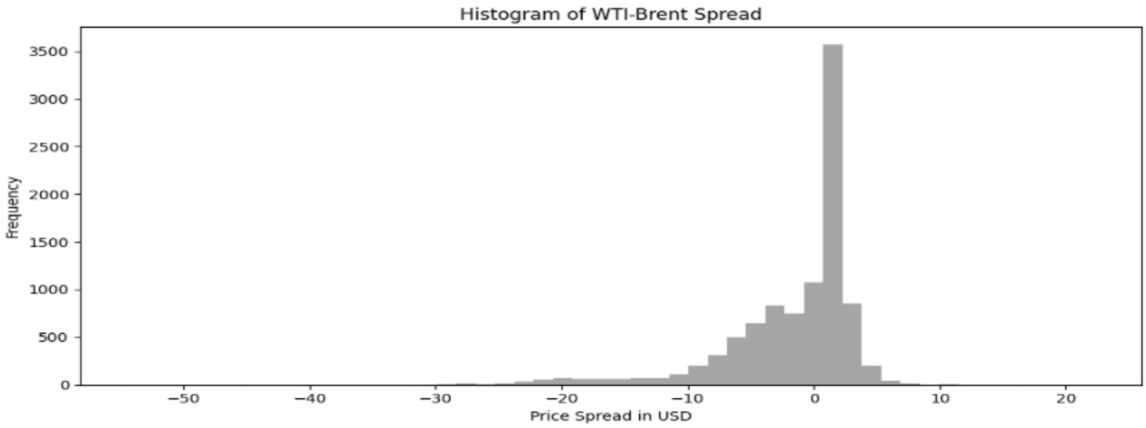

Figure 3: Histogram of Spread Distribution

The empirical distribution, literature, and historical episodes outlined in the analysis above demonstrate that large deviations between benchmarks are primarily driven by region specific changes to the fundamentals, as shown by localized supply expansions with the Shale-revolution on the U.S. mainland. For this reason, Brent is widely considered the most representative benchmark price for the international crude oil market. This thesis will therefore exclusively focus on Brent prices, allowing the study to examine predictive uncertainty within the benchmark that most consistently captures global oil market dynamics.

3 Theoretical Framework

3.1 Time Series Forecasting

A time series is a sequence of observations indexed by time or time-steps (Shumway & Stoffer, 2017, 1–2). Unlike standard supervised learning problems where observations are assumed to be independent, time series data exhibits temporal dependence, meaning that past observations carry information about future values (Shumway & Stoffer, 2017). This fundamental property shapes how forecasting models are constructed, trained, and evaluated. This thesis focuses on one-step-ahead forecasting of Brent crude oil prices, which reduces the forecasting problem to predicting y_{t+1} from information available at time t. Let the forecasting procedure be expressed as;

$$y_{t+1} = f(y_{1:t}, X_{1:t}) + \varepsilon_{t+1}$$

Equation 1:One-Step-Ahead Time-Series Forecasting

Where $f(\cdot)$ denotes the forecasting model, $X_{1:t}$ denotes the exogenous variables available up to time t, and ε_{t+1} denotes the forecast error. As expressed by Bontempi et al. (2013), the time series forecasting problem is represented as a supervised learning problem by constructing input-output pairs. For each forecast step t, the input vector X_t contains lagged Brent-price information and variables observed at time t, while the target is the next-day Brent price y_{t+1}. This allows tree-based regression models to be used for one-step ahead forecasting.

3.1.1 Look-Ahead Bias

A critical pitfall in time series forecasting is look-ahead bias, which occurs when information from the future is inadvertently used to make predictions about the past or present. This can arise in several ways, such as using future values when constructing features, shuffling data before splitting into training and test sets, or applying normalization across the full sample before the split is made. In financial forecasting, look-ahead bias is particularly consequential because it produces artificially optimistic results that do not reflect real-world predictive performance.

A fundamental requirement for valid forecast evaluation is therefore that the test set always follows the training period in time, ensuring that the model is only ever assessed on observations it could not have seen during training. When constructing features derived from the price series itself, all lagged and rolling variables must be computed using information available at the time of prediction. For a one step-ahead-forecast of y_{t+1}, the feature vector may therefore only use information from $t,\ t-1,\ t-2, \ldots$ and preceding periods (Hyndman & Athanasopoulos, 2018).

3.1.2 Inherent Non-Stationarity

A time series is stationary if its statistical properties, namely mean, variance, and autocovariance, remain constant over time. Financial time series are non-stationary, meaning that these properties change over time due to trends, structural breaks, and regime shifts (Tsay, 2005). Non-stationarity presents a fundamental challenge in forecasting because a model trained on historical data may encounter a materially different statistical environment during evaluation (Hyndman & Athanasopoulos, 2018). This is also important for uncertainty quantification, as uncertainty is often estimated from historical forecast errors, but under non-stationarity the magnitude and error distribution might change over time.

3.1.3 Temporal Dependence

In standard supervised learning, observations are typically assumed to be independent and identically distributed. Time series data violates this assumption, as consecutive observations are correlated over time (Shumway & Stoffer, 2017). This has direct consequences for model evaluation, as randomly shuffling observations before splitting would break the temporal structure of the data and produce misleading performance estimates.

It also has consequences for uncertainty quantification, as methods that assume independence between observations may produce poorly calibrated intervals when applied to dependent data (Cont, 2001). Evaluating forecasting performance on data that is temporally separated from the training period is therefore a necessary condition for valid out-of-sample inference (Hyndman & Athanasopoulos, 2018).

3.2 Supervised Machine Learning for Regression

Machine Learning (ML), understood as a subset of artificial intelligence (AI), refers to a group of data-driven algorithms designed to learn relationships from observed data. ML involves the construction of models that are able to identify patterns in data, generalize to unseen observations, and produce predictions or decisions without being explicitly programmed for each individual outcome. There is a common distinction between supervised, unsupervised, and reinforcement learning, which all have different approaches and use-cases. In supervised learning, the model's objective is to estimate a mapping from input variables to a known, pre-defined target-variable based on labeled observations (Cunningham et al., 2008).

In this thesis, the relevant supervised learning task is regression, as the objective is to predict a known continuous target variable from observed input features; the one-step-ahead Brent crude spot price. Let the training dataset be denoted as:

$$D = \{(x_i, y_i)\}_{i=1}^{m}$$

Equation 2: Training Dataset

Where m denotes the number of observations, x_i denotes the vector of explanatory variables for observation i , and y_i denotes the corresponding target value (Zhou, 2021). Each instance x_i can be represented as:

$$x_i = (x_{i1}, x_{i2}, \ldots, x_{id})$$

Equation 3: Feature Vector Definition

where x_{ij} denotes the value of feature j for observation and d is the number of features (Zhou, 2021). Using this notation, the regression problem is expressed as:

$$y_i = f(x_i) + \varepsilon_i$$

Equation 4: Supervised Regression

Where $f(\cdot)$ denotes the underlying unknown relationship between input features and the target variable, and ε_i is the error term denoting the unexplained variation, or random noise (James et al., 2023). For this scenario, the fitted forecasting model estimates $f(\cdot)$ while the conformal prediction layer uses the resulting forecast errors to construct prediction intervals.

3.3 Tree Based Learning

Tree-based learning refers to a family of algorithmic methods used for both classification and regression tasks. The original framework was formalized by Breiman et al. (1984) through Classification and Regression Trees (CART), which recursively partition the observations into increasingly homogeneous groups based on selected predictor variables and threshold values.

Since the thesis is concerned with forecasting Brent crude prices, a continuous outcome variable, the discussion is limited exclusively to regression trees and their role as the basis for subsequent, more advanced tree-based ensemble methods like Random Forest, LightGBM and XGBoost.

As also described by Loh (2011), a decision tree regressor predicts a numerical outcome by iteratively partitioning the input data through a series of binary splits. Starting from the top root node, the algorithm selects the input feature and the split threshold that best reduces the prediction error (e.g., mean squared error or variance). Each split divides the observations into two child nodes, each node individually representing a smaller subset of the original data, and the splitting process continues until either maximum tree depth, minimum number of observations or observations per leaf, or no further meaningful reduction in prediction error is reached, in a manner called recursive partitioning.

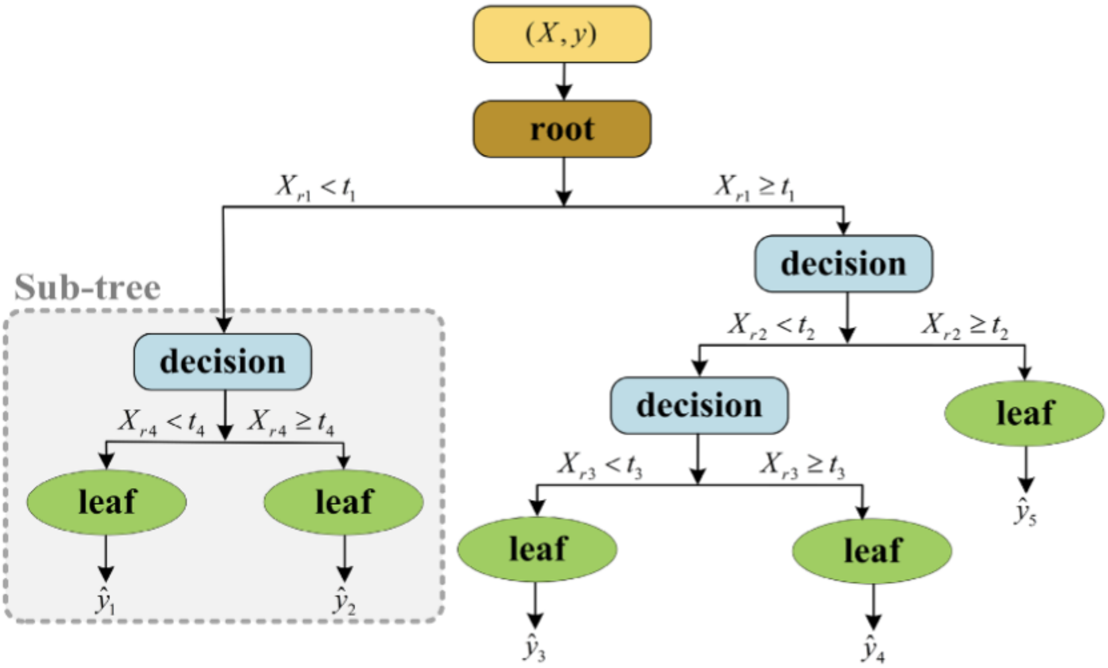

Figure 4: Decision Tree Structure

Starting above the root, X denotes feature matrix and y denotes the target variable. In each split, $X_{r_j} < t_j$ denotes the split rule applied at node j, where X_{r_j} denotes the predictor selected for node j , and t_j denotes the corresponding threshold value. The threshold determines whether an observation is assigned to the left or right branch; observations satisfying $X_{r_j} < t_j$ are assigned to the left branch, while observations satisfying $X_{r_j} \geq t_j$ are assigned to the right branch. The terminal leaf nodes generate the final prediction $\hat{y}$, where $\widehat{y_1}$.......$\widehat{y_5}$

represents all the predicted values associated with the terminal leaf nodes (Yazdani et al., 2023).

In the context of Brent crude oil prices, the same mechanism can be understood as follows; if the lagged Brent crude oil price is selected as the splitting variable and the threshold is given as 82.5 USD/barrel, observations with Brent crude below 82.5 are sent to the left branch, while observations with Brent crude equal to or above 82.5 are sent to the right branch. After repeated splits, the final leaf node provides the one-step ahead forecast for $\widehat{y_{t+1}}$.

In practical implementation scenarios, stopping criteria are usually controlled through user-defined hyperparameters, which includes, but not limited to; *max_depth, min_samples_split, min_samples_leaf,* and *max_leaf_nodes.*
These variables determine the maximum depth of the tree, minimum number of observations that are needed to split a node, minimum number of observations that need to remain in each leaf, and maximum number of terminal nodes allowed. This can be explained as:
Max_depth: adjusted to limit how many splits the tree can make. A deeper tree might capture a more nonlinear structure for complicated tasks, but it also increases the risk of overfitting.
Min_samples_split: adjusted to prevent the model from splitting nodes that contain too few samples from the original data. If a split is completed on a small subset, it increases the risk of reflecting random variation instead of a predictive signal.
Min_samples_leaf: adjusted to ensure that each terminal leaf contains a minimum number of observations. This can avoid leaves that were only based on a handful of observations.
Max_leaf_nodes: adjusted to cap the total leaf nodes the tree can create. This limits how many separate regions the model can split into.

3.4 Ensemble Learning: Bagging and Boosting

Ensemble learning is a common term for combining multiple base-learners. These methods significantly improve predictive performance compared to using a single model. This is accomplished by aggregating the outputs of multiple inducers, a process where the individual errors of a single base-learner are likely to be compensated for by others in the ensemble (Sagi & Rokach, 2018). For this to work, the base-learners must be reasonably accurate and make uncorrelated errors (Dietterich, 2000).

Dietterich (2000) identifies three reasons why ensembles tend to outperform single learners. The first is statistical. When training data is limited, averaging across multiple accurate hypotheses reduces the risk of selecting a poor one. The second is computational. Combining learners that start from different points helps overcome the tendency of local search algorithms to get stuck in suboptimal solutions. The third is representational. Weighted combinations of base-learners can approximate functions that lie outside the hypothesis space of any individual model.

The expected error of a regression model can be decomposed into bias, variance, and irreducible noise (Hastie, Tibshirani & Friedman, 2009). The two dominant ensemble paradigms target different parts of this decomposition: bagging primarily reduces variance, while boosting primarily reduces bias (Sagi & Rokach, 2018). The next two sections develop these paradigms in detail.

3.5 Bagging and Random Forest

Bagging, short for bootstrap aggregating, trains each base-learner independently on a different bootstrap sample of the training data (Sagi & Rokach, 2018). A bootstrap sample is drawn with replacement from the original dataset and typically contains the same number of observations, meaning some training instances appear multiple times while others are left out. The final prediction is obtained by averaging the outputs across all learners for regression, or through majority voting for classification (Sagi & Rokach, 2018). Because the base-learners are trained independently, bagging can be implemented in parallel. The method works best with unstable learners such as decision trees, where small changes in the training data can produce substantially different models (Dietterich, 2000).

Random Forest, originally formalized by Breiman (2001), extends bagging by further reducing the correlation between the trees in the ensemble (Hastie et al., 2009). Like bagging, it trains a large number of unpruned decision trees, each on its own bootstrap sample. The key difference lies in how each tree is built; at every split, only a random subset of the input features is considered as candidates, rather than searching across all features (Sagi & Rokach, 2018). Different trees therefore end up using different features at different points, which reduces how correlated their predictions are. When predictions from less correlated trees are averaged, the variance of the ensemble drops further than with plain bagging (Hastie et al., 2009). Random Forest has become one of the most widely used ensemble methods in practice, largely because it requires little tuning and remains robust across a wide range of problems (Sagi & Rokach, 2018).

3.6 Boosting and Gradient Boosted Trees

Boosting trains base-learners sequentially, with each new learner fit to correct the errors of the current ensemble (Sagi & Rokach, 2018). In each iteration, the training examples that were poorly handled by the current ensemble are given more weight, so that the next learner focuses on the parts of the problem that have not yet been solved well (Dietterich, 2000). The final prediction is a weighted combination of all the learners added along the way.

Gradient Boosted Trees is a regression-oriented formulation of boosting (Sagi & Rokach, 2018). The method builds an additive model by fitting a sequence of regression trees, where each new tree is trained to predict the residual errors of the current ensemble. More formally, each tree is fit to the negative gradient of a chosen loss function, which is why the method is called gradient boosting (Sagi & Rokach, 2018). Squared error is the standard loss for regression problems. Predictions are combined additively, with each tree contributing a scaled update to the running prediction.

Gradient boosted trees typically use many shallow trees, in contrast to Random Forest, which uses fewer but deeper trees (Sagi & Rokach, 2018). The number of trees is an important tuning parameter: setting it too low leads to underfitting, while setting it too high can cause the model to overfit, which is why it is typically chosen using a validation set (Sagi & Rokach, 2018). The final update step by the model, as shown by Friedman (2001), is given as;

$$F_m(x) = F_{m-1}(x) + \rho_m h(x; a_m)$$

Equation 5: Gradient Boosting Update Equation

Where $F_m(x)$ denotes the updated model after boosting iteration m, and $F_{m-1}(x)$ denotes the model from the previous iteration. The index m refers to the current boosting stage, while $h(x; a_m)$ denotes the new base learner added at iteration with a_m representing the fitted parameters of that learner. Finally, ρ_m denotes the step size, which determines how strongly the new learner contributes to the updated ensemble (Friedman, 2001).

3.7 LightGBM

LightGBM is an efficient implementation of Gradient Boosting Decision Trees (GBDT), and was developed to improve computational efficiency and scalability in settings when training tree-based boosting models. Standard GBDT algorithms can become computationally expensive as many central split points must be evaluated during tree construction. This is addressed by LightGBM through several design choices like leaf-wise tree growth, Gradient-based One-Side-Sampling (GOSS) and Exclusive Feature Bundling (EFB), (Ke et al., 2017). GOSS reduces the number of training observations used when estimating split gains. The method keeps observations with large gradients, because these observations contribute more strongly to the current model error and therefore contain more information about where the model needs to improve. Observations with smaller gradients are not all used; LightGBM draws a random sample from them and reweighs the sampled observations to compensate for the sampling approach. By doing this, GOSS reduces computational cost while still preserving the majority of the information needed to identify useful splits (Ke et al., 2017).

EFB reduces computational cost by merging mutually exclusive features in order to reduce the dimensionality of the datasets, especially in sparse datasets. LightGBM also uses a histogram-based algorithm to identify split points during tree construction. Contrary to other similar and related models like XGBoost who typically grows decision trees level-wise, LightGBM grows trees leaf-wise by expanding the leaf that yields the largest improvement in model performance. This can reduce loss more aggressively, but may also increase the risk of overfitting if the algorithm is not properly regularized (Ke et al., 2017).

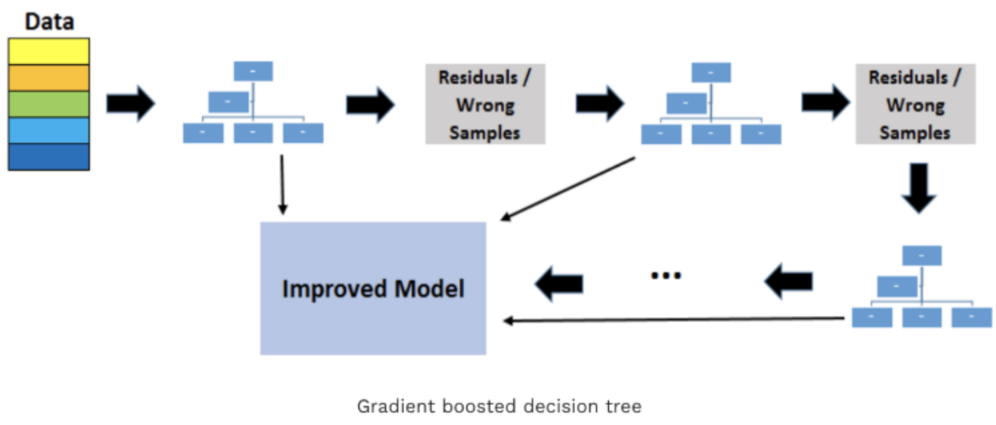

Figure 5: Gradient Boosted Decision Tree Structure

The boosting process begins with an initial prediction, often a constant value such as the mean of the target variable under squared error loss. The model then evaluates the remaining prediction errors, or the pseudo-residuals, from the current ensemble. A new decision tree is fitted to these remaining errors, so that it captures parts of the signal not yet explained by the ensemble. Throughout this iterative process, each successive tree contributes a small correction to the current model. As with other gradient boosting machines, LightGBM constructs the final prediction as the aggregated contribution of multiple fitted trees, as further described mathematically in section 3.8.

3.8 XGBoost

XGBoost, short for Extreme Gradient Boosting, is a regularized implementation of GBDT (Chen & Guestrin, 2016). As with other boosting algorithms, XGBoost builds an ensemble sequentially, where each new tree is added to improve the current model by reducing the remaining prediction error. What separates XGBoost is that the boosting objective includes explicit regularization terms that penalize model complexity, which can make the model less exposed to overfitting than an unrestricted boosting tree. For observation i , the final prediction is defined as the sum of the contributions from K fitted trees, and can be written as:

$$\hat{y}_i = \phi(x_i) = \sum_{k=1}^{K} f_k(x_i), \qquad f_k \in \mathcal{F}$$

Equation 6: XGBoost Predictive Function

Where $\hat{y}_i$ denotes the final prediction for observation i, x_i denotes the input feature vector, K denotes the number of trees in the ensemble, $f_k(x_i)$ denotes the prediction contribution from tree k, and F denotes the space of possible regression trees. Each tree maps the input vector to a terminal leaf through a sequence of decision rules, and each terminal leaf contains a numerical score. The final output is obtained by aggregating all these leaf scores across all fitted trees (Chen & Guestrin, 2016).

4 Uncertainty Quantification

4.1 Distribution-Free Predictive Inference

A single point prediction, as generated by algorithms in many real-world forecasting scenarios today, is insufficient for decision making. Forecasts such as "Brent crude oil will trade at 110 USD/barrel tomorrow" may be informative, and from time to time, also correct, but it does not communicate or indicate how uncertain that prediction is, nor what range of other alternative outcomes remains plausible. In high-stakes settings such as finance, energy, healthcare or public policy, decision makers require more than a point estimate. An assessment of predictive uncertainty is also needed in order to evaluate risk, up-and downside, allocate resources, and plan for adverse outcomes. As emphasized by Manokhin (2025), uncertainty quantification becomes especially important when overconfident or uninformative predictions can lead to costly operational or strategic errors.

This broader problem motivates predictive inference, which raises the concern of not only forecasting future observations, but also with quantifying the uncertainty attached to those forecasts. In forecasting settings, the predictive uncertainty is closely related to the model's forecast errors. If the model historically missed the true Brent crude price with 1-2 USD/barrel, a narrow interval can be appropriate. However, if the model's residuals become larger, especially during periods of stress, the interval must widen in USD/barrel terms in order to maintain the same empirical coverage rate. For a prediction made at time t, the forecast error can be expressed as:

$$\varepsilon_t = y_t - \hat{y}_t$$

Equation 7: Forecasting Error

Where ε_t denotes the forecast error, y denotes the actual Brent crude price at timestep t, and $\hat{y_t}$ denotes the predicted Brent crude price. The difficulty is that the true distribution of future forecast errors is unknown, especially for non-stationary series such as Brent crude oil. This, in practice, raises questions such as; "*how large are the errors likely to be*?", "*are the errors symmetrical*", "*do the errors contain fat tails*", and "*how is the variance changing over time*?".

As formalized by Lei et al. (2018), distribution-free predictive inference can be understood as the task of predicting a future Y_{n+1} from a new feature vector X_{n+1} without specifying the true

relationship between the input features and the outcome variable. Given a nominal miscoverage level of $\alpha = 0.10$, the corresponding nominal coverage level is 1 - $\alpha = 0.90$. The aim is to create a prediction interval consisting of a lower and upper bound, where the actual future value is contained within the interval with a probability of at least $1 - \alpha$.
The coverage criterion, also known as the *validity* of the intervals, can then be expressed as (Shafer & Vovk, 2008; Manokhin, 2025):

$$\mathbb{P}\left(Y_{n+1} \in C(X_{n+1})\right) \geq 1 - \alpha$$

Equation 8: Conformal Prediction Validity Criterion

Where Y_{n+1} denotes the future outcome, X_{n+1} denotes the feature observation, and $C(X_{n+1})$ denotes the prediction set constructed.

4.2 Conformal Prediction

Conformal prediction (CP) is a distribution free framework for constructing prediction intervals or prediction sets with finite-sample marginal coverage under the assumption of exchangeable data, meaning that the coverage guarantee holds in finite data-samples and does not require a large sample size to be theoretically valid.

The CP framework was formalized by Vovk et al. (2005), while earlier work by Papadopoulos et al. (2002) introduced inductive CP as a more computationally scalable variant based on a proper training set and a separate calibration set. This allowed CP to be applied as a top layer on any pre-trained model, thus making it model agnostic. Vovk later examined the validity properties of conformal predictors, showing that under finite-sample marginal validity, the true value falls outside the prediction set with a probability no greater than a pre-determined user significance level (Vovk, 2012).
A key advantage of CP is therefore that these coverage guarantees do not require the data to follow a specific probability distribution, such as a normal distribution, in order to construct valid prediction intervals.

Early work on CP formalized the distinction between point predictions and set-valued predictions. Given an underlying point prediction model, conformal methods construct a prediction region such that the true outcome is contained within the region with probability at least $1 - \alpha$. In regression settings, this region takes the form of an interval around the point estimate, while in classification it corresponds to a subset of possible labels (Shafer & Vovk, 2008). Simultaneously, classical CP relies on exchangeability, an assumption that is difficult to justify in financial and energy-related time series, where observations usually exhibit temporal dependence, non-stationarity, and regime shifts. The formal validity guarantees from standard CP therefore do not directly carry over to such circumstances. This limitation has motivated later extensions of the framework for non-exchangeability or settings with distribution shifts, as seen by Barber et al. (2023). As a result of this violation of exchangeability, CP is employed in this thesis as an empirical calibration layer on top of pre-trained ML models, where the empirical coverage and efficiency of the intervals are evaluated instead of assuming that CP guarantees exact validity.

4.2.1 Exchangeability

The statistical validity of classical CP relies on the assumption of exchangeability. This assumption is central to the CP framework, as it supports the finite-sample coverage guarantees that distinguish CP from many other approaches to uncertainty quantification. Exchangeability is a weaker assumption than the usual independent and identically distributed (i.i.d) assumptions.

While i.i.d implies independence and identical distribution for all data points, exchangeability only requires that the joint distribution of the data points remains the same under any arrangement of the sequence of the samples (Vovk et al., 2005; Shafer & Vovk, 2008). Formally, this is shown as:

$$\mathbb{P}(Z_1 = z_1, \ldots, Z_n = z_n) = \mathbb{P}(Z_{\pi(1)} = z_{\pi(1)}, \ldots, Z_{\pi(n)} = z_{\pi(n)})$$

Equation 9: Exchangeability Assumption

This means that a sequence of random variables $(Z_1, Z_2 \ldots, Z_n)$ is exchangeable if its joint distribution remains unchanged under any arrangement of the indices, implying that the ordering of the observations does not affect the joint distribution of the data, making any arrangement of the sequence statistically equivalent (Manokhin, 2025).

4.2.2 Marginal Coverage vs Conditional Coverage in Conformal Prediction

In conformal prediction, marginal coverage refers to the proportion of forecasted values that fall within the constructed prediction intervals when evaluated across the full set of forecasts. At a nominal coverage level of (often $1 - \alpha = 0.90$), marginal coverage implies that approximately 90% of all realized observations are expected to lie inside their corresponding prediction intervals when performance is assessed over the entire test set/evaluation period. Under exchangeability, this is the central validity target of classical CP.

Marginal coverage, however, only describes average performance across all observations taken together. It does not imply that the same coverage level is achieved across different parts of the data distribution. For financial time series such as Brent crude oil, this distinction is especially important, as observations are not drawn from a stable, similar environment but from a non-stationary environment and regime-dependent environment where volatility clustering, temporal dependence, and regime shifts all exists. In settings like this, a method may appear well calibrated but still systematically failing under specific market conditions or market states.

This motivates a clear distinction between marginal and conditional coverage. Conditional coverage refers to predictive coverage evaluated within a particular subset of observations, being conditional on an underlying specific period, state, or regime. The pre-determined volatility regimes fall within these conditions for the thesis. The fundamental question is therefore not only whether 90% conformal intervals achieve 90% empirical coverage overall, but whether they continue to do so during stress periods.

The distinction can easily be illustrated; Assume an evaluation period containing 1000 observations, of which 900 belong to non-stress periods and 100 belong to stress periods. Let's further assume that prediction intervals achieve 92% empirical coverage across non-stress periods, but only 72% coverage during stress periods. The final resulting coverage is then defined as:

$$(0.9 \cdot 0.92) + (0.1 \cdot 0.72) = 0.90$$

Equation 10: Marginal vs Conditional Coverage

Which exactly matches the nominal 90% target. However, the intervals clearly underperform during stress periods. The method would therefore appear well-calibrated when estimated only by marginal coverage, while failing the conditional, volatility regime. Empirical coverage, although individually informative, is not sufficient on its own.
Coverage must therefore also be evaluated separately across stress and non-stress regimes in order to determine whether conformal calibration remains robust under the whole sample period.

4.2.3 Nonconformity Score

One of the main pillars of CP is the nonconformity score. The role of this measure is to quantify how abnormal or "nonconforming" a new observation is measured relative to the calibration scores obtained from previously observed data. More formally, it tries to assess how poorly a data observation "conforms" to the previously observed data points. The models point prediction is used together with the observed value to construct a nonconformity score. A high nonconformity score will then indicate that the data point is more "nonconforming" or abnormal (Shafer & Vovk, 2008). Two relevant residual-based nonconformity scores are the absolute residual and the locally weighted residual:

a) **Absolute Residual**: Defined as the absolute difference between the true value and the models point prediction, given by $|y - \hat{y}|$), where y_i denotes the true value, and $\hat{y}_i$ the point prediction. A progressively larger absolute residual indicates a wider gap between the actual and predicted values, e.g., a higher nonconformity score (Lei et al., 2018). The formula can be expressed as:

$$a_i = |y_i - \hat{y}_i|$$

Equation 11: Absolute Residual

b) **Locally weighted residual**: Given heteroskedasticity, a locally weighted residual, given by |y - $\hat{y}| / \hat{s}$), where $\hat{s}$ denotes the models estimated uncertainty (or the spread of absolute residuals in any given situation), may be more appropriate because some observations will be naturally more volatile than others. A raw comparison between true and predicted values might therefore be less informative. This, in practice, means that the residuals or deviations are evaluated relative to how uncertain the model itself is, making the nonconformity measure a better fit for settings with time-varying uncertainty, such as time-series settings (Lei et al., 2018). Formula can be expressed as:

$$R_i = \frac{|Y_i - \hat{\mu}(X_i)|}{\hat{\rho}(X_i)}, \quad i \in I_2$$

Equation 12: Locally Weighted Residual

Where R_i denotes the locally weighted residual for observation i, $\hat{\mu}(X_i)$ denotes the fitted prediction and $\hat{\rho}(X_i)$ denotes the estimated error spread. The index I_2 denotes the calibration observation used to compute the residual scores (Lei et al., 2018).

4.2.4 Calibration and Quantile Selection

Calibration links the non-conformity scores defined above in Section 4.2.3 to the final prediction interval. In split conformal regression, the available data is divided into a training set, a calibration set, and a test set. The forecasting model is first fitted on the training set. It is then used to generate predictions for the calibration observations, after which the corresponding nonconformity scores are computed. These scores yield the empirical calibrated residual threshold, denoted as $\hat{q}_{1-\alpha}$, which later determines the size of the upper and lower bounds of the interval.

As a rule of thumb, the observations within the calibration set should be representative of the distribution encountered during evaluation. If the data exhibit stress or volatility clustering, the calibration set should therefore reflect these patterns as well, since calibration scores are used to determine the residual threshold applied to future predictions (Manokhin, 2025).

The purpose of calibration is not to refit the underlying model, but to estimate how large the correction term around the point prediction should be. Given a nominal miscoverage level α, split conformal regression selects an upper quantile of the calibration scores. Let $a_1, \dots, a_{(n_{\text{cal}})}$ denote the previously computed nonconformity scores from the calibration set, where n_{cal} represents the number of calibration observations. These are then sorted in ascending order, expressed as:

$$\alpha_{(1)} \leq a_{(2)} \dots, \leq a_{(n_{\text{cal}})}$$

Equation 13: Sorted Calibration Scores

The conformal cutoff, as given by Lei et al. (2018) can then be expressed as;

$$d = \text{the } k\text{th smallest value in } \{R_i : i \in I_2\}, \qquad k = \left\lceil \left(\frac{n}{2} + 1\right)(1 - \alpha) \right\rceil$$

Equation 14: Conformal Quantile Cutoff

Where R_i denotes the calibration residuals computed on I_2 and d denotes the selected cutoff value. Following Lei et al. (2018), the cutoff is chosen as the k-th smallest residual after sorting calibration residuals in ascending order. The expression returns a dataset that is split into two equally sized subsets therefore yielding a calibration subset of $n/2$. The use of $n/2 + 1$ shows the finite-sample adjustment, which means that the cutoff is chosen slightly higher than the standard 90th percentile of the calibration residuals. The reason is that the next, unseen observation is also accounted for when deciding which residual to use as the cutoff, making the selected cutoff used for bounds slightly more conservative than the straight percentile, especially for smaller datasets (Lei et al., 2018).

For example, let's assume a nominal miscoverage level of α = 0.10, yielding 0.90, and suppose the calibration residuals consists of 100 ordered absolute residuals increasing from 0.1USD/barrel to 10.0 USD/barrel in increments of 0.1. The 91st ordered calibration residual is then chosen. In this simplified example, the selected cutoff returns;

$$d = 9.1 USD/barrel$$

Equation 15: Calibration Residual in USD/barrel

This links past prediction errors from the calibration set to the threshold used for future intervals. The cutoff is then used for the interval construction described below in Section 4.2.5.

4.2.5 Construction of Prediction Intervals

After the calibration residual cutoff d has been chosen, the final split conformal prediction interval is constructed around the point prediction. Following Lei et al. (2018), the interval can be expressed as;

$$C_{\text{split}}(x) = [\hat{\mu}(x) - d, \hat{\mu}(x) + d], \quad \text{for all } x \in \mathbb{R}^d$$

Equation 16: Split Conformal Prediction Interval

Where $\hat{\mu}(x)$ denotes the model's point prediction, x denotes the information available at forecast time, and d denotes the calibrated cutoff selected previously. The cutoff is then added and subtracted from the point prediction, thus forming the upper and lower bounds of the interval. Following Manokhin (2025), larger calibration errors produce a larger d and wider intervals, while smaller errors produce a smaller d, and therefore more narrow intervals.

Using the simplified example from Section 4.2.4, if assumed that d = 9.1 USD/barrel and the point prediction is 85 USD/barrel, the split conformal prediction interval can be given as;

$$[85 - 9{,}1, 85 + 9.1] = [75.9, 94.1]$$

Equation 17: Split Conformal Interval Example

Giving a lower bound of 75.9 USD/barrel and an upper bound of 94.1 USD/barrel, returning a total interval width of 18.2 USD/barrel.

4.3 Adaptations for time-series

In a time-series setting, the calibration problem differs from the static split-conformal setting because forecast errors are observed sequentially and may change over time. This is particularly relevant when the data-generating process moves between calmer and more volatile regimes. Instead of relying only on a fixed calibration residual from a past calibration set, time-series adaptations use recent forecasting errors to update the interval structure as new observations become available.

4.3.1 Ensemble Batch Prediction Intervals

To handle non-stationarity and complex temporal dependency, Xu and Xie (2023) introduces a framework known as Ensemble Batch Prediction Intervals (EnbPI) which builds prediction intervals for dynamic time-series settings and targets marginal coverage under weaker assumptions than classical CP. Prediction intervals generated from EnbPI are centered around a point prediction and the width of the intervals are decided by a quantile of past residuals. One feature of EnbPI is also its ability to generate a adapted both lower and upper bound of the quantile, e.g., the conformal method can increase the numerical value of the upper or lower bound as it sees fit, to adapt to sharp upwards and downwards movement for the given time-series.

EnbPI works by first training several B bootstrap models on resampled versions of the training data and then aggregates predictions across models. A "*leave-one-out*" approach is then used, where, for each observation i in the dataset, the prediction is formed by aggregating the results from only those bootstrap models that were not trained on the observation, rather than retraining the model from scratch. Following Xu and Xie (2023), let the prediction interval be expressed as;

$$C_t^{\alpha} := \left[\hat{f}_{-t}(x_t) + \hat{\beta} \text{ quantile of } \{\hat{\epsilon}_i\}_{i=t-T}^{t-1},\ \hat{f}_{-t}(x_t) + (1-\alpha+\hat{\beta}) \text{ quantile of } \{\hat{\epsilon}_i\}_{i=t-T}^{t-1}\right]$$

Equation 18: EnbPI Prediction Interval

Where C_t^{α} denotes the prediction interval at time t, Tdenotes the length of the residual history, or the backwards window, used for calibration. The parameter α denotes the miscoverage level, e.g., α = 0.10 gives 90% coverage, and $\widehat{f^{-t}}(x_t)$ denotes the leave-one-out point prediction made at time *t.* The residuals $\hat{\epsilon}_i$, following Xu and Xie (2023), can be expressed as;

$$\hat{\epsilon}_i := y_i - \hat{f}_{-i}(x_i)$$

Equation 19: EnbPI Residual Computation

Where $\widehat{\epsilon_i}$ denotes the leave one-out prediction residual for observation i ,and y_i denotes the observed value, and $\widehat{f^{-i}}(x_i)$ is the leave-one-out prediction for observation i. The residual keeps it sign ($\pm$) and not is not changed to absolute as EnbPI constructs the interval using the lower and upper empirical quantiles of past residuals. The quantile-selection parameter $\hat{\beta}$, as expressed by Xu and Xie (2023), can be expressed as;

$$\hat{\beta} := \underset{\beta \in [0,\alpha]}{\arg\min} \left[(1-\alpha+\beta) \text{ quantile of } \{\hat{\epsilon}_i\}_{i=t-T}^{t-1} - \beta \text{ quantile of } \{\hat{\epsilon}_i\}_{i=t-T}^{t-1} \right]$$

Equation 20: EnbPI Quantile Selection

Where $\hat{\beta}$ determines the upper and lower residual quantiles computed by EnbP, selecting the pair of quantiles that minimizes the interval width for the specified significance level α.

4.3.2 Adaptive Conformal Inference

Another attempt to address the limitations of exchangeability in non-stationary time-series was proposed by Gibbs & Candès (2021), who introduced Adaptive Conformal Inference (ACI). ACI can be combined with any black-box prediction model and adds an adaptive conformal layer on top of the underlying predictor. In contrast to standard CP, ACI does not keep the miscoverage level fixed throughout the prediction sequence. Rather, it updates a time-varying miscoverage parameter, given by a_t, based on whether previous prediction sets covered the true value. The step-size parameter is called gamma and denoted as γ, which in return controls how strongly a_t reacts to past coverage errors.

ACI's mechanism, as expressed by Gibbs and Candès (2021), can then be expressed as;

$$\text{err}_t := \begin{cases} 1, & \text{if } Y_t \notin \hat{C}_t(\alpha_t), \\ 0, & \text{otherwise,} \end{cases} \quad \text{where } \hat{C}_t(\alpha_t) := \{y : S_t(X_t, y) \leq \hat{Q}_t(1-\alpha_t)\}$$

Equation 21: Adaptive Mechanism of ACI

Where $(\hat{C})_t(\alpha_t)$ denotes the prediction interval at time t, $(S_t(X_t, y)$ denotes the conformity-score of the candidate value *y, and* $(\hat{Q})_t(1 - \alpha_t)$ denotes the fitted quantile threshold. The variable err_t equals 1 when the observed value Y_t falls outside the prediction set, and 0 vice versa. This error indicator is then used to update the adaptive miscoverage level.

With this approach, ACI uses recent coverage errors as a feedback signal and sequentially updates α_t as data distribution shifts during periods with volatility and regime-shifts, which makes it fit for time-series data. This also allows ACI to respond to different data distribution regimes by widening intervals after repeated misses and narrowing them when intervals has been unnecessarily wide. This adaptive approach also enables ACI to adjust the lower and upper bound as it sees fit. Following Gibbs & Candès (2021), the sequential calibration formula can then be expressed as;

$$\alpha_{t+1} := \alpha_t + \gamma\left(\alpha - \mathrm{err}_t\right)$$

Equation 22: ACI Sequential Calibration Update

Where α denotes the target miscoverage level, a_t denotes the adaptive miscoverage level at time *t*, err_t indicates whether the previous interval failed to contain the observed value, and γ > 0 is step-size parameter. If err_t = 1, the update decreases a_t, which makes the next interval wider. If err_t = 0, the update increases a_t, which makes the next interval more narrow.

4.4 Prior work

This section review follows a concept-centric approach, as outlined by Webster and Watson (2002). Rather than reviewing studies individually, the literature is organized around three concepts that together structure the field of conformal prediction for time series:

i) *Theoretical relaxation of exchangeability.*
ii) *Sequential and adaptive calibration under temporal dependence.*
iii) *Regime-conditional and state-aware approaches.*

A subsequent section addresses domain-specific applications in finance and energy markets and presents a closely related study on volatility regime evaluation in commodity futures. A final subsection identifies the three gaps that motivate the present thesis.
The reviewed literature collectively establishes that conformal prediction is a viable and

flexible framework for financial time series, but that its performance under persistent and elevated volatility remains poorly understood, particularly for crude oil markets and particularly when coverage is evaluated conditionally across pre-defined regimes rather than pooled across the full evaluation period.

4.4.1 Theoretical Relaxation of Exchangeability

The classical conformal prediction framework relies on exchangeability, an assumption that is difficult to justify in financial time series characterized by volatility clustering and regime shifts. A first stream of literature has therefore worked to relax this assumption at a theoretical level.

Barber et al. (2023) explicitly relax the classical exchangeability requirement by introducing weighted quantiles and allowing nonsymmetric fitting algorithms. Observations considered more relevant for the test point can receive more weight in the construction of prediction intervals, which becomes particularly relevant under distribution drift, where more recent observations and residuals may be more informative than older ones. This contribution provides a theoretical bridge between classical conformal prediction and later time-series-specific methods, formalizing how weighting may reduce validity loss when the data-generating process evolves over time.

A complementary perspective is provided by Oliveira et al. (2024), who offer a theoretical and empirical re-evaluation of standard split conformal prediction under non-exchangeable data. Rather than proposing a new algorithm, they show that classical split conformal prediction can still maintain approximately valid marginal coverage, provided that the dependence structure is not too strong. Their framework is based on concentration inequalities and decoupling. Their relevance lies in challenging the assumption that conformal prediction must be fundamentally redesigned once exchangeability fails. This is directly relevant to the present thesis, which retains split conformal prediction as a baseline despite the presence of temporal dependence in Brent crude oil returns.

A related contribution from Chernozhukov et al. (2021) addresses exchangeability violations through distributional conformal prediction, which incorporates models of the conditional distribution of the target given exogenous features. Multiple quantiles are estimated, allowing prediction intervals to remain approximately valid in settings with heteroskedasticity. While the present thesis does not employ quantile regression directly, this work is included because it shifts the discussion from purely exchangeability-based guarantees toward conditional distribution modelling under realistic forecasting conditions.

Taken together, these three contributions establish that the formal CP framework can accommodate temporal dependence and distributional instability to varying degrees, which motivates using CP as an empirical calibration layer in the present thesis rather than relying on its exact theoretical guarantees.

4.4.2 Sequential and Adaptive Conformal Prediction

A second stream of research moves from theoretical relaxation toward operational forecasting frameworks designed for sequential data. The central problem is how to maintain coverage when residuals exhibit temporal dependence and the data-generating process evolves over time.

Xu and Xie (2023) introduce Ensemble Batch Prediction Intervals (EnbPI), which addresses the case where data arrives sequentially and exchangeability cannot be assumed. EnbPI relies on an ensemble of bootstrap predictors trained once on the original data and a sliding window of recent residuals to recalibrate intervals over time. Leave-one-out aggregation across bootstrap models avoids repeated retraining, while the rolling residual window allows interval widths to adapt to recent error patterns. The authors demonstrate that EnbPI maintains target coverage in solar and wind energy data where competing methods fail, including settings with missing data, change points, and anomaly detection.

Gibbs and Candès (2021) introduce Adaptive Conformal Inference (ACI), which can be combined with any black-box prediction model and adds an adaptive conformal layer on top of the underlying predictor. In contrast to classical conformal prediction, ACI does not keep the calibration parameter fixed. It updates a single miscoverage parameter sequentially based on whether previous prediction intervals covered the true value, using recent coverage errors as running feedback. This allows ACI to widen intervals after repeated misses and narrow them when coverage has been excessive.

A theoretical and empirical analysis of ACI in time series settings is provided by Zaffran et al. (2022), who examine the role of the learning rate γ on both validity and efficiency. They prove that in the exchangeable case, ACI degrades interval efficiency relative to standard split conformal prediction, with interval width increasing linearly with γ. In contrast, under autoregressive residual dependence, a structure consistent with the volatility clustering documented in Brent crude oil returns, a strictly positive γ can reduce average interval width compared to $\gamma = 0$. Their simulation study further shows that EnbPI loses coverage as temporal dependence increases, while ACI with a well-chosen γ maintains valid coverage across a wider range of dependence structures. This finding is directly relevant to the present thesis: it provides empirical support for evaluating both EnbPI and ACI under the persistent volatility dynamics characteristic of crude oil markets, and motivates the expectation that the two methods may diverge in performance during stress periods. Zaffran et al. (2022) also demonstrate that even when marginal coverage is maintained, conditional coverage varies substantially across subgroups, in their electricity price application, empirical coverage ranges from 88% on weekends to 93% on weekdays. This finding reinforces the core motivation of the present thesis: aggregate coverage metrics can conceal systematic undercoverage within specific regimes, making regime-conditional evaluation a necessary complement to marginal coverage reporting.

Beyond EnbPI and ACI, several extensions to adaptive conformal prediction have been proposed. Angelopoulos et al. (2023) introduce Conformal PID control, borrowing from control theory to stabilize coverage over time and decompose ACI into proportional, integral, and derivative components. Zhang and Zhou (2025) propose adaSECP-ACI for industrial time series, addressing the lag problem that arises when adaptive conformal methods respond too slowly to sudden distribution shifts. Their framework combines signed-error nonconformity scores with a dual-median adjustment strategy, reporting 95.6% coverage while reducing prediction interval width by up to 15.7% compared to benchmark models. Zong et al. (2025) introduce Dual Adaptive Conformal Prediction (DACP), where a dual regulator system simultaneously adjusts the significance level and the training-calibration split based on recent historical coverage statistics, dynamically reallocating information between model fitting and calibration. While the present thesis does not employ these specific variants, they collectively illustrate that adaptive conformal calibration is an active research area, and that aggregate coverage alone is insufficient when interval stability and width matter.

4.4.3 Regime-Conditional and State-Aware Conformal Prediction

A third research stream moves beyond reactive recalibration and conditions calibration explicitly on regimes, latent states, or structural breaks in the time series. This direction is conceptually close to the present thesis, which evaluates conformal performance separately across volatility regimes.

Auer et al. (2023) propose HopCPT, a regime-based conformal method where historical residuals from similar regimes are given more weight, with regimes identified through Modern Hopfield Networks. Across datasets ranging from solar radiation to air quality, HopCPT produces narrower prediction intervals than EnbPI and standard conformal prediction while preserving comparable coverage. Sun and Yu (2025) take a more explicit regime-shift perspective with their change-point-aware method, which combines conformal prediction with switching dynamical systems to anticipate regime changes rather than merely react to them. Across six datasets at 90% nominal confidence, their method attains 91.2% coverage compared to 89.8% for ACI and 84.1% for Sequential Predictive Conformal Inference.

These methods build regime-awareness directly into the conformal mechanism. The present thesis takes a different position: rather than embedding regime-detection in the calibration procedure, it evaluates conformal performance ex post on regimes defined externally through GARCH(1,1) and rolling standard deviation thresholds. This distinction matters because it isolates whether calibration deteriorates under elevated volatility before asking how it might be fixed. The approach taken here is therefore diagnostic rather than corrective, and complements the methods above by establishing a baseline understanding of how existing conformal frameworks behave under different volatility states in crude oil markets. The distinction between marginal and conditional coverage that motivates this diagnostic approach was formalized by Christoffersen (1998), who showed that an interval forecast may achieve correct unconditional coverage while exhibiting systematic miscoverage during other statistical environments of the same time-series, precisely the failure mode that regime-conditional evaluation is designed to detect.

4.4.4 Domain Applications in Finance and Energy

A growing body of work applies conformal prediction in finance and energy markets, where uncertainty quantification has direct decision-making consequences.

An early and foundational application of CP to electricity price forecasting is provided by Kath and Ziel (2021), who evaluate inductive and normalized CP across three short-term power markets using linear and machine learning point forecast models. Their results show that CP produces valid and sharp prediction intervals in electricity markets, with normalized

CP performing better than standard CP at the 90% interval level in more volatile markets. Normalized CP differs from standard CP in that the nonconformity score is scaled by the model's own estimated prediction error, making the interval width sensitive to local uncertainty rather than fixed across all observations. Kath and Ziel explicitly identify structural breaks and regime changes as a fundamental limitation of CP: when the data-generating process shifts, the exchangeability assumption fails and calibration based on pooled historical residuals may no longer reflect the current error distribution. Their path-dependent analysis further decomposes CP performance into three components, being symmetry, normalization, and sampling, finding that symmetric nonconformity scores and normalization by estimated errors are the most consequential design choices for interval performance in volatile commodity markets. These findings support two design decisions in the present thesis: the use of absolute residual nonconformity scores as a transparent baseline, and the evaluation of conformal coverage separately across volatility regimes rather than pooled across the full test period.

Bastos (2024) applies gradient boosting machines and conformalized quantile regression (CQR) to option price prediction. The results show that conformal intervals achieve empirical coverage close to the nominal level, while non-conformal quantile intervals systematically undercover. For all options aggregated, CQR achieves 90% empirical coverage at 6% relative interval width, whereas non-conformal quantile regression attains only 69% coverage at 3% width. This pattern reinforces a central principle for the present thesis; narrower intervals are not useful if they systematically fail to achieve required coverage. This principle is formalized by Gneiting and Raftery (2007), who show that a proper scoring rule for interval forecasts must penalize both excessive width and missed observations, rewarding neither sharpness nor coverage alone, and propose the Interval Score as a utility function that addresses both dimensions simultaneously.

O'Connor et al. (2025) compare conformal approaches including EnbPI, split conformal prediction, and sequential predictive conformal inference in the Irish day-ahead and balancing electricity markets. Their results show that adapted conformal methods deliver both stronger Winkler scores and stronger trading returns than alternatives, suggesting that conformal calibration adapted to time-series data may improve not only statistical performance but also downstream economic usefulness. Luna et al. (2025) apply conformal prediction with XGBoost to Atlantic salmon price hedging, using interval widths to drive dynamic hedge ratios. Tested on real-world cases over two and a half years, they show that conformal-driven hedging produces more stable income and reduces financial loss during price drops. Both contributions illustrate that conformal intervals can translate predictive uncertainty into operational action in commodity markets.

4.4.5 Closely Related Work

A particularly close recent contribution is Juan et al. (2026), who develop a leakage-free, regime-conditioned framework for forecasting silver futures returns and evaluating conformal calibration under volatility shock regimes. Using daily silver futures data, they construct log returns and a 20-day realized-volatility proxy, classify shock regimes through a rolling quantile threshold estimated strictly from past observations, and evaluate random walk, ARX-Ridge, and XGBoost models in a rolling out-of-sample design. Online rolling conformal intervals are formed using absolute residuals and a 252-day calibration window. Their results are directly relevant to the present thesis: conformal intervals widen materially during volatility shock regimes, with average width increasing from 7.011 to 9.229 at horizon 1 for XGBoost, while empirical coverage simultaneously deteriorates from 0.950 to 0.860. The authors interpret this as evidence that distribution shift can outpace finite-window calibration, degrading the practical usefulness of classical conformal guarantees precisely when reliable uncertainty quantification is most valuable. Whether analogous coverage deterioration arises in Brent crude oil remains an open empirical question, and represents the primary motivation for the empirical design of the present thesis.

4.4.6 Identified Gaps

Three gaps are identified that the present thesis addresses. First, in terms of domain, no prior study evaluates conformal prediction interval performance under conditional volatility regimes for crude oil markets specifically. Existing applications cover electricity prices (Kath and Ziel, 2021; O'Connor et al., 2025), salmon prices (Luna et al., 2025), silver futures (Juan et al., 2026), and option prices (Bastos, 2024). Crude oil, the most internationally referenced commodity benchmark, is absent.

Second, in terms of model class, tree-based models with conformal layers are underrepresented in the time-series literature, which is dominated by recurrent neural network and Transformer baselines. Bastos (2024) and Juan et al. (2026) are notable exceptions. The present thesis evaluates four tree-based models under three conformal frameworks, contributing to an emerging but still limited body of evidence on how gradient-boosted and ensemble methods behave as base learners within the CP framework.

Third, in terms of the broadness of different evaluation scopes, most studies report aggregate marginal coverage as the primary performance criterion. Kath and Ziel (2021) explicitly warn that exchangeability fails under regime shifts, yet their own evaluation remains pooled across the full sample. Zaffran et al. (2022) demonstrate that marginal coverage can mask systematic undercoverage within subgroups, with coverage ranging from 88% to 93% across days of the week in their electricity price application. Juan et al. (2026) show that coverage deteriorates specifically during shock regimes, but their regime classification is based on a single rolling volatility measure without cross-validation against a model-based volatility estimate.

The present thesis addresses this gap by evaluating conformal coverage separately across stress and non-stress regimes defined through a joint GARCH(1,1) and rolling standard deviation threshold, combining model-based and non-parametric volatility signals to produce a more conservative and reproducible regime classification than any single measure alone.

5 Methodology

5.1 Research Design

The study applies a sequential out-of-sample design to investigate whether conformal prediction intervals remain reliable and efficient in Brent price oil forecasting methods during periods of elevated volatility. The empirical design applies the chronological structure of financial time series by training prediction algorithms on earlier observations, calibrating conformal intervals on subsequent data, and evaluating both single point forecasts and prediction intervals on unseen future data. The setup is intended to mirror a realistic, industry-applicable forecasting environment and to avoid look-ahead bias. As Brent crude prices contain temporal dependence, instability and volatility clustering, conformal prediction is used in this experiment as an interval framework rather than a method meant to guarantee coverage. Emphasis on empirical coverage and interval efficiency across stress and non-stress regimes is meant to give an empirical answer to how conformal intervals behave during turbulent markets.

5.2 Data Collection and Feature Selection

The empirical analysis is based on a multivariate daily time series dataset assembled from publicly available and well-established institutional sources. Brent spot price (FOB Europe) constitutes the primary variable of interest and serves as the target series throughout the study. In addition to Brent, the dataset includes a set of explanatory variables selected on economic grounds to reflect key dimensions of oil price formation, including benchmark crude market dynamics, supply-side conditions, substitution effects within energy markets, financial market sentiment, macro-financial conditions, commodity price co-movements, and geopolitical uncertainty.

Brent and WTI spot prices were obtained publicly from the U.S. Energy Information Administration (EIA), allowing the thesis to analyze both differences and similarities of the two commodities, laying grounds for the rest of the analysis, although WTI prices is not applied in the analysis post the WTI-Brent comparison. OPEC production volume was included as a direct measure of supply-side developments in the global oil market and was likewise collected from the EIA. Natural gas prices were incorporated to account for interactions across energy commodities and were retrieved through the FRED API.

Broader financial indices were represented by the S&P 500 Index and the VIX Index, which shows general market performance and perceived financial short-term uncertainty. These variables were included to reflect the extent to which movements in oil prices are rooted in wider risk-on/risk-off dynamics. The dataset also contains macro-financial indicators expected to be relevant for the pricing of internationally traded commodities. The U.S. 10-year Treasury yield was included as an indicator of interest rate conditions and macroeconomic expectations, while the U.S. Dollar Index was used to capture exchange-rate-related effects associated with dollar-denominated commodity pricing. Gold, silver, and copper prices were also incorporated to account for broader cross-commodity linkages and common movements in global raw material markets, while assets like gold also function as a "safe haven" during turbulent geopolitical periods.

Geopolitical risk was proxied by the Geopolitical Risk Index (GPR) developed by Caldara & Iacoviello (2022) and was included to reflect the role of political tension and international conflict in shaping oil market uncertainty. As documented by Li et al. (2022), the GPR index performed better to predict crude oil volatility than other relevant uncertainty features, thus supporting its inclusion. The index works by being based on searches in the archives of 10 leading global newspapers, primarily from the western hemisphere. It therefore also works as a "text-derived geopolitical risk indicator". The methodology behind the index is worked by aggregating the number of written articles related to certain geopolitical incidents in each newspaper each month. Eight distinct categories are included in the search, ranging from: *War Threats (1), Peace Threats (2), Military buildups (3), Nuclear threats (4), Terror Threats (5), Beginning of War (6), Escalation of War (7), Terror acts (8).*

Table 2: Exogenous Variables and Data Sources

Variable	Economic Category	Data Source
Brent Spot Price (FOB Europe)	Energy – Oil Benchmark	U.S. EIA (RBRTE.D)
WTI Spot Price (Cushing)	Energy – Oil Benchmark	U.S. EIA (RWTC.D)
OPEC Production Volume	Energy – Supply Side	U.S. EIA
Natural Gas Price	Energy – Substitutes	FRED API
S&P 500 Index	Financial Markets – Equity	Yahoo Finance (GSPC)
VIX Index	Financial Markets – Volatility	CBOE
US 10Y Treasury Yield	Macroeconomic – Interest Rates	FRED API
US Dollar Index	Macroeconomic	FRED API
Gold Price	Commodities	Yahoo Finance (GC=F)
Silver Price	Commodities	Yahoo Finance (SI=F)
Copper Price	Commodities	Yahoo Finance (HG=F)
Geopolitical Risk Index (GPR)	Sentiment / Political Risk	Caldara & Iacoviello

The table reports the exogenous variables used in the empirical analysis, their economic classification, and the corresponding data source.

5.2.1 Data Preprocessing, Cleaning and Normalization

After combining all exogenous variables and relevant technical price indicators into a unified dataset, further preprocessing was necessary to obtain a complete and aligned dataset suitable for the thesis. Although Brent crude oil prices were available as far as 1987-05-20, comparable historical prices were not available for all collected exogenous variables. A common starting date therefore had to be imposed, and to retain the full selected set of macroeconomic and financial variables while avoiding missing and otherwise unusable observations, the sample set was set to begin on 2000-09-13 and ending on 2025-10-01. The cutoff removed missing values from Gold, Silver, and Copper prices.

To construct a continuous weighted US dollar index over the full sample period, two different indices were gathered and merged, both obtained through FRED API. The newer broad trade-weighted dollar index (DTWEXBGS) is available from 2006 and onwards, while the older series (DTWEXB) was used to maintain data integrity during the earlier period. As the two series are not expressed with the same numerical values on the same scale, a rescaling was made to the older series before splicing. This was done by; identifying the overlapping period where both series were available and calculating the median ratio between the new and the old index. A median ratio was used to reduce sensitivity to temporary fluctuations or outliers in the overlapping window.

The next step was to multiply the older series with the scaling factor to align the numerical value with the newer series. A continuous index was created by merging the rescaled older series with the newer series. When possible, series were collected at daily frequency when available and subsequently aligned on a common date index. If series were not available in a daily format (as with the GPR index), a forward-fill technique was applied to the remaining calendar days to ensure alignment with the common date index.

5.2.2 Feature Construction

In addition to the exogenous variables added, a set of 6 technical indicators was constructed directly from the Brent price series and included in the final feature set. These variables were constructed to capture short-term price dynamics that might not be fully captured by the external and macroeconomic variables alone. The final feature set then included the daily log return, a one-day lag of the Brent price, one-and two-day lags of the log return, and rolling volatility measures computed from log returns over 5 and 10 trading days. These indicators were meant to provide information to the models about recent price direction, short-run fluctuations, and local time-step volatility conditions and were included after several pre-runs to strengthen the model's ability to capture near-term price movements, yielding better results than runs done on only exogenous variables, which empirically justified including them in the final dataset.

Table 3: Technical Indicators Included in the Feature Set

Variable	Type	Description
`log_return`	Return	Daily log return of Brent crude oil price
`price_lag_1`	Lagged price	Brent price lagged by one trading day
`log_return_lag_1`	Lagged return	Log return lagged by one trading day
`log_return_lag_2`	Lagged return	Log return lagged by two trading days
`rolling_vol_5`	Volatility	Rolling 5-day volatility of log returns
`rolling_vol_10`	Volatility	Rolling 10-day volatility of log returns

The table reports the technical indicators constructed from the Brent price series and included as model input variables.

5.3 Quantifying Volatility and Stress-Episodes

5.3.1 Time-varying volatility and conditional heteroskedasticity

Earlier research on the underlying statistical properties of financial time series has identified a large set of recurring regularities, also known as stylized facts. These include; excess volatility, heavy tails, absence of autocorrelation in return series, volatility clustering, and volume/volatility correlation (Cont., 2001, 2007). As described early by Mandelbrot (1963), volatility clustering is defined as when large changes are followed by large changes, and small changes are usually followed by small changes. In econometric terms, this indicates that the conditional variance of returns depends on lagged information, rather than remaining fixed and static throughout the full sample.

As the thesis focuses on elevated, conditional volatility, defined as stress-periods, the conditional variance is directly relevant for the problem at hand. The objective is not only to identify days with high or low prices, but to identify longer periods where the uncertainty surrounding the price series deviates from baseline expectations. A high price isolated does not have to imply market stress, and vice versa, a low price does not have to imply market stability. Classification of stress episodes therefore needs to be based on strength of the price movements instead of binary high/low flagging. The following sections operationalize this logic in three steps;

i) log-return transformation of raw prices,
ii) quantifying volatility,
iii) define stress episodes through a joint threshold approach.

5.3.2 Log-Return Transformation

Adhering to standard practice in financial time series analysis, the Brent spot price series was transformed into daily logarithmic returns before measuring volatility (Tsay, 2005). As logarithmic returns measure relative price movements rather than the absolute price level, this was an appropriate transformation. Any given change in USD/barrel has different economic meanings when the price is low compared with when it's high, therefore giving a more comparable measure of daily price variation across the gathered sample. The formula is expressed as;

$$r_t = \log(P_t) - \log(P_{t-1})$$

Equation 23: Log Return

Where P_t denotes the Brent spot price on timestep t, and P_{t-1} denotes the Brent spot price on the previous day. The variable r_t therefore represents the daily logarithmic return, which measures the relative change in Brent price from one trading day to the next.

5.3.3 Quantifying volatility with Rolling Standard Deviation

Rolling standard deviation (RSTD) is used to quantify local volatility in a time series by computing the standard deviation within a fixed length moving window. Unlike the unconditional standard deviation, which summarizes dispersion from the entire sample, RSTD updates as the window progresses over time, enabling identification of changing volatility levels. A window length of $m = 30$ trading days was applied to the log-return series. For each time step t, volatility was estimated as:

$$\sigma_t = \sqrt{\frac{1}{m-1} \sum_{i=t-m+1}^{t} (r_i - \bar{r}_t)^2}$$

Equation 24: Rolling Standard Deviation

Where r_i denotes the log return at time i , $\bar{r}_t$ denotes the mean log return over the rolling window ending at time t , and m represents the window length of 30. Thus, σ_t represents the rolling standard deviation of log returns computed over the most recent m trading days. This measure provides a backward-looking estimate of realized volatility but remains a descriptive rolling measure of log-return distribution within a fixed window. It is therefore complemented by a GARCH-based volatility estimate, which explicitly models volatility persistence through past return shocks and previous volatility levels.

5.3.4 Quantifying volatility with Generalized Autoregressive Conditional Heteroskedasticity

Originally introduced by Engle (1982) as the autoregressive conditional heteroskedasticity (ARCH) model and later generalized (G) by Bollerslev (1986), Generalized Autoregressive Conditional Heteroskedasticity (GARCH) models are used as a statistical framework to model and estimate the time-varying volatility of a given financial market or instrument. This is done by assuming that tomorrow's volatility is dependent on three things: a baseline level of inherent risk to the market or financial instrument, yesterday's squared return, and how volatile the market already was. The structure of GARCH captures two empirical structures of financial markets, being persistent volatility, and large movements are often followed by large movements, clustering volatility in periods before stabilizing into lower-volatility regimes, making it a good fit for measuring how the log-return series of Brent are distributed. The equation of a GARCH (1,1) model is given as:

$$\sigma_{t+1}^2 = \omega + \alpha_1 r_t^2 + \beta_1 \sigma_t^2$$

Equation 25: GARCH(1,1)

Where σ_{t+1}^2 indicates the volatility tomorrow, r_t^2 shows the squared return observed today, and σ_t^2 is the variance estimated for today. A pre-defined threshold of values exceeding the 90th empirical percentile of the estimated volatility distribution was flagged, returning the following graphical overview of the two chosen methods.

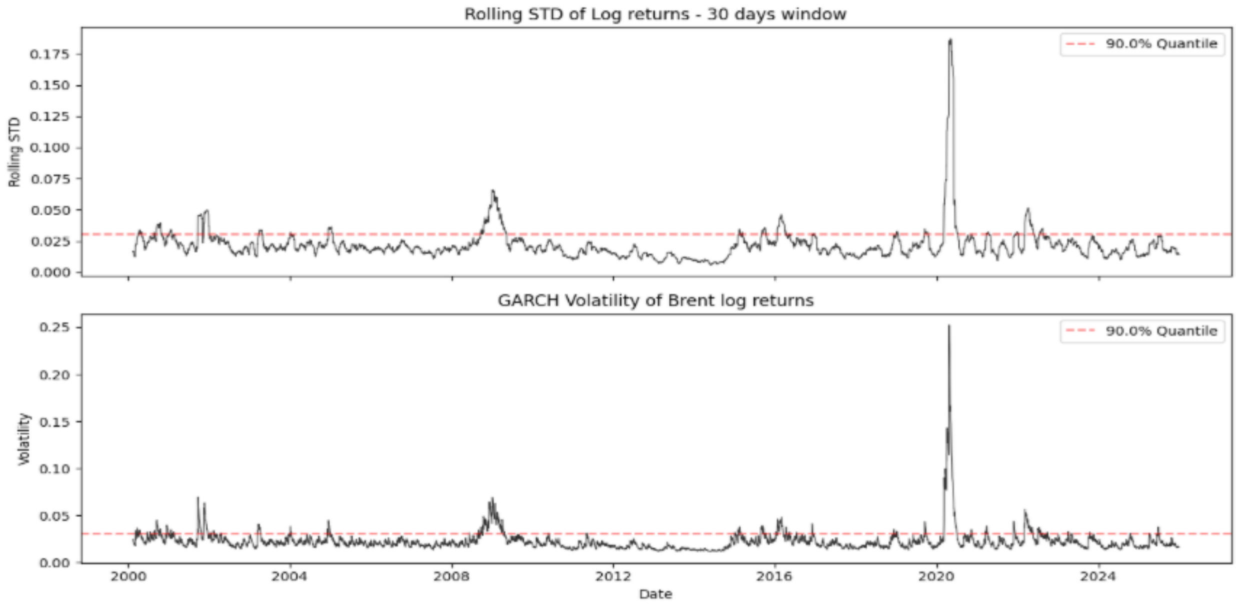

Figure 6: Rolling STD and GARCH Flagged Periods above the 90th Percentile

5.3.5 Operationalizing the Uncertainty Framework under a Joint Threshold Rule

To operationalize the volatility framework into discrete and reproducible stress episodes, periods are classified as stress regimes when both volatility measures exceed their respective 90th empirical percentile thresholds. A stress date is defined when the conditional volatility estimated by the GARCH (1,1) model and the 30-day rolling standard deviation simultaneously indicate extreme dispersion relative to the full sample distribution.

This raw AND condition identified 489 stress-flagged days. To avoid treating isolated volatility spikes as meaningful stress regimes, consecutive stress blocks were then grouped into episodes, allowing short gaps of up to three days between blocks. Only episodes containing at least ten stress-flagged days were kept for further analysis. The final stress indicator, *'stress_m10'*, therefore represents persistent volatility episodes rather than individual high-volatility days. This procedure resulted in ten identified stress episodes, covering 422 days in total.

Table 4: Stress Classification Based on 90th Percentile Volatility Thresholds

Classification criterion	Number of days
GARCH $\geq$ 90th percentile	657
Rolling STD $\geq$ 90th percentile	657
Final stress-regime days after episode filtering	422

The table reports the number of observations flagged by each volatility threshold and the final number of stress-regime days retained after filtering.

By requiring agreement between a model-based volatility measure (GARCH 1,1) and the rolling standard deviation to signal unusually high volatility at the same time, the identified stress periods reflect periods of elevated and persistent market uncertainty rather than short lived fluctuations. It is also important to underscore that the market episodes were only used for regime evaluations and were not created as explanatory input features. The stress indicator is therefore used as an ex-post evaluation variable, not as a real-time forecasting signal or model input. Its role is to separate the test observations into volatility regimes so that interval coverage and efficiency can be evaluated conditionally.

Table 5: Identified Stress Episodes Based on the Joint Volatility Criterion

Episode	Start date	End date	Duration (trading days)
1	2000-09-13	2000-10-26	32
2	2001-09-24	2001-10-24	23
3	2001-11-14	2001-12-20	27
4	2003-03-27	2003-04-10	11
5	2004-12-16	2004-12-31	12
6	2008-10-13	2009-04-22	132
7	2015-09-03	2015-09-24	16
8	2016-01-22	2016-03-18	40
9	2020-03-09	2020-07-01	79
10	2022-03-09	2022-05-20	50

Stress episodes are identified from consecutive observations satisfying the joint volatility criterion after applying the episode-filtering rule. Duration refers to the number of trading-day observations retained in each episode.

The identified stress episodes indicate that elevated volatility in the Brent market is not evenly distributed over time but instead concentrated in a limited number of distinct periods. Several of these episodes appear simultaneously as well-known episodes of market disruption, including the global financial crisis, the 2015–2016 oil market downturn, the COVID-19 shock, and the 2022 energy crisis. These findings further support the use of a regime-based evaluation framework, where conformal prediction performance can be assessed separately during sustained stress conditions and more stable market periods.

5.3.6 Sampling Uncertainty

50/954 of the observations in the test set are classified as stress days. As empirical coverage is estimated as the proportion of observations captured by the prediction interval, small changes in coverage or 'hits' can alter the coverage percentage. Each individual observation represents 1/50 = 0.02, meaning that one additional 'hit'or 'miss' changes the estimated stress coverage by two percentage points. Let's assume that 25/50 observations are covered, which gives 50% empirical coverage, whereas 26/50 returns 52% empirical coverage. Stress coverage can, and should, therefore, be interpreted as a rough estimate rather than a precise estimate of long-run conditional coverage and is therefore supplemented with other performance metrics such as interval width and Winkler score.

5.4 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted primarily on a transformed log-return of the target variable, except for when analysis of the raw price series was statistically sound. The findings are meant to give an overview of the dataset (exogenous variables included), and to support the main research themes of the thesis; namely volatility, fat tail behavior, and regime changes. The EDA focuses on three main areas: visual inspection of the target series and explanatory variables, volatility and distribution of Brent log returns, and stationarity testing of the Brent price level combined with return-based volatility analysis.

5.4.1 Initial Inspection of Target and Exogenous Variables

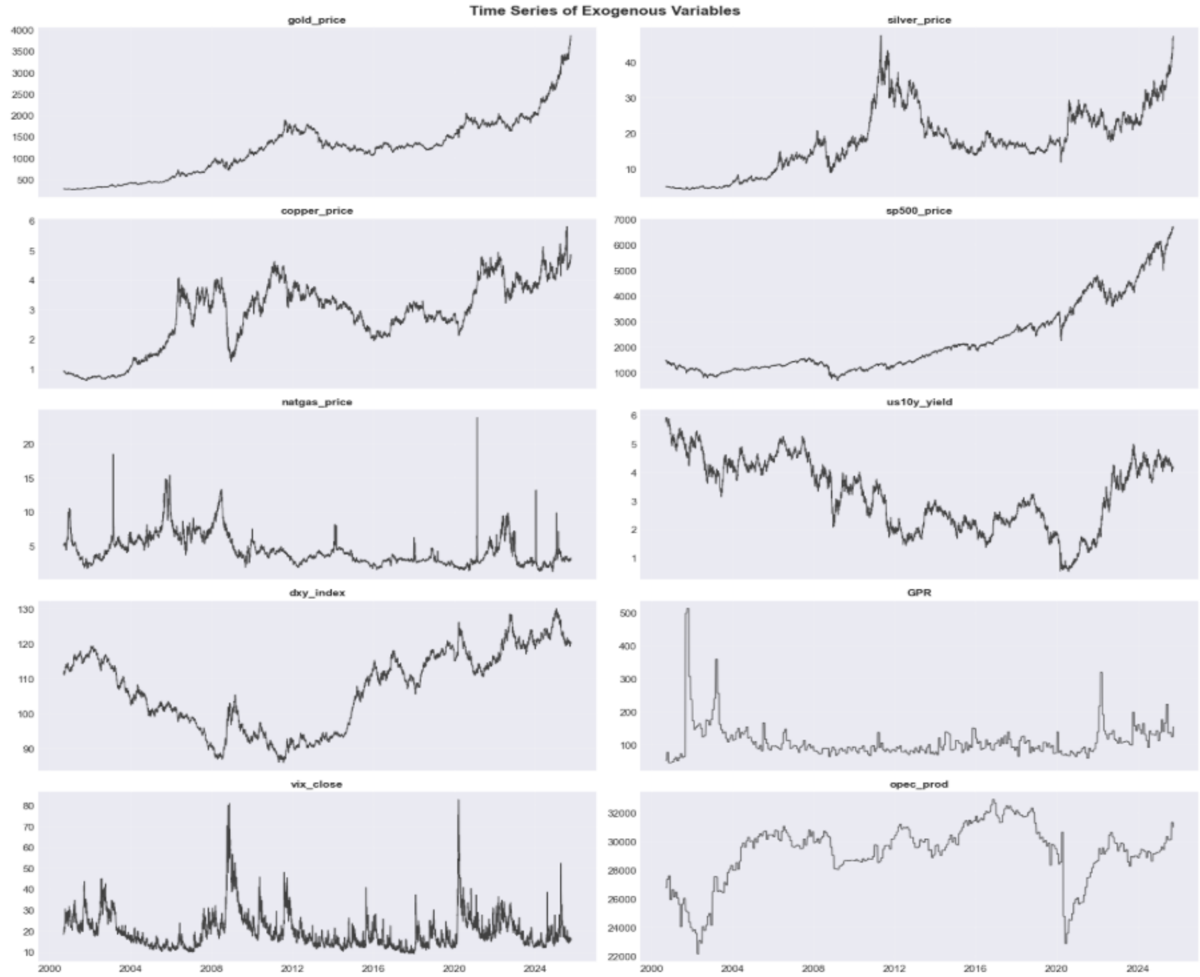

Figure 7: Time Series Plot of Exogenous Variables

As seen in the time-series plot of all numerical exogenous variables, several of the features (namely 'vix_close', 'dxy_index', 'us10_yield' and 'GPR') all show persistent trends and level shifts. Long-term upwards trends are also visible for commodity prices like gold, silver, & copper, while the same is visible for equities like the broader SP500 index.

Large shocks can be seen for several of the explanatory features during periods of heightened volatility (2008 & 2020), suggesting a spill-over effect of market stress and volatility across financial instruments. Such movements may indicate that the forecasting problem at hand can involve time-varying relationships and regime-dependent uncertainty, rather than a stable relationship between the target variable and the explanatory features.

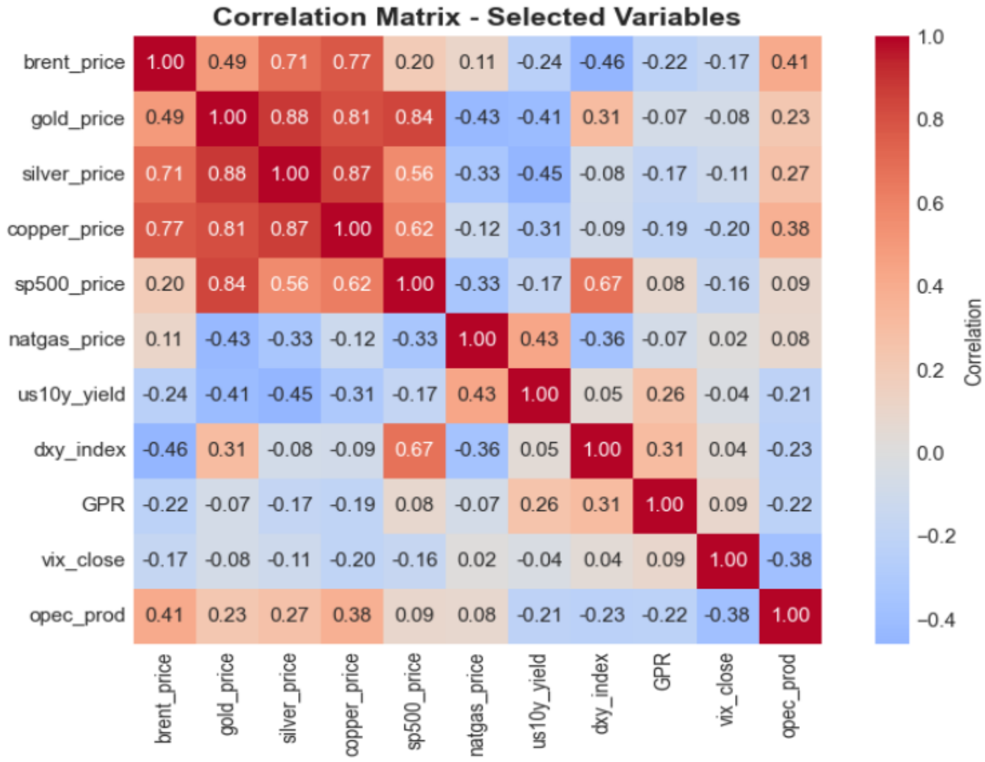

Figure 8: Correlation Matrix of Exogenous Variables

The correlation matrix provides an initial overview of linear relationships between Brent crude oil prices and the selected numerical features. The matrix works by giving the linear relationships values between *r –1 to +1 (where r = the correlation coefficient, usually Pearson's)*, where r = 0 equals no linear correlation. Brent prices can be seen to positively correlate with several commodity-related features, particularly copper (*r = 0.77),* silver (*r = 0.71),* gold (*r = 0.49), and* OPEC production numbers (*r = 0.41*). Contrastingly, Brent prices show negative correlations with the dollar index (*r = -0.46*), the U.S. 10-year Treasury yield (*r = -0.24*), GPR-index (*r = -0.22*), and the VIX index (*r = -0.17*). The role of the U.S. dollar in commodity pricing shows up given that the most pronounced negative correlation is the dollar index, explained by a stronger dollar makes oil more expensive for buyers using other currencies, which may reduce demand for oil and put downward pressure on the price.

5.4.2 Distribution of Brent returns

A time-series graph of daily log returns, expressed in percentage terms, together with the major identified stress periods defined by the joint GARCH (1,1) and RSTD criteria was first plotted. The series fluctuates around zero for most of the sampling period, but large absolute returns are concentrated in specific periods rather than being evenly distributed. This pattern exhibits clear volatility clustering, where large movements tend to be followed by large movements. Shaded periods show recognizable episodes of oil-market disruption, further showcasing the effect of geopolitical turmoil on the market, as the Great Financial Crisis (GFC), 2014-2016 oil market downturn, COVID-19 shock and the Russo-Ukraine war are all visible.

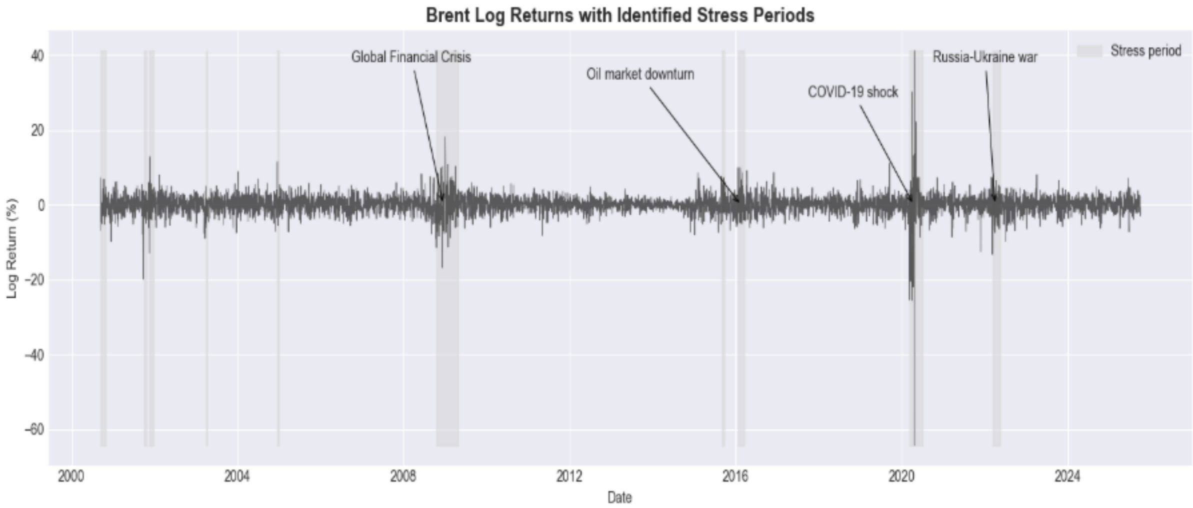

Figure 9: Log Returns with Shaded Stress Periods

Descriptive statistics for the two sample periods show clear differences in the underlying data distribution. The stress regime not only shows higher variance, but also a different return environment. The mean return remains close to zero in both regimes, but turns negative for the ‘Stress’ period, indicating that elevated volatility as defined by GARCH(1,1) and RSTD is associated with weaker average day-to-day return. What also sticks out is the distribution;

Table 6: Descriptive Statistics for Brent Log Returns Across Non-Stress and Stress Regimes

Statistic	Non-Stress	Stress
Mean	0.0002	-0.017
Standard deviation	0.0204	0.0666
Skewness	-0.1675	-1.6629
Excess kurtosis	2.0283	25.5628
Observations	5937	422

The table reports descriptive statistics for daily Brent log returns across non-stress and stress regimes.

Standard deviation increases from 0.0204 to 0.0666, meaning that the size of the distributional values increases threefold under stress, indicating that the framework of defining regimes is working as intended. Skewness also changes from mildly negative (-0.1675) to strongly negative (-1.6629), indicating a stronger left, fat tail (where *mean* < *median*). Kurtosis also supports the interpretation that stress periods contain more extreme observations; stress-kurtosis is 25.5628, while non-stress kurtosis is 2.0283, indicating a much higher concentration of extreme observations during stress episodes. As stress periods exhibit substantially fatter tails, prediction intervals calibrated on a pooled historical distribution may underestimate uncertainty during turbulent regimes which can result in lower empirical coverage.

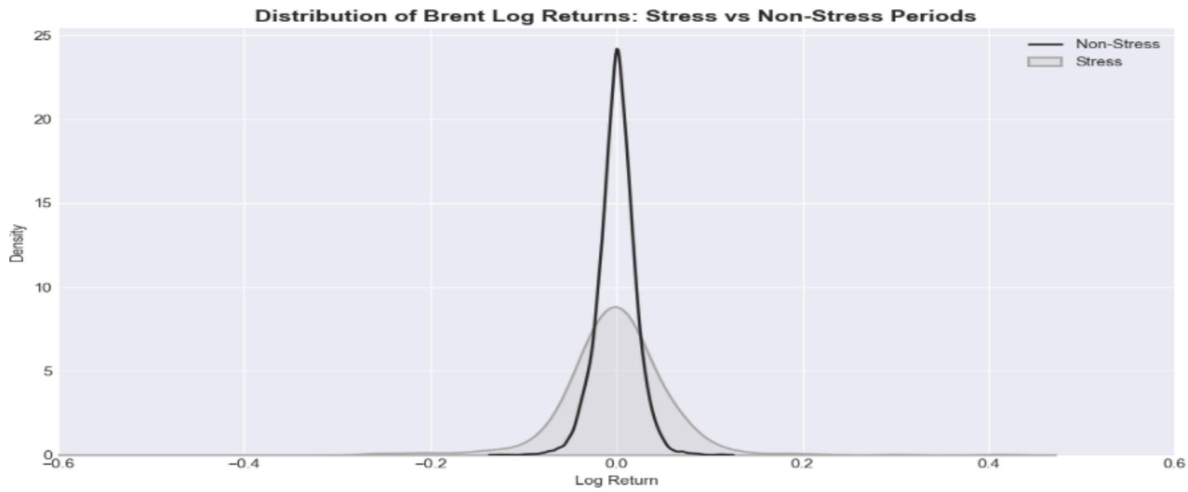

Figure 10: Distribution of Log Returns

Log returns for stress vs non-stress compare the density of both periods. Non-stress clearly indicate a concentration around zero, indicating that most returns are smaller during calmer periods. The stress distribution, however, is flatter and wider, showing that both large positive and negative returns occur more frequently during periods of elevated volatility.

5.4.3 ACF Analysis

Autocorrelation Function (ACF) was applied to the Brent log return series and squared Brent log returns as a descriptive diagnostic method to examine linear dependence in the Brent return series across different lags. The ACF of log returns was used to assess whether the log return series itself displays linear autocorrelation, while ACF of squared log returns was used to examine whether high-volatility days tend to cluster over time. As volatility is related to the size of price movements and not their direction, the inclusion of squared log returns is important. In the log return series, positive and negative movements may cancel each other out, which in return leads to weaker autocorrelation even when the market is volatile. By squaring the returns, the $\pm$ sign is removed, meaning that both positive and negative returns contribute to the measure of return magnitude.

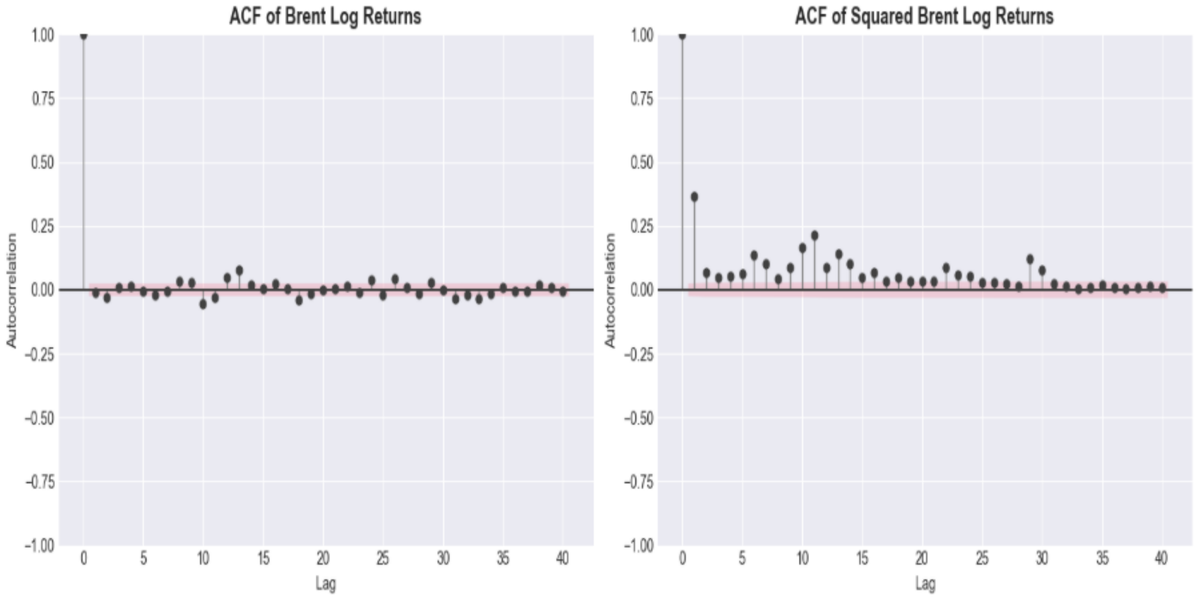

Figure 11: ACF of Log Returns and Squared Log Returns

ACF of Brent log Returns shows little evidence of linear autocorrelation after lag *0* (where the series is compared to itself, thus returning *1.00*). This suggests that the return series itself contains limited linear dependence, e.g., past returns may provide little information about the direction of future returns.

In the context of the thesis, this will be insufficient as this return series will for instance be flagged as low ACF; *+4%, -3%, +5%, -4%, +3%* when in practice the series show high volatility. By contrast, ACF of squared Brent log returns provide insights into the movement of absolute values. A stronger and more persistent autocorrelation is shown, especially at shorter and medium lags (where lags indicate timesteps/daily observations). As the sign of the return is removed for squared returns, the patterns indicate that large movements tend to cluster over time, especially in the short to medium-term. Results are therefore consistent with volatility clustering and support the usage of RSTD and GARCH(1,1) for identifying stress-periods.

5.4.4 Stationarity testing

The Augmented Dickey-Fuller test is used to assess whether a time series contains a unit root. A unit root indicates that the given time-series is non-stationary, meaning that the underlying statistical properties is likely to change over time. With the presence of a unit root, forecasting can become more unreliable and deviate model performance. The lag length was selected automatically using The Akaike Information Criterion (AIC) approach, which balances model fit automatically against complexity. The ADF test provides the following H_0 and H_a;

$H_0 =$ *Series contain a unit root (non-stationary series)*

$H_a =$ *Series does not contain a unit root (stationary series)*

Table 7 reports the ADF results for raw Brent price. ADF statistic is –2.549958, while critical values are –3.431382 at 1%, -2.861996 at 5% level, and –2.567013 at 10% level. To reject the null hypothesis, the ADF statistic must be more negative than the given critical value. As shown below, this criterion is not fulfilled for any of the critical value levels. A p-value of 0.103787 indicates the same conclusion of non-stationarity, as it is above the 5% significance level. ADF test therefore fails to reject H_0, indicating non-stationarity for the raw Brent price series.

Table 7: ADF Test Results for Brent Price Level

Test component	Value
ADF statistic	-2.54958
p-value	0.103787
Lags used	16
Observations	6,342
Critical value (1%)	-3.43182
Critical value (5%)	-2.861996
Critical value (10%)	-2.567013

The Augmented Dickey-Fuller test is applied to the Brent price level.

5.4.5 Summary of exploratory findings

The EDA points to a forecasting setting where the main difficulty is not only predicting the next Brent price, but doing so under changing market conditions. The Brent price level is used as the forecasting target, but the return-based analysis reveals why uncertainty around this target cannot be treated as constant. Large daily movements are concentrated in specific historical episodes rather than spread evenly across the sample. The stress regime captures this directly: the standard deviation of Brent log returns rises from 0.0204 in non-stress periods to 0.0666 in stress periods, while skewness becomes strongly negative and excess kurtosis increases sharply. This means that stress periods are not just more volatile; they also contain more extreme downside movements.

The surrounding market variables follow the same pattern of instability. Commodity prices, equity markets, the dollar index, interest rates, VIX, GPR, and OPEC production all contain visible trends, shifts, or shocks. The correlation matrix confirms that Brent is connected to broader commodity and macro-financial conditions, especially copper, silver, gold, OPEC production, and the dollar index, but these relationships are unlikely to remain constant across the full period. The ACF plots support this interpretation: returns themselves show weak directional persistence, while squared returns remain autocorrelated, meaning volatility clusters even when return direction is hard to predict. Since the ADF test also fails to reject non-stationarity in Brent prices, the EDA motivates the thesis focus directly: conformal intervals calibrated on past residuals may be reliable on average, but their behaviour during high-volatility regimes must be evaluated separately.

5.5 Evaluation Metrics

The empirical evaluation is based on two categories of performance metrics. For point forecasts generated by ML-models, the thesis uses Mean Absolute Error (MAE) and Root Mean Square Error (RMSE). For prediction intervals, the analysis applies empirical coverage, mean interval width, and the Winkler score. Interval performance is reported for the full test set, as well as separately for stress and non-stress periods for comparison.

5.5.1 Point Forecast Metrics

Mean Absolute Error (MAE) is applied as an evaluation metric to evaluate the average size of the Brent price forecast errors. With a forecasting target of *t + 1,* MAE measures how far the predicted next-day price is from the actual observed next-day price, expressed directly in the same units as the Brent price, thus making the metric straightforward to interpret. An MAE of 2.25 for instance, means that the models' predictions are, on average, 2.25 USD/barrel away from the actual registered price on that timestep. Hyndman and Athanasopoulos (2018) describe MAE as one of the most commonly used scale-dependent forecasting measures. A lower MAE indicates more accurate model performance, with a value of zero representing a perfect fit. The formula is given as:

$$\mathrm{MAE} = \frac{1}{n}\sum_{t=1}^{n}|y_t - \hat{y}_t|$$

Equation 26: Mean Absolute Error

Where y_i represents the observed Brent price, and $\hat{y}_i$ is the forecasted value, and n is the number of observations in the evaluation sample. As MAE uses absolute errors, positive and negative errors do not cancel each other out. Unlike other metrics such as RMSE, MAE does not square the errors before averaging them. Larger errors therefore affect MAE in direct proportion to their size, rather than receiving additional weight from squaring.

Root Mean Squared Error (RMSE) is applied as an evaluation metric to evaluate the magnitude of the Brent price forecast errors, with a stronger emphasis on large deviations compared to the MAE metric. As with MAE, RMSE is a scale-dependent measure and is therefore also expressed in the same unit as the Brent price (Hyndman & Athanasopoulos, 2018). What RMSE does differently is to square each forecast error before averaging and then taking the square root, making the metric more sensitive to large errors compared to MAE. The formula is given as:

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{t=1}^{n}(y_t - \hat{y}_t)^2}$$

Equation 27: Root Mean Squared Error

Where y_i represents the observed Brent price, $\hat{y}_i$ represents the forecasted value, and n is the number of observations in the sample. The inclusion of RMSE as a metric is justified because large errors during volatile periods are not merely average deviations, as they indicate that forecasting models experience heightened struggle when oil prices move abruptly.

5.5.2 Interval Metrics

Regression Coverage Score (RCS) or empirical coverage measures the proportion of observed values that fall within the prediction interval. For a $1 - \alpha$ interval, often at 0.90, the empirical coverage indicates how closely the interval procedure matches the intended nominal coverage level in practice (MAPIE, 2026). Coverage is important for the thesis and subsequent analysis because the main objective is to assess whether conformal prediction intervals maintain their nominal level during periods that contain elevated conditional volatility. The formula can be expressed as;

$$\mathrm{RCS} = \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}\left(\hat{y}_i^{\mathrm{low}} \leq y_i \leq \hat{y}_i^{\mathrm{up}}\right)$$

Equation 28: Regression Coverage Score

Where n denotes the number of observations, y_i denotes the observed value for observation i, $\hat{y}_i^{\mathrm{low}}$ and $\hat{y}_i^{\mathrm{up}}$ denotes the lower and upper prediction interval bounds for observation i.

Regression Mean Width Score (RMWS) is used to measure the average distance between the upper and lower bounds of the prediction intervals (MAPIE, 2026). It lays out an indication of interval sharpness, where narrower intervals are preferred, provided that they still maintain adequate coverage. Interval alone, however, is not sufficient, as very narrow intervals can lead to systematic undercoverage, and too wide intervals may achieve adequate coverage by becoming uninformative. Due to this, both mean interval width is interpreted in symbiosis with both empirical coverage and the Winkler score, as one performance metric alone is not sufficient to answer what this thesis seeks to analyze. The formula is given as:

$$\mathrm{RMWS} = \frac{1}{n}\sum_{i=1}^{n} \left(\hat{y}_i^{\mathrm{up}} - \hat{y}_i^{\mathrm{low}}\right)$$

Equation 29: Regression Mean Width Score

Where n denotes the number of samples in the test sample, while $\widehat{y_i^{low}}$ and $\widehat{y_i^{up}}$ denotes the upper and lower bounds of the generated prediction interval for the $i-th$ observation.

The Winkler score, originally proposed by Winkler (1972), evaluates prediction intervals by combining interval width with a penalty for non-coverage. For a $(1 - \alpha)$ prediction interval with lower bound l and upper bound u , the formula is given as follows:

$$W_\alpha = \begin{cases} (u-l) + \frac{2}{\alpha}(l-y), & \text{if } y < l, \\ u - l, & \text{if } l \leq y \leq u, \\ (u-l) + \frac{2}{\alpha}(y-u), & \text{if } y > u. \end{cases}$$

Equation 30: Winkler Score

The formula for Winkler score defines the value for each individual prediction interval. In the empirical evaluation, the mean score is computed across all observations, thus returning a comparable value across regimes. y denotes the true value, u the upper bound, and l the lower bound, and α is the miscoverage level. Lower values of the Winkler score indicate better interval forecasts. If the observed value falls within the interval, the score equals the interval width, $u - l$, which rewards more precise intervals. If the observed forecasted value falls outside the interval, an additional penalty is added, corresponding to the distance between true value y and the violated bound and scaled by $2/\alpha$. The Winkler score then favors narrow intervals and intervals that are well calibrated.

5.6 Train, Calibration and Test Splits

The baseline version for CP demands a training set, calibration set, and lastly a test set, adhering to the regular train/test split applied for ML-forecasting. And as all periods need to include pre-determined stress periods, a split was set up to ensure that all three different datasets included volatility periods. To ensure a rigorous and comprehensive comparison pipeline between the three chosen methods of baseline CP, EnbPI and ACI, all models were compared on the same test set, including the same stress period contained within the test set.

Train set, calibration set and test set were inspected, and yielded 4450 rows in the train set, spanning from 2000-09-13 to 2018-03-19, which included 293 stress-days. The calibration set used to calculate non-conformity scores for baseline CP yielded 954 rows, spanning from 2018-03-20 to 2021-12-17, which included 79 stress-days. Lastly, the test set also yielded 954 rows, spanning from 2021-12-20 to 2025-09-30, and included 50 stress days. The 50 stress days in the test were ultimately the stress period which was supposed to be compared across all four machine-learning algorithms and all three CP methods.

5.7 Forecasting Models and Architecture Setup

The selected tree-based forecasting models were set limited to Decision Tree, Random Forest, LightGBM and XGBoost. These models were chosen for two reasons;

(i) their ability to capture non-linear relationships in the input data,

(ii) their compatibility with the MAPIE-framework, the Python library used to implement baseline CP, EnbPI and ACI.

The applied models were also hyperparameter-tuned individually, as model tuning were treated as a model selection problem, where the objective is not only to minimize training error, but to choose configurations that generalize to unseen observations and avoid overfitting as effectively as possible for each individual model (Guyon et al., 2010).

As the thesis uses a one-step-ahead forecasting approach, the target variable of Brent crude was defined as the Brent price at time $t + 1$. This was implemented methodologically by applying a *shift (-1)*, aligning the predictive signals available to the model at time t with the actual value at $t + 1$. To ensure reproducibility, *random state=42* was applied consistently across all model configurations. All models were evaluated using a nominal coverage level set to 0.90 (α = 0.10), while absolute residuals used as the conformity score for Baseline CP.

5.7.1 Decision Tree Architecture

Decision Tree was included as the baseline model against which the ensemble and boosting methods could be evaluated. The model was configured with a maximum depth of 6 and a minimum of 10 observations required per leaf node. These constraints limited tree complexity and reduced the risk of overfitting to noise in the training data.

5.7.2 Random Forest Architecture

Random Forest was configured with 300 estimators, a maximum depth of 12, and a minimum of 5 observations per leaf node. The higher number of estimators was motivated by the need for stable out-of-bag residual estimates.

5.7.3 LightGBM Architecture

LightGBM was configured with 800 estimators, a learning rate of 0.03, a maximum depth of 5, 30 leaves, and a minimum of 30 observations per leaf node. Stochastic subsampling of 0.9 was applied to both observations and features at each iteration. The lower learning rate required a proportionally higher number of estimators to allow the model sufficient capacity to converge.

5.7.4 XGBoost Architecture

XGBoost was configured with 800 estimators, a learning rate of 0.03, a maximum depth of 5, and a minimum child weight of 3, with subsampling of 0.9.

5.7.5 EnbPI Architecture

As EnbPI does not require an individual calibration set, the training dataset defined as '*X_pretest'* was defined to include all observations up to the start of the test set and used as input data. The method was set to *BlockBootStrap,* meaning that EnbPI draws blocks of observations rather than drawing individual observations, making it more appropriate for time-series data. *N_samplings* were set to 30, which fed the conformal method with 30 block-based training sets. Further, *n_blocks* were set to 10, so that each bootstrap resample was composed of 10 blocks of observations. This was done to preserve parts of the temporal structure and not to divide training data in any non-meaningful way, making it appropriate for time-series. The *agg_function* was set to *mean,* meaning that the final prediction was obtained by averaging the predictions across the ensembles.

5.7.6 ACI Architecture

As with EnbPI, *n_samplings=30* and *n_blocks=10* were applied to ACI for all machine learning methods. Predictions were produced in a sequential one-step ahead setting. This meant that the first observation was predicted before any test outcome was observed, before looping through the test-set one observation at a time. After each true value became available, the conformal procedure was updated with the previous feature-target pair, and the gamma-step was applied using the latest prediction error. The study applied a value of $y = 0.005$ for all ACI runs to maintain comparability between model specifications. The value was treated as a conservative implementation choice rather than a model-specific optimization target.

6 Results

6.1 Overview of the Empirical Evaluation

This section reports the empirical performance of the forecasting models and conformal prediction methods across the full test set and the two defined volatility-regimes. As the research questions are concerned with interval precision and efficiency, the main emphasis of subsequent analysis is therefore placed on empirical coverage, mean interval width, and the Winkler score. MAE and RMSE are included and reported to document point forecast accuracy, but the central comparison will concern whether conformal intervals remain calibrated during stress episodes.

6.2 Overall Test Set Performance

Table 8: Total Test-Set Results

Model	Method	MAE	RMSE	Coverage	Mean Width	Winkler Score
Decision Tree	Baseline CP	2.246535	3.131656	0.791405	6.529391	15.314482
Decision Tree	EnbPI	1.969823	2.759376	0.853249	6.895499	13.463565
Decision Tree	ACI	1.969823	2.759376	0.903564	9.403529	13.005364
Random Forest	Baseline CP	1.804344	2.534045	0.786164	5.312112	12.821066
Random Forest	EnbPI	1.791568	2.535121	0.850105	6.237199	12.602438
Random Forest	ACI	1.791568	2.535121	0.916143	7.721429	10.900399
LGBM	Baseline CP	2.767986	3.488156	0.683438	6.850440	17.972379
LGBM	EnbPI	2.311190	3.263909	0.850105	7.693821	16.688309
LGBM	ACI	2.311190	3.263909	0.920335	8.082257	10.640828
XGB	Baseline CP	4.219846	5.164063	0.557652	8.647369	30.806846
XGB	EnbPI	3.482102	4.668761	0.836478	10.476614	23.034288
XGB	ACI	3.482102	4.668761	0.933962	10.213951	13.315370

Results are computed over the full test set, including both stress and non-stress observations.

In terms of point forecast accuracy, Random Forest gives the overall strongest performance, with the lowest MAE of 1.804 for baseline CP and 1.791 for EnbPI and ACI and RMSE of 2.534 for baseline CP and 2.535 for EnbPI and ACI, while XGB on the contrary returns the largest forecasting errors with an MAE of 4.219 for baseline CP and 3.482 for EnbPI and ACI, and an RMSE of 5.164 for baseline CP and 4.668 for EnbPI and ACI. The reasoning for different MAE and RMSE for Baseline CP and EnbPI/ACI stems from the fact that the latter models had access to more training data, as they did not require a calibration set.

These differences matter for the interval results as conformal prediction intervals are calibrated from past realized forecasting errors. Larger and less stable residuals make it harder for a conformal method to obtain the desired nominal coverage level without increase in interval width. The weaker point forecasts from XGB for instance helps to explain why its baseline CP intervals achieve low coverage despite the wider average intervals compared to other models.

Across the full test set, Baseline CP undercovers for all models, with coverage ranging from 0.557 for XGB, 0.683 for LGBM, 0.786 for Random Forest to 0.791 for Decision Tree, indicating that the calibration intervals are too narrow to capture the fluctuations in the test periods. EnbPI improves coverage substantially, returning 0.850 for LGBM, 0.836 for XGB, 0.850 for Random Forest and 0.853 for Decision Tree, but still remains below the 90% target for all models. ACI is the only method that consistently reaches or exceeds the target, with coverage above 0.900 for all selected models with the lowest being Decision Tree with 0.903 and highest 0.933 for XGB, although this improvement comes with wider intervals, particularly for Decision Tree (9.403) and XGB (10.213).

The Winkler scores indicate that improved coverage does not have to imply weaker interval efficiency, as ACI also returns the lowest Winkler Score across all four models; 13.005 for Decision Tree, 10.900 for Random Forest, 10.640 for LGBM and 13.315 for XGB. However, as the aggregate results represent marginal coverage over the full test set, hidden conditional coverage failures within stress regimes might be present, which motivates subsequent separate regime analysis.

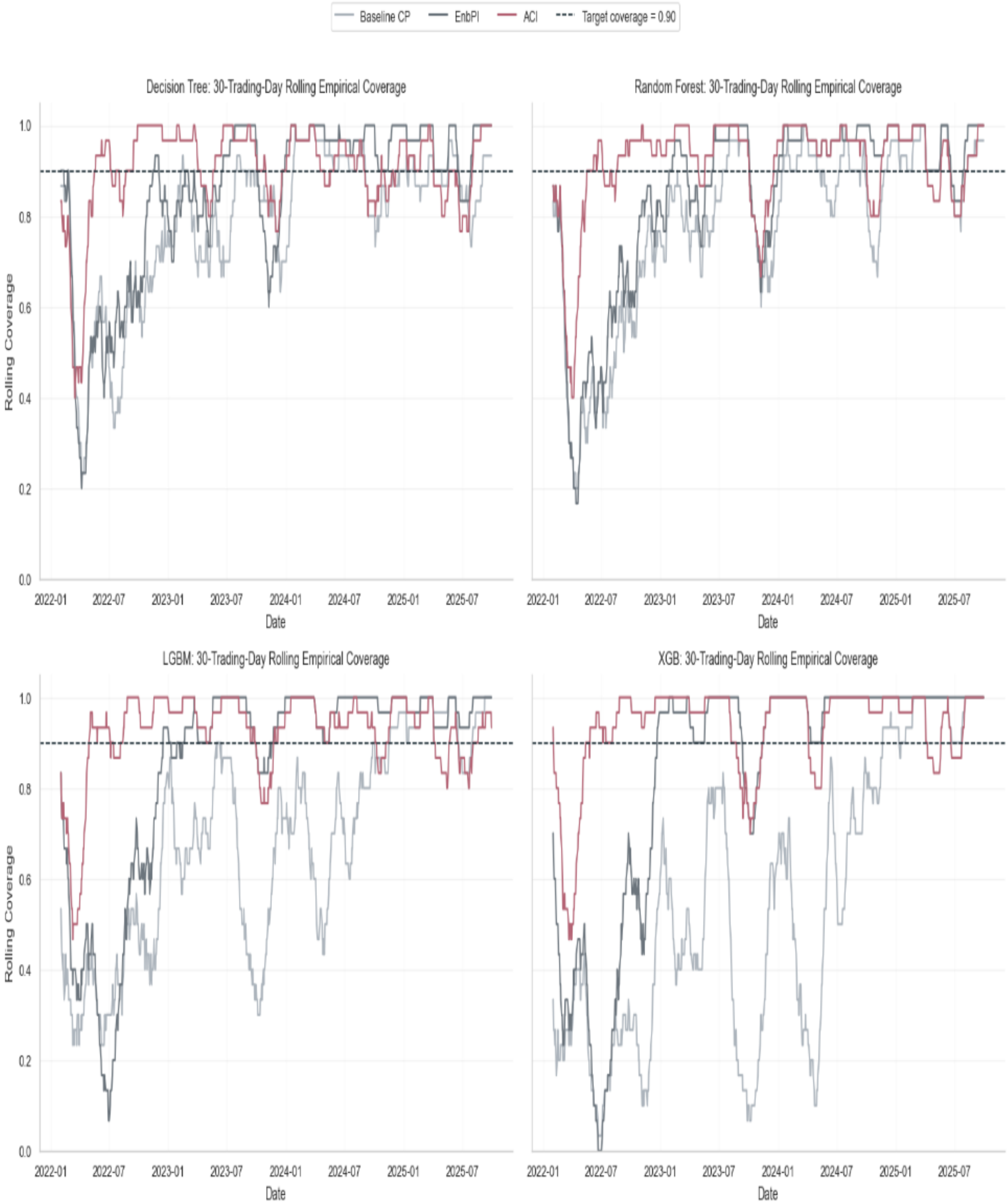

Figure 12: Rolling 30-Day Empirical Coverage

6.3 Non-Stress Performance

Table 9: Non-Stress Results

Model	Method	MAE	RMSE	Coverage	Mean Interval Width	Winkler Score
Decision Tree	Baseline CP	2.090150	2.824827	0.810841	6.529391	13.072156
Decision Tree	EnbPI	1.813543	2.494272	0.877212	6.910577	11.525862
Decision Tree	ACI	1.813543	2.494272	0.908186	8.625860	11.648540
Random Forest	Baseline CP	1.653372	2.258507	0.813053	5.312112	10.775316
Random Forest	EnbPI	1.637070	2.283075	0.877212	6.250042	10.786875
Random Forest	ACI	1.637070	2.283075	0.925885	7.315474	9.679624
LGBM	Baseline CP	2.654169	3.316144	0.701327	6.850440	16.271032
LGBM	EnbPI	2.151968	3.013677	0.873894	7.713123	14.507254
LGBM	ACI	2.151968	3.013677	0.923673	7.730583	10.090433
XGB	Baseline CP	4.052667	4.915022	0.570796	8.647369	27.910230
XGB	EnbPI	3.261901	4.335352	0.862832	10.518341	19.923248
XGB	ACI	3.261901	4.335352	0.941372	9.876905	12.309303

Results are computed only over non-stress observations in the test set.

To comprehensively assess the thesis' main research question, non-stress results will be used as a benchmark to analyze how the results under elevated conditional volatility impact interval efficiency and validity. MAE, RMSE and mean interval-width metrics are reported in USD/barrel to make the results directly interpretable in economic terms.

With a nominal coverage level of 0.900, the non-stress results show that conformal performance is significantly better outside volatility-defined stress-periods. Baseline CP still undercovers all models, but the degree of undercoverage varies between selected models. Decision Tree and Random Forest returns coverage of 0.810 and 0.813 between them, while LGBM and XGB return weaker coverage with 0.701 and 0.570. Winkler score for all baseline CP also returns the highest scores, ranging from 10.775 for Random Forest to 27.910 for XGB. The weaker non-stress performance of baseline CP indicates why more adaptive methods are needed in this setting.

EnbPI improves non-stress coverage across all models, with coverage ranging from 0.863 to 0.877 for Random Forest and Decision Tree. Even though EnbPI returns coverage below the nominal target of 0.900, this brings the interval coverage closer while only moderately increasing mean interval width, thus returning more favorable scores than baseline CP. ACI is the only conformal method that consistently reaches or exceeds the nominal level, with coverage ranging from 0.908 for Decision Tree to 0.941 for XGB. This significant coverage-improvement also comes with wider intervals, especially compared to the static intervals from baseline CP, but the non-stress Winkler scores remain relatively low. Random Forest + ACI combination achieves the lowest non-stress Winkler score at 9.679, followed by LGBM + ACI at 10.090.

6.4 Stress Performance

Table 10: Stress Results

Model	Method	MAE	RMSE	Coverage	Mean Interval Width	Winkler Score
Decision Tree	Baseline CP	5.073972	6.546044	0.440000	6.529391	55.855731
Decision Tree	EnbPI	4.795356	5.726719	0.420000	6.622896	48.497245
Decision Tree	ACI	4.795356	5.726719	0.780000	20.921040	37.739432
Random Forest	Baseline CP	4.533793	5.504309	0.300000	5.312112	49.809794
Random Forest	EnbPI	4.584503	5.326755	0.360000	6.007119	45.409718
Random Forest	ACI	4.584503	5.326755	0.740000	15.060635	32.952405
LGBM	Baseline CP	4.825801	5.773084	0.360000	6.850440	48.732742
LGBM	EnbPI	5.189915	6.249333	0.420000	7.344843	56.121783
LGBM	ACI	5.189915	6.249333	0.860000	14.440529	20.591966
XGB	Baseline CP	7.242428	8.488237	0.320000	8.647369	83.177664
XGB	EnbPI	7.463341	8.722061	0.360000	9.722178	79.281883
XGB	ACI	7.463341	8.722061	0.800000	16.307755	31.505069

Results are computed only over stress observations in the test set.

As the stress-period evaluation contains only 50 observations, stress coverage should be interpreted as a discrete empirical estimate. For example, LGBM + ACI covers 43 out of 50 stress observations, Decision Tree + ACI covers 39/50, Random Forest + ACI covers 37/50, and XGB + ACI covers 40/50, which makes small changes in the number of covered observations affect the reported coverage rate.

Results from the stress period show a significant deterioration in empirical coverage. Baseline CP and EnbPI both achieve low empirical coverage far below the nominal target despite relatively narrow intervals, indicating that their calibrated residuals do not compensate for the larger forecast errors generated during stress. This is directly related to the increase in MAE and RMSE during stress periods; as larger point forecast errors require wider prediction intervals to maintain sufficient coverage.

ACI attempts to handle this by increasing interval width, albeit at the cost of interval efficiency. The lowest coverage is returned by baseline CP + Random Forest, with coverage of 0.300. The remaining baseline CP models range between 0.320 for XGB to 0.440 with Decision Tree showing the highest return. All selected models paired with the EnbPI framework also remain below the target of 0.900, as coverage is 0.360 for both Random Forest and XGB, while Decision Tree and LGBM both return a coverage of 0.420, meaning that the intervals capture the same number of observations.

ACI is an improvement in this regard, reaching coverage between 0.740 for Random Forest and 0.860 for LGBM, but this improvement is achieved through greatly wider intervals as Random Forest returns a mean interval width of 15.060 and LGBM returns a mean interval width of 14.440. The two subpar models here, being Decision Tree and XGB, returned coverage of 0.780 and 0.800, with mean interval width yielding 20.921 for Decision Tree and 16.307 for XGB.

Winkler score also sees a substantial increase across all models and conformal methods, confirming vastly weaker overall interval quality. LGBM + ACI yields the most favorable stress-period Winkler score at 20.591, combining the highest stress coverage (0.860) with the narrowest ACI interval width (14.440) among the stress ACI results. XGB performs weakest overall during stress, with the point forecasts (*MAE > 7, RMSE > 8*), and the highest Winkler Scores under baseline CP and EnbPI (83.177 & 79.281)

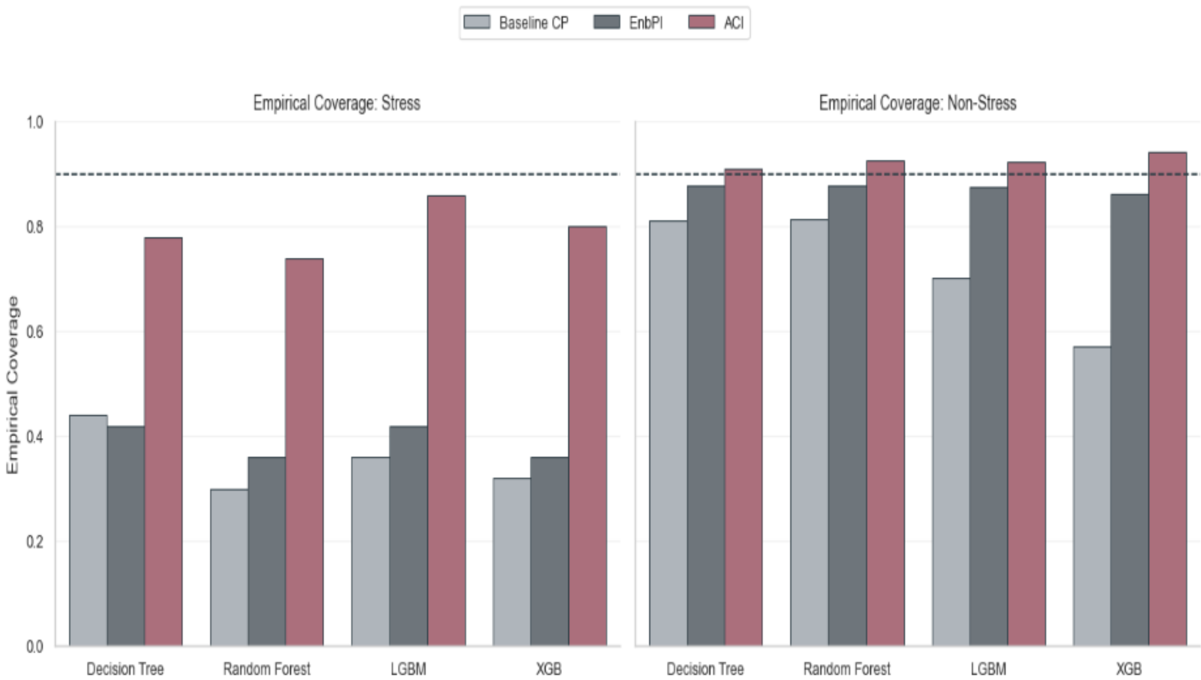

Figure 13: Empirical Coverage across Stress and Non-Stress

6.5 Regime Dependent Interval Width

As with coverage, mean interval width clearly differs between stress and non-stress observations, especially for the ACI framework. Baseline CP shows identical interval width across regimes as it uses a static, fixed calibration width. This approach returns widths of 6.529 for Decision Tree, 5.312 for Random Forest, 6.850 for LGBM, and 8.647 for XGB. EnbPI shows only limited regime-dependent adaptiveness, with stress-periods width close to the non-stress interval widths, indicating subpar model performance on the given observations.

ACI displays the highest widening during stress, as its mean interval increases from 8.885 to 20.921 for Decision Tree, from 7.315 to 15.061 for Random Forest, from 7.731 to 14.405 for LGBM, and from 9.877 to 16.308 for XGB.

Figures 15 and 16 illustrates the behaviour of ACI during the stress episode from the test set. Across all four models, the prediction intervals widen substantially around periods of sharp price movement, which is consistent with the higher stress-period mean interval widths reported in Table 10. With the presence of missed observations and a max coverage of 0.860 for LGBM + ACI, this indicates that the adaptive widening still is not aggressive enough to fully capture price fluctuations within the Brent crude series. This is also reflected in the stress-period coverage results for other models, where the ACI coverage ranges between 0.740 to 0.800.

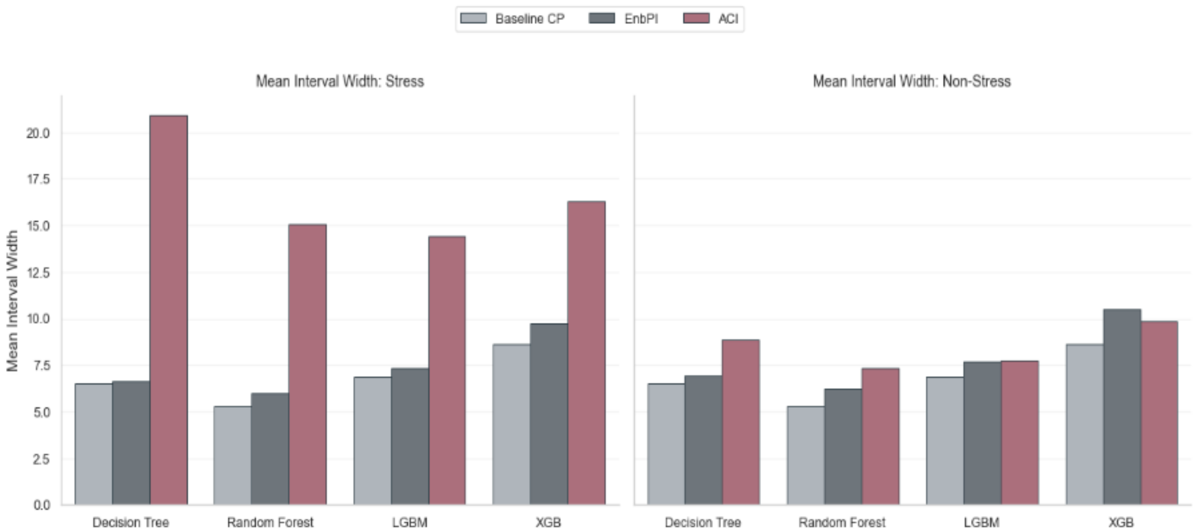

Figure 14: Mean Interval Width across Stress and Non-stress

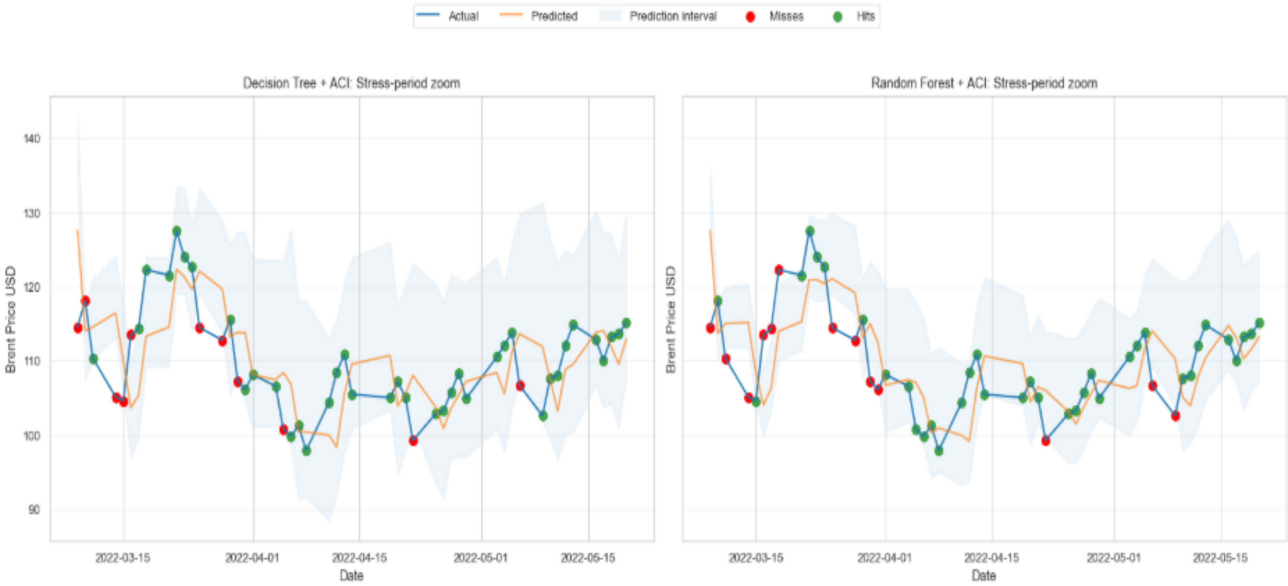

Figure 15: ACI Decision Tree and Random Forest under Stress

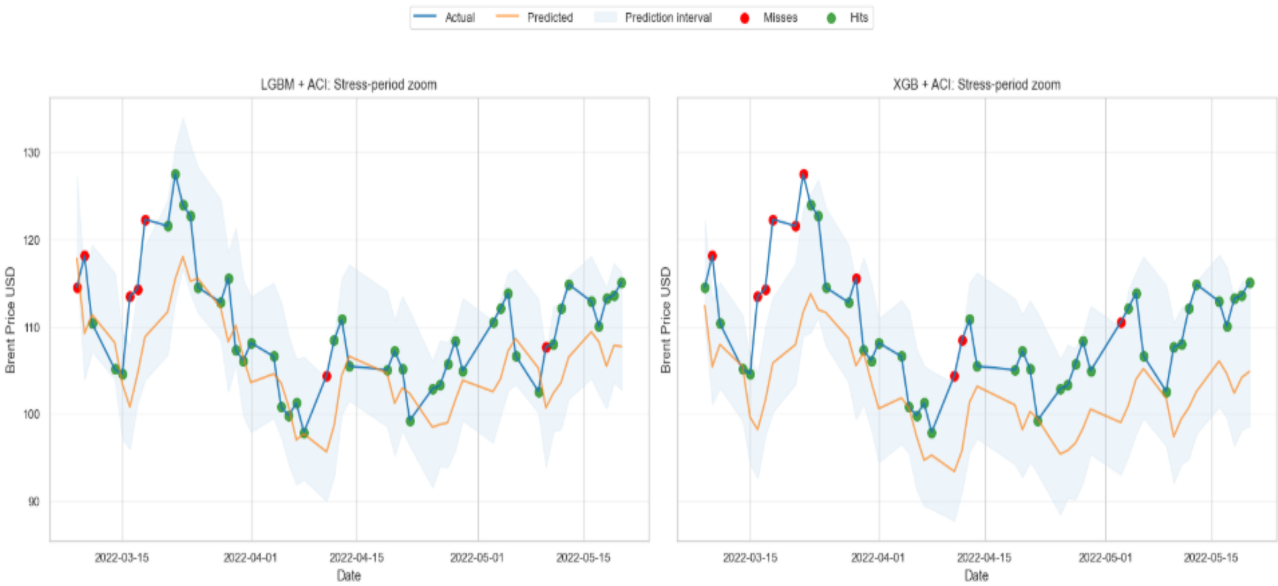

Figure 16: ACI LGBM and XGB under Stress

6.6 Regime-Dependent Interval Quality

In the non-stress regime, Winkler scores are lower and more stable across methods, reflecting a less frequent error frequency during calmer periods for the Brent crude series. Random Forest + ACI returns the lowest non-stress Winkler score at 9.679, followed by LGBM + ACI at 10.090. Decision Tree returns 11.648, and XGB returns 12.309, indicating subpar interval quality. For Decision Tree, EnbPI performs marginally better than ACI, with a Winkler score of 11.526 compared with 11.648, which might indicate that the adaptiveness and additional

width of ACI intervals can potentially backfire during stable regimes.
In the stress-regime, elevated Winkler scores can be observed across all model and method combinations. This sharp deterioration is the sum of larger forecasting errors, lower empirical coverage and the need for widened intervals. Baseline CP and EnbPI perform poorly during this regime, with XGB + Baseline CP returning the highest Winkler score for 83.177 and XGB + EnbPI returning a Winkler score of 79.282. On the contrary, ACI significantly improves the quality of prediction intervals for all models, with the lowest stress-Winkler seen by LGBM + ACI at 20.591, followed by XGB + ACI at 31.505. The need for wider intervals is indicated by the returned Winkler scores, hinting that the widening is an efficient expansion and needed to actually capture more of the fluctuations seen during stress, which further sparks an analysis of the coverage-width trade-off.

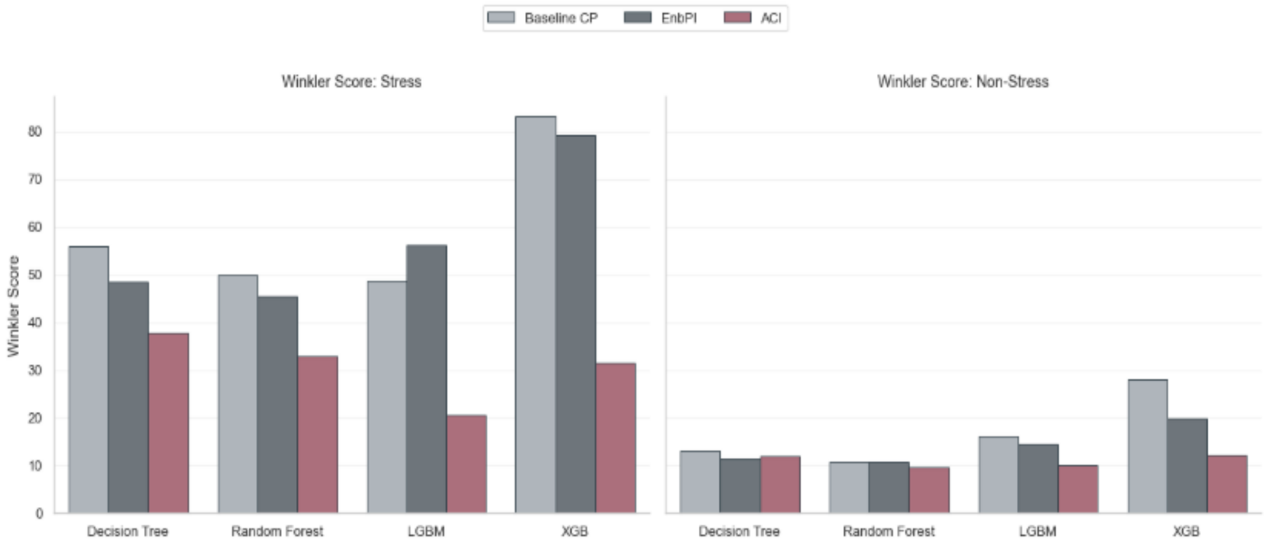

Figure 17: Winkler Score Stress and Non-stress

6.7 Coverage-Width Trade Off

A preferable interval method should achieve empirical coverage close to the nominal 0.90 level while keeping mean interval width as low as possible. The coverage-width relationship is central because it shows whether higher coverage is obtained through informative calibration or merely through excessively wide intervals. Scatter plot below illustrates this relationship for all machine-learning methods and all conformal prediction frameworks during stress-periods, non-stress periods, and for the total test. As stress-observations are deemed to be the most critical, the stress plot reports that none of the applied frameworks return a coverage equal or better than the nominal coverage level of 0.900. Models also fail in fundamentally different ways.

Baseline CP and EnbPI are centered in the lower-left region of the plot, producing relatively narrow intervals while achieving empirical coverage in the ranges of 0.300 and 0.440. Intervals are returned as narrow, which in isolation is positive, but they are systematically miscalibrated during stress. As previously established in the thesis, narrow intervals are deemed useless if they systematically fail to achieve required coverage.

ACI data points are consistently positioned higher on the coverage axis, showing improved calibration relative to Baseline CP and EnbPI. This improvement also comes naturally with wider intervals, given by larger and less stable forecasting errors, and more fluctuations within the Brent crude series itself.
With a coverage of 0.860 for LGBM + ACI, a possible takeaway is that the obtained interval width could also expand even further and still be deemed more effective, where a small increase in USD-interval width could substantially increase coverage rate, effectively showcasing the coverage-width trade-off in practice. The LGBM + ACI stress zoom plot shows that several misses occur near the interval bounds, suggesting that further widening could improve stress-period coverage.

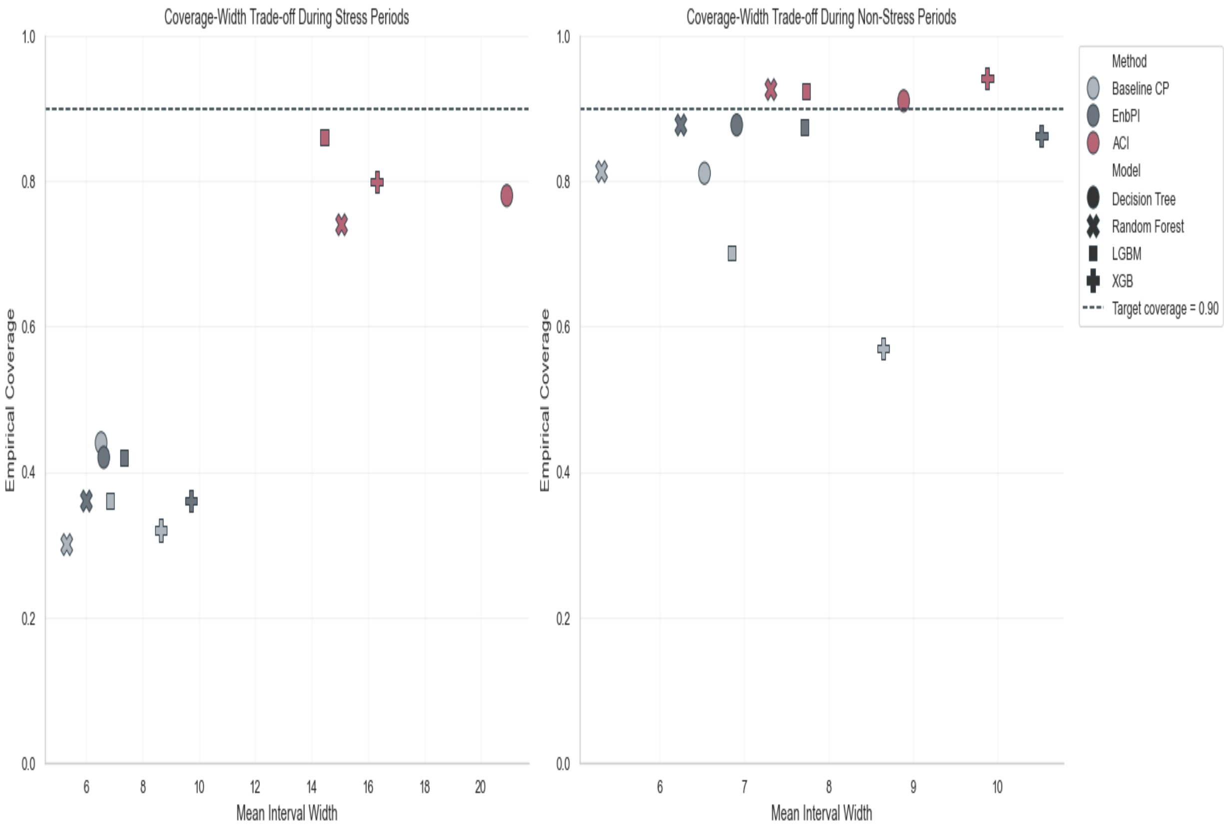

Figure 18: Scatterplot Coverage-Width Trade off

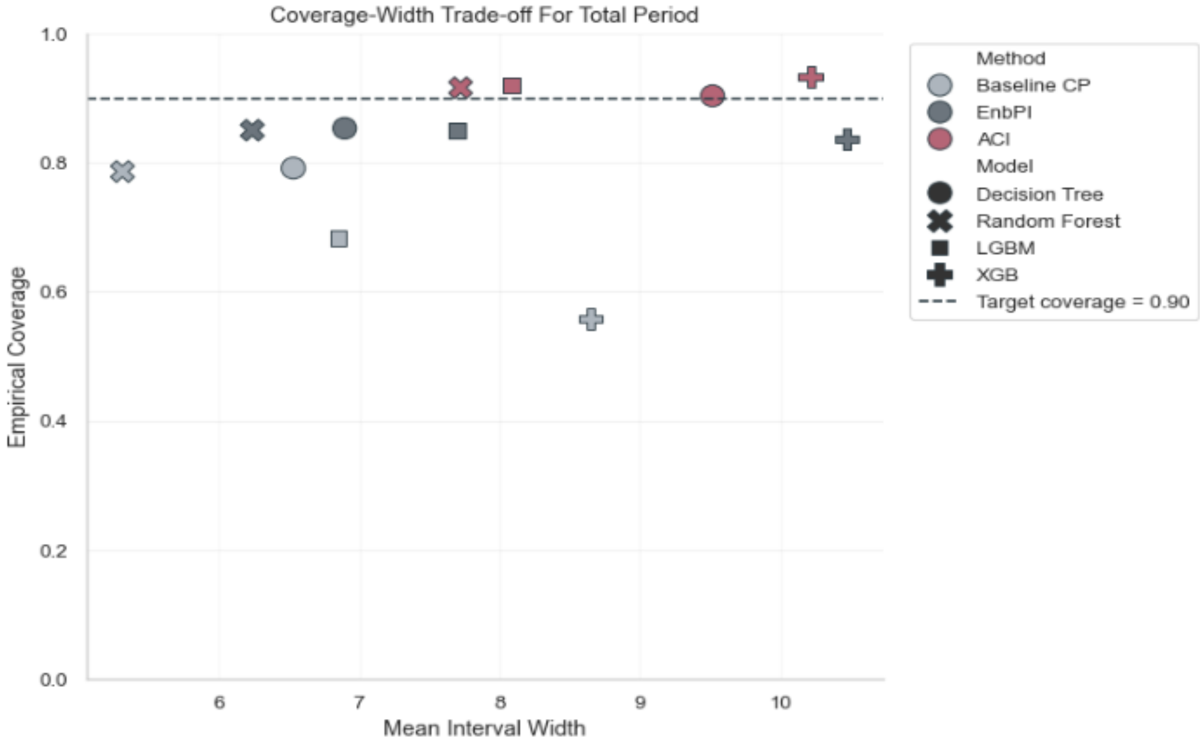

Figure 19: Scatterplot Coverage-Width Trade-off

7 Discussion

This final section discusses the empirical findings and presents the concluding remarks of the thesis. The discussion begins by examining how the reliability of the conformal prediction methods changed between stress and non-stress periods. Particular attention is given to the effect of conditional volatility on empirical coverage, interval width, and interval quality. The section then discusses the central coverage-efficiency trade-off observed in the empirical evaluation, where improved calibration during volatile periods generally required wider prediction intervals.

Methodological pitfalls and limitations of the study are also considered, including the restricted number of stress observations and the implications that this has for regime-specific analysis. The section then outlines our suggested directions for further research, and alternative angles that could be explored in the future to further improve prediction intervals during elevated conditional volatility. Finally, the section presents the overall conclusion of the study.

7.1 Coverage Collapse as a Calibration-Transfer Problem

The coverage collapse present in stress results can be interpreted as a calibration-transfer problem from the volatility parts of the calibration-sample to the 2022 stress episode. Although the calibration period was ensured to include 79 stress observations compared with 50 stress observations in the test set, the residual structure observed during the test period appears to have been more severe, or at least statistically different, from what was captured by the calibration information. The issue was therefore not simply that the calibration set lacked stress observations, but that the type of stress observed during calibration was not fully representative of the forecast-error distribution observed during the 2022 volatility regime. Market stress should therefore not be interpreted as one homogeneous statistical state.

To further interpret this, market stress in the calibration period included the COVID-19 shock, while the test period included the 2022 energy crisis following the Russo – Ukraine war. Both periods contained elevated volatility, but not necessarily statistically equivalent from a forecasting-error perspective, as the raw price movements could show different directional movements and sharper reversals.

This interpretation is supported by the stress-period point forecast results, as MAE and RMSE increased substantially for all models during stress compared to non-stress, indicating that the underlying forecasting task increased in difficulty under elevated conditional volatility. As conformal intervals are calibrated from realized forecast errors, larger and less stable residuals require wider intervals to preserve coverage.

For Baseline CP, this limitation affects the results directly as it uses a fixed calibration quantile after calibration. Once the test-period forecast-errors became larger than the calibrated residual quantile computed, the method had no mechanism for widening the intervals, thus clearly showcasing the weaknesses of Baseline CP for non-stationary data; its interval widths remained static while the realized forecast errors increased.

EnbPI, less static by design, showed only limited improvement over Baseline CP during stress. The stress-period interval widths remained close to non-stress widths, indicating that the residual updating mechanism did not adjust aggressively enough to the underlying distributional shift during stress. This might indicate a calibration-lag problem; even methods possessing adaptive structure can react too slowly when price movements occur abruptly.

ACI adapted more effectively by substantially widening intervals, which improved stress-period coverage relative to both Baseline CP and EnbPI. ACI did not fully restore nominal coverage during stress, indicating that the sequential adaptation improved calibration but did not fully compensate for abrupt price movements. These results taken into account suggest that elevated conditional volatility within time-series should be interpreted not only as a forecasting problem, but also a calibration problem for conformal inference.

7.2 Marginal vs Conditional Coverage

The results show why aggregate-test coverage returns an incomplete picture of interval reliability. Across the full test set, ACI appeared well calibrated, reaching or exceeding the nominal 0.900 coverage level for all four models. A pitfall here is however to interpret this without assessing marginal versus conditional coverage, as intervals were not equally reliable across all regimes. This aggregate result was strongly influenced by the composition of the test set. Of the 954 test observations, 904 belonged to non-stress periods and only 50 belonged to stress periods. The total coverage estimate was therefore dominated by observations from the calmer regime. This is important to highlight as regime-specific results return a different picture. ACI reached 0.903 to 0.934 coverage across the full test set, and 0.908 to 0.941 during non-stress periods, but only 0.740 to 0.860 during stress, thus yielding empirical evidence that strong aggregate coverage can coexist with substantial undercoverage in certain periods within an aggregated result.

The central issue is whether coverage remains stable across volatility regimes, not only whether it reaches the nominal level in aggregate. Baseline CP and EnbPI failed both overall and during stress, while ACI reached satisfactory full-sample coverage but still remained below 0.900 during stress. Its aggregate performance was therefore mainly driven by strong non-stress coverage, partly masking stress-period undercoverage.

This is directly relevant to the research question. If evaluation stopped at total coverage, ACI could appear to solve the calibration problem. The regime-specific results show instead that ACI improved reliability but did not fully preserve coverage under elevated conditional volatility. Thus, marginal coverage can hide conditional coverage failure in non-stationary financial time series.

7.3 Interval Adaptation, Bounds and the Price of Coverage

The stress-period results also show that interval width should not be interpreted only as a measure of inefficiency. During periods of sharp Brent price movements, wider prediction intervals may be necessary because the forecast-error distribution becomes larger, less stable, and more difficult to transfer from earlier calibration data. In a setting like this, the relevant issue should therefore not simply be whether an interval is narrow, but whether the upper and lower bounds provide a plausible uncertainty range around the forecast when prices move abruptly. This means that a wider distance between the lower and upper bound can be a necessary representation of the uncertainty surrounding the next-day price, rather than a loss in sharpness.

The stress-period plot in figures 15 and 16 supports this interpretation. Several missed observations, for ACI, occur close to the interval bounds, suggesting that the problem was not only poor point forecasting, but also insufficient interval expansion around abrupt price movements. This is particularly relevant for adaptive methods, where the lower and upper bounds can change over time as the method reacts to recent errors. The main issue is therefore not whether the interval is wide in absolute terms, but whether the widening occurs quickly enough and strongly enough to capture the changed forecast-error environment.

ACI illustrates this trade-off most clearly. Its wider intervals improved stress-period coverage and reduced Winkler penalties relative to the narrower alternatives, indicating that the additional width carried useful information. However, the remaining misses show that the adjustment was still incomplete. Under elevated conditional volatility, the relevant standard is therefore not minimum interval width, but whether the interval bounds are sufficiently responsive to large and rapidly changing forecast errors

7.4 Model Choice and Conformal Performance

The results show that point forecast accuracy influenced conformal interval performance, but did not determine it alone. Random Forest produced the strongest point forecasts in several parts of the evaluation, including during stress, but this did not translate into the best stress-period interval quality. Instead, LGBM combined with ACI produced the most favorable interval performance under stress, despite having weaker point forecast accuracy than Random Forest.

The reason is that conformal methods depend not only on the average size of forecast errors, but also on the stability and regime-transferability of the residual distribution. A model with low point forecast errors can still produce weak intervals if its residual structure changes substantially between calibration and stress periods. Conversely, a model with somewhat weaker point forecasts may produce better intervals if the conformal method can adapt its residual pattern into more reliable bounds.

XGB illustrates the opposite case. Its larger and less stable forecast errors made conformal calibration more difficult, especially during stress. Even with ACI, the interval method had to compensate for a less stable error structure, which reduced overall interval quality. The model comparison therefore suggests that strong point forecasting is useful, but the best conformal performance depends on the interaction between the forecasting model, the residual distribution, and the adaptive calibration method.

7.5 Contextualization Within Existing Literature

Findings from the study are consistent with recent work showing that conformal intervals can deteriorate under volatility shocks even when intervals widen. Juan et al. (2026) report a similar pattern for silver futures, where shock regimes were associated with wider intervals and weaker empirical coverage. The Brent results follow the same logic; elevated volatility increased forecast-error instability, and wider intervals were not always sufficient to preserve nominal coverage levels.

The undercoverage of Baseline CP during stress also aligns with the work on conformal prediction under non-exchangeability. Oliveira et al. (2024) show that split conformal prediction becomes more fragile when dependence and distributional instability weaken the representativeness of the calibration sample. Present results from the study confirm this, as the 2022 Brent stress episode appears to have produced a residual structure not fully captured by the calibration period.

The results further support the motivation behind adaptive conformal methods. Gibbs and Candès (2021) propose sequential adaptation because fixed calibration can become unreliable under distributional change, while Angelopoulos et al. (2023) emphasize that the practical issue is how quickly and efficiently adaptive methods respond to changes. In this thesis, ACI improved stress-period coverage relative to baseline CP and EnbPI, but did not fully restore nominal coverage, which is consistent with adaptive methods being useful but incomplete under abrupt price movements.

The model comparison also supports the broader point that conformal interval quality depends on the interaction between the base learner and the calibration method, as discussed by Nam et al. (2025). The applied relevance of this also connects to O’Connor et al. (2025) and Luna et al. (2025), where prediction intervals are treated as decision-support tools rather than only statistical output.

7.6 Implications

7.6.1 Implications for Practical Risk Management

The results have direct implications for the use of prediction intervals in energy-market risk management. During stress periods, Baseline CP and EnbPI produced coverage far below the nominal 90% level, meaning that the reported intervals systematically understated realized uncertainty in Brent prices. This is more serious than ordinary point forecast error: a point forecast can be wrong without claiming to describe uncertainty, while a prediction interval explicitly communicates a range of plausible outcomes. When such intervals repeatedly fail during stress, they create a misleading impression of calibrated risk.

This matters in settings where prediction intervals inform exposure management, hedging, capital allocation, or risk limits. If intervals are too narrow during volatile regimes, decision makers may underestimate both the probability and magnitude of adverse price movements. The issue is therefore not only statistical undercoverage, but misrepresentation of the risk environment in the periods where uncertainty estimates have the highest decision value. Poor stress-period coverage weakens the operational usefulness of the intervals, even if the same method performs acceptably during calmer periods.

The contrast between aggregate and regime-specific performance is therefore central. Across the full test set, ACI appeared reliable because it reached or exceeded nominal coverage for all models. During stress, however, this reliability did not fully transfer to the most volatile regime. For practitioners, satisfactory full-sample coverage is therefore insufficient evidence that a conformal method is suitable for risk management. A method can appear calibrated on average while still underestimating uncertainty in the market states where robust intervals are most needed.

The practical implication is that conformal prediction methods should be evaluated conditionally before being used in applied energy-market forecasting
Total empirical coverage should be supplemented with regime-specific coverage, interval width, and interval score analysis. In this thesis, static and weakly adaptive methods were not reliable during stress, while ACI provided a more useful but still incomplete adjustment. The conclusion is therefore cautious: conformal intervals can support Brent crude oil risk management, but only if their reliability is tested directly under volatility regimes rather than inferred from aggregate test-set performance.

7.6.2 Implications for Conformal Prediction in Financial Time Series

The results suggest that conformal prediction in financial time series should be evaluated conditionally, not only marginally. When the underlying series contains volatility regimes, aggregate empirical coverage can overstate interval reliability because performance in calmer periods may dominate the full test-set estimate. Regime-specific reporting of empirical coverage, interval width, and interval scores is therefore necessary to assess whether conformal intervals remain reliable when the forecast-error distribution changes. This is particularly important in financial applications, where the practical value of uncertainty quantification depends heavily on performance during unstable market conditions.

7.6.3 Contribution to Existing Literature

This study contributes to the existing literature in three main ways. First, it provides empirical evidence from Brent crude oil forecasting, a setting that remains less examined in the conformal prediction literature than electricity prices, option prices, and other financial or energy-market applications. The results extend the evidence on volatility-driven calibration failure by showing that conformal intervals can widen during stress periods while still failing to preserve nominal coverage. This supports the broader argument that volatility shocks can weaken conformal reliability even when intervals become less sharp.

Second, the thesis contributes by evaluating conformal prediction under explicitly defined volatility regimes rather than only over the full test period. This allows the analysis to assess whether interval reliability changes when the market moves from ordinary conditions into elevated conditional volatility. The contribution is therefore not the stress-classification procedure itself, but the use of regime-specific evaluation to reveal coverage failures that would be hidden in aggregate results.

Third, the study compares four tree-based forecasting models across three conformal prediction methods under the same sequential out-of-sample design. This makes it possible to examine how interval performance depends both on the base forecasting model and the conformal calibration method. The results show that stronger point forecasts do not automatically produce better conformal intervals under stress, since interval quality also depends on residual stability and the method's ability to adapt to regime changes.

The main contribution is therefore more specific than showing that conformal prediction performs worse during stress. The thesis shows that aggregate coverage can conceal regime-dependent undercoverage, that static and weakly adaptive conformal methods are especially vulnerable under elevated conditional volatility, and that ACI improves stress-period performance without fully restoring nominal coverage. These findings add empirical evidence on conformal prediction under non-stationarity in the specific context of Brent crude oil price forecasting.

7.7 Limitations and Future Work

7.7.1 Stress Episode Coverage and Comparability

All models and conformal methods were evaluated against a single stress episode in the test set, spanning March to May 2022, to ensure a consistent and comparable evaluation framework across all three conformal methods. This constraint arose from the requirement that Baseline CP relies on a fixed calibration set computed prior to the test period, meaning that the stress episode had to fall within the shared test window. EnbPI and ACI, which calibrate sequentially without requiring a dedicated calibration set, could in principle have been evaluated across multiple stress episodes from earlier periods.

This limitation affected the precision of stress-period coverage estimates, as 50 stress observations meant that each observation represented two percentage points in the empirical coverage estimate. Future work could address this by applying a walk-forward evaluation design with multiple stress windows, or by evaluating EnbPI and ACI independently across the entire sample of stress-periods.

7.7.2 Single Asset and Alternative Targets

The empirical analysis was conducted exclusively on Brent crude oil spot prices, which limited the generalizability of the findings. While Brent is the most internationally representative crude oil benchmark, the stress episodes identified in the sample reflected oil-market-specific dynamics. Whether the observed patterns of coverage deterioration generalize to other markets remains an open empirical question.

Two directions for future work emerged from this limitation. First, replicating the evaluation framework on WTI crude oil would provide a natural robustness check within the same commodity class, given the documented differences in regional pricing dynamics between the two benchmarks. Second, applying the framework to asset classes characterized by lower levels of conditional volatility, for example, less fluctuating stocks like Walmart within stable sectors such as the Consumer Defensive sector, or other financial activa classes like government bonds. This comparison would provide a more complete picture of when conformal prediction intervals remain reliably calibrated and when they do not.

7.7.3 Regime Classification Sensitivity

The stress regime classification relied on a joint threshold rule requiring both GARCH(1,1) and rolling standard deviation to simultaneously exceed their respective 90th percentile thresholds. While this approach reduced sensitivity to isolated volatility spikes, the specific threshold choice directly influenced which observations were classified as stress days and therefore affected the regime-specific coverage estimates. The robustness of the findings to alternative threshold specifications, such as the 80th or 95th percentile, was not formally evaluated. As the field of volatility classification is vast, future work could implement other more technical ways to assess stress versus non-stress periods before implementing conformal methods.

7.7.4 Evaluation Metrics

Interval performance was evaluated using empirical coverage, mean interval width, and the Winkler score. While the Winkler score provided a useful combined measure of validity and efficiency, more advanced evaluation metrics could supplement these for further diagnostics. Adhering to for instance MAPIE documentation, general metrics such as Size-Stratified Coverage, Hilbert-Schmidt Independence Criterion and calibration metrics such as Expected Calibration Error, Kuiper's Test and Spiegelhalter's Test could be implemented. Including these metrics in future work would strengthen the empirical basis for conclusions about regime-dependent interval quality beyond what was done in this thesis.

7.7.5 Model Scope and Deep Learning

The forecasting models were restricted to tree-based methods; Decision Tree, Random Forest, LightGBM, and XGBoost, due to the compatibility requirements of the MAPIE framework. A naive random-walk benchmark is also not included in the present version of the empirical design. The point forecast results should therefore be interpreted as relative comparisons between the selected tree-based models, rather than as evidence that these models outperform a simple persistence forecast. This limitation is less central to the main research question, since the thesis primarily evaluates conformal interval behavior across volatility regimes.

LSTM and GRU recurrent neural networks were initially considered as candidate models given their capacity to capture sequential dependencies in financial time series, but were excluded due to this compatibility constraint. Future work could address this by implementing conformal prediction for deep learning models through alternative frameworks that are not restricted to scikit-learn compatibility, allowing a direct comparison between tree-based and sequence-based architectures under regime-dependent volatility conditions.

8 Conclusion

When oil markets are calm, conformal prediction intervals look reliable. When they are not, the picture changes. This thesis was motivated by exactly that gap: the question of whether uncertainty estimates that appear calibrated on average remain useful when market conditions become unstable.

The starting point was the observation that crude oil markets are prone to abrupt regime shifts. Geopolitical events, supply disruptions, and financial shocks have repeatedly produced periods where price movements outpace historical norms. In these periods, decision-makers need reliable uncertainty estimates the most. The thesis asked whether conformal prediction can deliver them.

Two research questions guided the analysis:

i) "*How does elevated conditional volatility affect the validity and efficiency of conformal prediction intervals in Brent crude oil price forecasting*"

ii) "*Which conformal prediction method achieves the strongest coverage-efficiency trade-off during volatility-defined stress regimes?*"

8.1 The effect of elevated conditional volatility

On aggregate, ACI reached or exceeded the nominal 0.900 coverage level for all four models. Taken at face value, this looked like a success. But 904 of 954 test observations were non-stress observations. The aggregate result was dominated by calm market conditions.
During the stress episode, coverage collapsed across all methods:

- Baseline CP achieved empirical coverage between 0.300 and 0.440. More than half of all stress observations fell outside the reported 90% intervals.
- EnbPI improved marginally, reaching between 0.360 and 0.420. Still far below the nominal target.
- ACI reached between 0.740 and 0.860. The best outcome of the three, but the nominal level was not restored for any model.

Elevated conditional volatility therefore degraded interval validity for every method evaluated. The degree of degradation was concealed by aggregate coverage statistics. A method can appear well-calibrated across the full test period while failing during the exact market conditions where reliable uncertainty estimates matter most.

8.2 The strongest coverage-efficiency trade-off under stress

Among all method and model combinations, LGBM combined with ACI produced the most favorable stress-period result. It achieved empirical coverage of 0.860 with a mean interval width of 14.440 USD/barrel and a Winkler score of 20.592.
This was not because LGBM had the lowest point forecast errors under stress. Random Forest did. The advantage came from how ACI calibration interacted with LGBM residuals during the episode. The result shows that the best coverage-efficiency trade-off under stress cannot be identified from point forecast accuracy alone. It depends on the interaction between the forecasting model, the residual distribution, and the calibration mechanism.

8.3 What this means

The thesis set out to ask whether conformal prediction delivers useful uncertainty estimates when the market becomes unstable. The honest answer is: partially, and only if evaluated correctly.
Three findings stand out:

- Aggregate coverage conceals regime-dependent failure. Reporting only total test-set coverage overstates the reliability of static and weakly adaptive conformal methods. This is not a minor qualification; it is the central finding.
- Adaptive calibration helps, but does not solve the problem. ACI improved stress-period coverage substantially compared to Baseline CP and EnbPI. Even so, the nominal 0.900 target was not reached under stress for any model. Adaptation narrows the gap. It does not close it.
- Point forecast quality is necessary but not sufficient. Stronger point forecasts produce better residuals to calibrate on, but the model that achieved the best stress-period Winkler score was not the one with the lowest forecast errors.

For practitioners, the implication is direct. A 90% interval that captures fewer than half of realized observations during a stress episode does not provide a reliable basis for hedging decisions, capital allocation, or risk limits. Conformal prediction can support uncertainty quantification in Brent crude oil forecasting, but only if stress-period performance is tested explicitly rather than inferred from aggregate results.

For researchers, regime-specific evaluation should be part of the primary evaluation framework when conformal prediction is applied to non-stationary financial data. Marginal coverage alone is not enough.

The broader contribution of this thesis is therefore not simply that conformal methods perform worse during stress. That is theoretically expected when exchangeability is violated. The contribution is more specific: the thesis shows, empirically in a Brent crude oil setting, exactly how large that deterioration is, where it occurs, and which method-model combinations best limit the damage. In a market defined by geopolitical uncertainty and recurrent volatility regimes, knowing where uncertainty estimates break down is at least as important as knowing where they hold.

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Appendix

Table 11: Forecasting and Prediction Interval Performance by Model, Method, and Regime

Model	Method	Regime	MAE	RMSE	Coverage	Mean Interval Width	Winkler Score
Decision Tree	Baseline CP	Total	2.246535	3.131656	0.791405	6.529391	15.314482
Decision Tree	Baseline CP	Stress	5.073972	6.546044	0.440000	6.529391	55.855731
Decision Tree	Baseline CP	Non-Stress	2.090150	2.824827	0.810841	6.529391	13.072156
Decision Tree	EnbPI	Total	1.969823	2.759376	0.853249	6.895499	13.463565
Decision Tree	EnbPI	Stress	4.795356	5.726719	0.420000	6.622896	48.497245
Decision Tree	EnbPI	Non-Stress	1.813543	2.494272	0.877212	6.910577	11.525862
Decision Tree	ACI	Total	1.969823	2.759376	0.903564	9.516713	13.235739
Decision Tree	ACI	Stress	4.795356	5.726719	0.780000	20.921040	37.739432
Decision Tree	ACI	Non-Stress	1.813543	2.494272	0.910398	8.885943	11.880446
Random Forest	Baseline CP	Total	1.804030	2.533770	0.786164	5.312112	12.818795
Random Forest	Baseline CP	Stress	4.533793	5.504309	0.300000	5.312112	49.809794
Random Forest	Baseline CP	Non-Stress	1.653047	2.258171	0.813053	5.312112	10.772833
Random Forest	EnbPI	Total	1.791388	2.534946	0.850105	6.237321	12.601016
Random Forest	EnbPI	Stress	4.584503	5.326755	0.360000	6.007119	45.409718
Random Forest	EnbPI	Non-Stress	1.636902	2.282978	0.877212	6.250054	10.786375
Random Forest	ACI	Total	1.791388	2.534946	0.916143	7.721440	10.898893
Random Forest	ACI	Stress	4.584503	5.326755	0.740000	15.060635	32.952405
Random Forest	ACI	Non-Stress	1.636902	2.282978	0.925885	7.315511	9.679119
LGBM	Baseline CP	Total	2.767986	3.488156	0.683438	6.850440	17.972379
LGBM	Baseline CP	Stress	4.825801	5.773084	0.360000	6.850440	48.732742
LGBM	Baseline CP	Non-Stress	2.654169	3.316144	0.701327	6.850440	16.271032
LGBM	EnbPI	Total	2.311190	3.263909	0.850105	7.693821	16.688309
LGBM	EnbPI	Stress	5.189915	6.249333	0.420000	7.344843	56.121783
LGBM	EnbPI	Non-Stress	2.151968	3.013677	0.873894	7.713123	14.507254
LGBM	ACI	Total	2.311190	3.263909	0.920335	8.082257	10.640828
LGBM	ACI	Stress	5.189915	6.249333	0.860000	14.440529	20.591966
LGBM	ACI	Non-Stress	2.151968	3.013677	0.923673	7.730583	10.090433
XGB	Baseline CP	Total	4.219846	5.164063	0.557652	8.647369	30.806846
XGB	Baseline CP	Stress	7.242428	8.488237	0.320000	8.647369	83.177664
XGB	Baseline CP	Non-Stress	4.052667	4.915022	0.570796	8.647369	27.910230
XGB	EnbPI	Total	3.482102	4.668761	0.836478	10.476614	23.034288
XGB	EnbPI	Stress	7.463341	8.722061	0.360000	9.722178	79.281883
XGB	EnbPI	Non-Stress	3.261901	4.335352	0.862832	10.518341	19.923248
XGB	ACI	Total	3.482102	4.668761	0.933962	10.213951	13.315370
XGB	ACI	Stress	7.463341	8.722061	0.800000	16.307755	31.505069
XGB	ACI	Non-Stress	3.261901	4.335352	0.941372	9.876905	12.309303

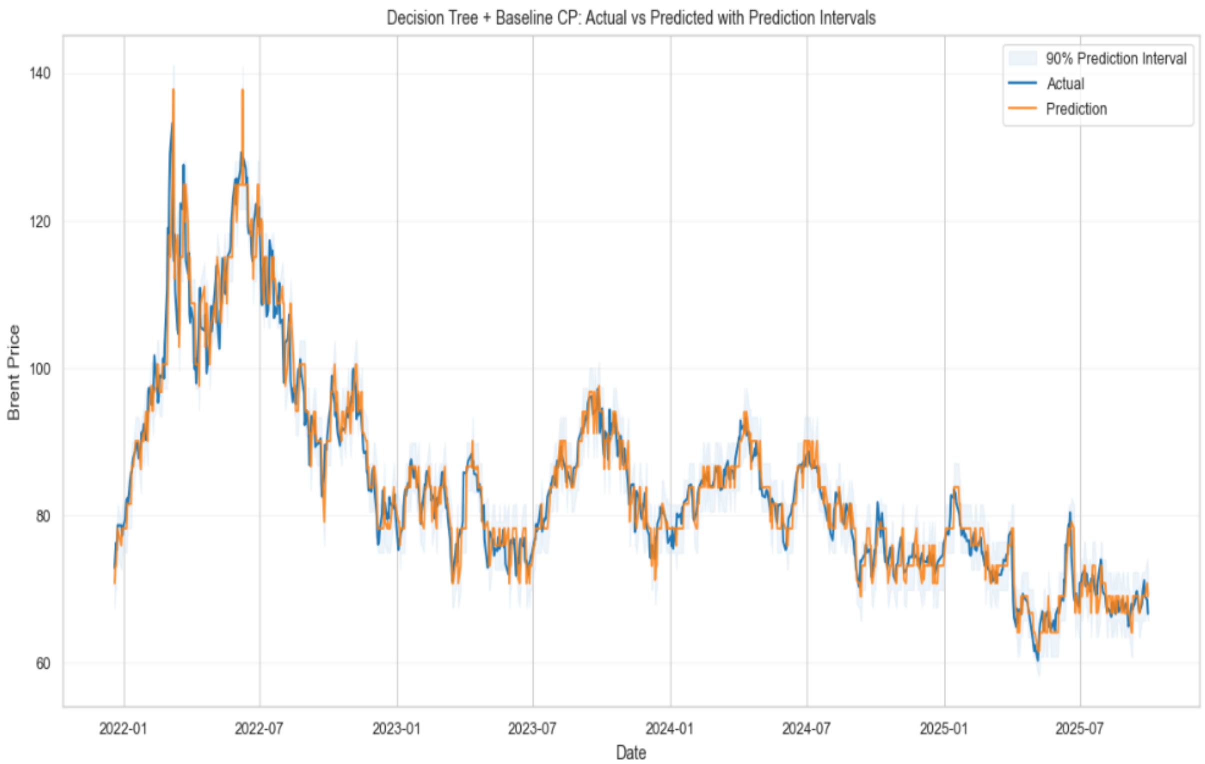

Figure 20: Test set Decision Tree + Baseline CP

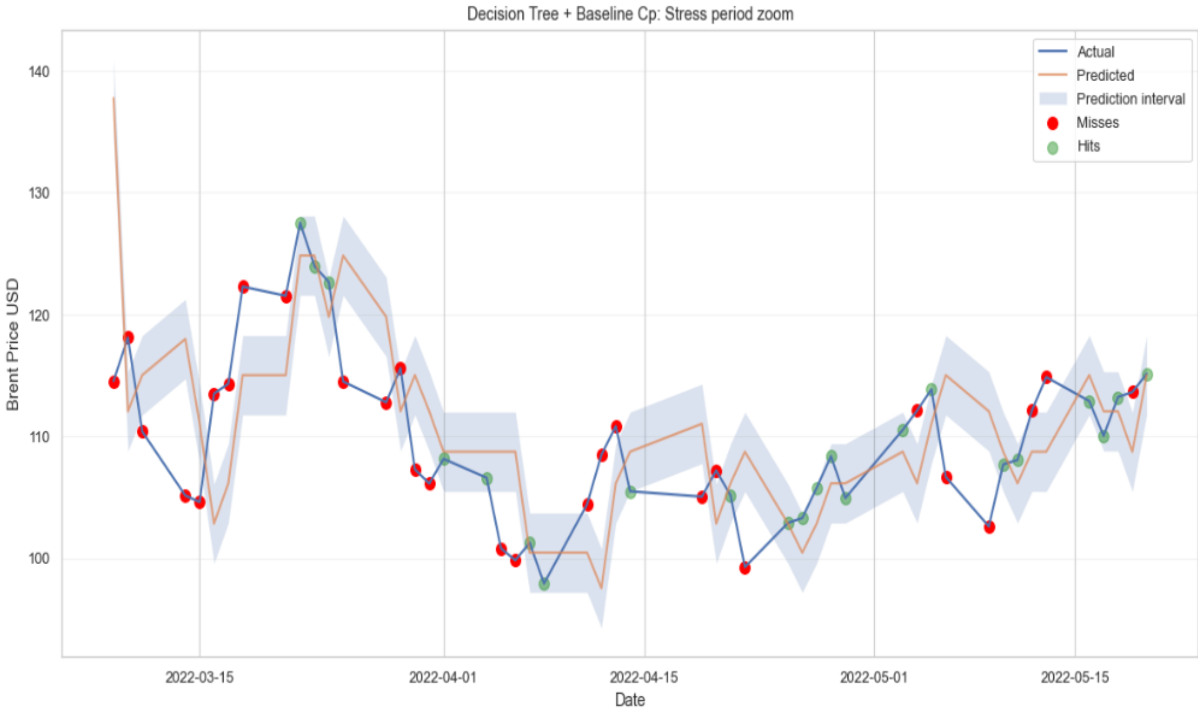

Figure 21: Decision Tree + Baseline CP under Stress

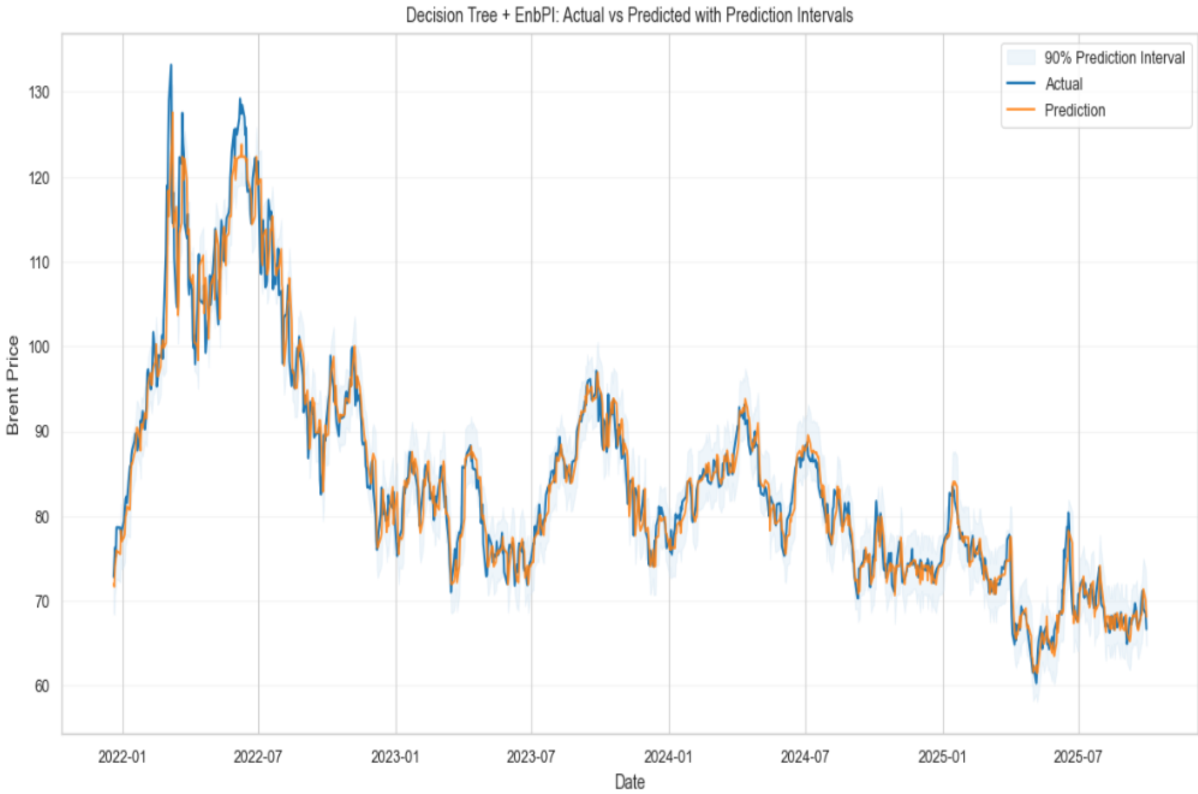

Figure 22: Test set Decision Tree + EnbPI

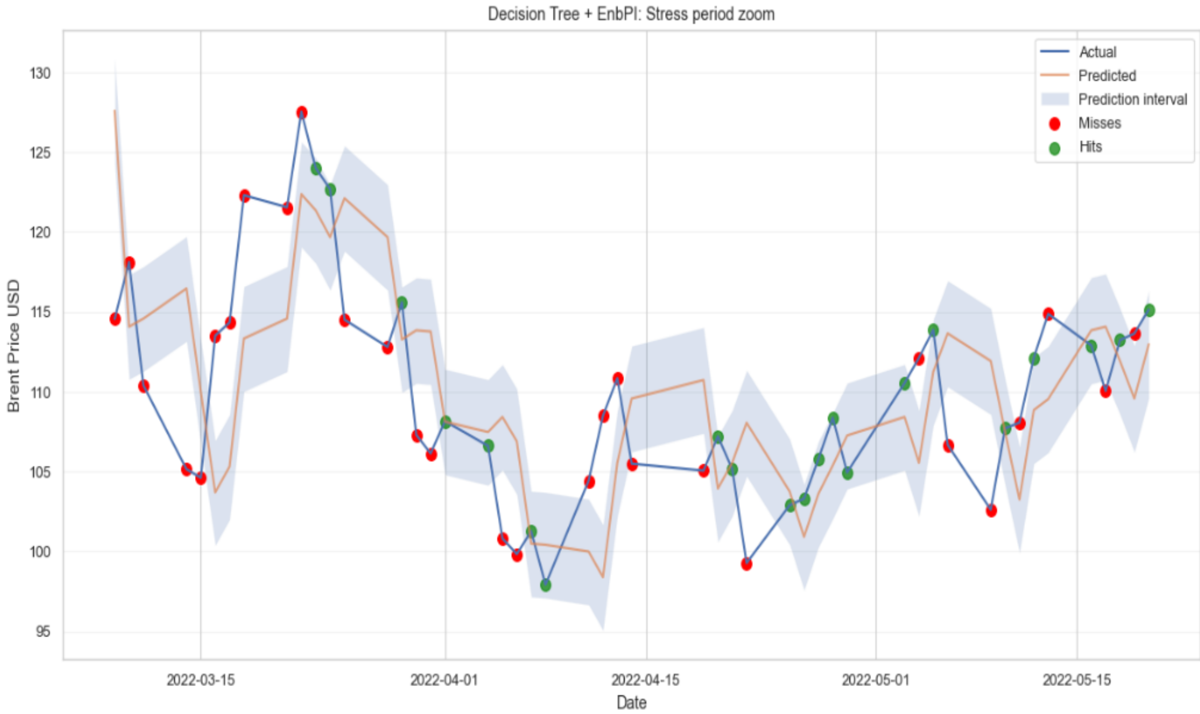

Figure 23: Decision Tree + EnbPI under Stress

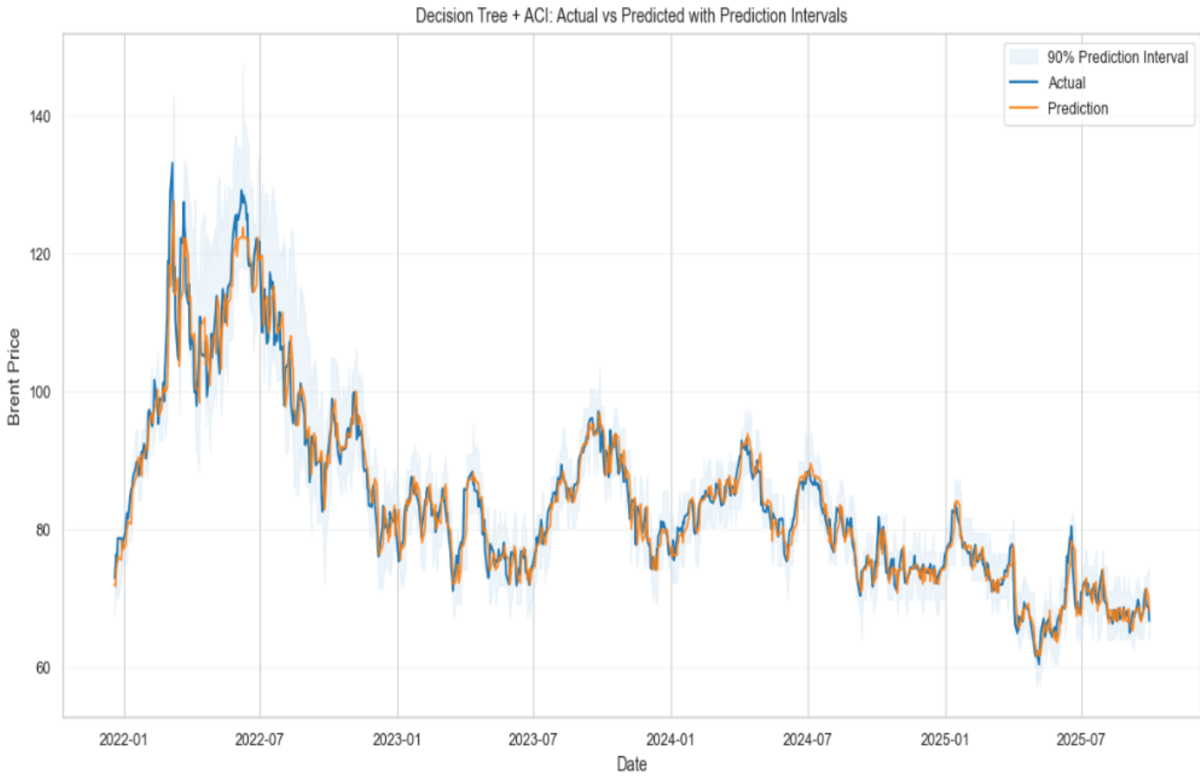

Figure 24: Test set Decision Tree + ACI

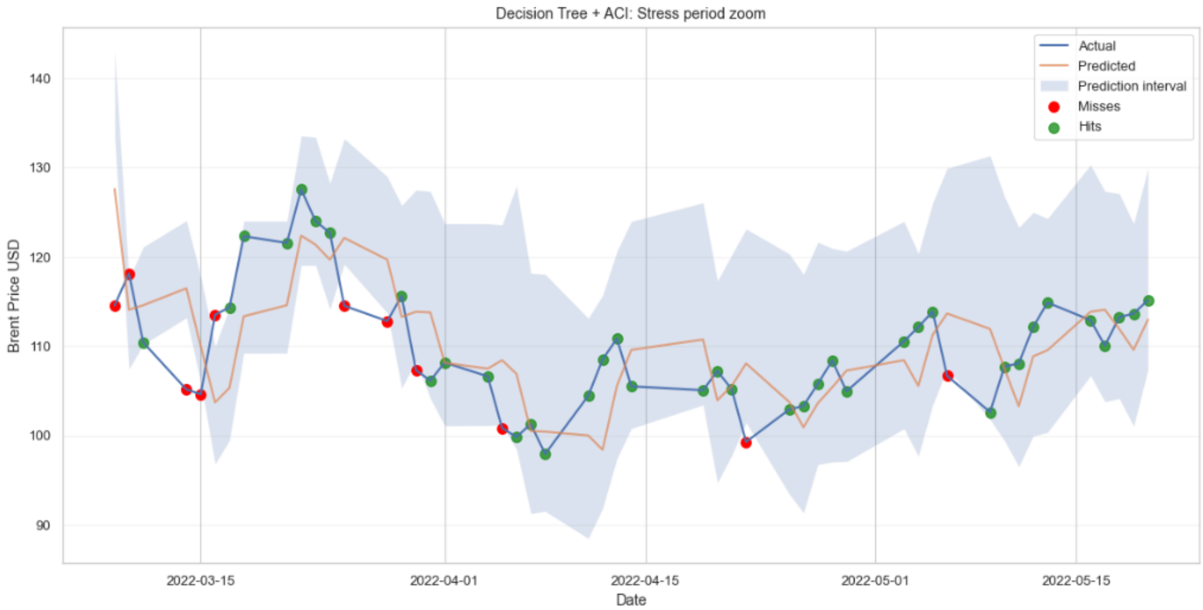

Figure 25: Decision Tree + ACI under Stress

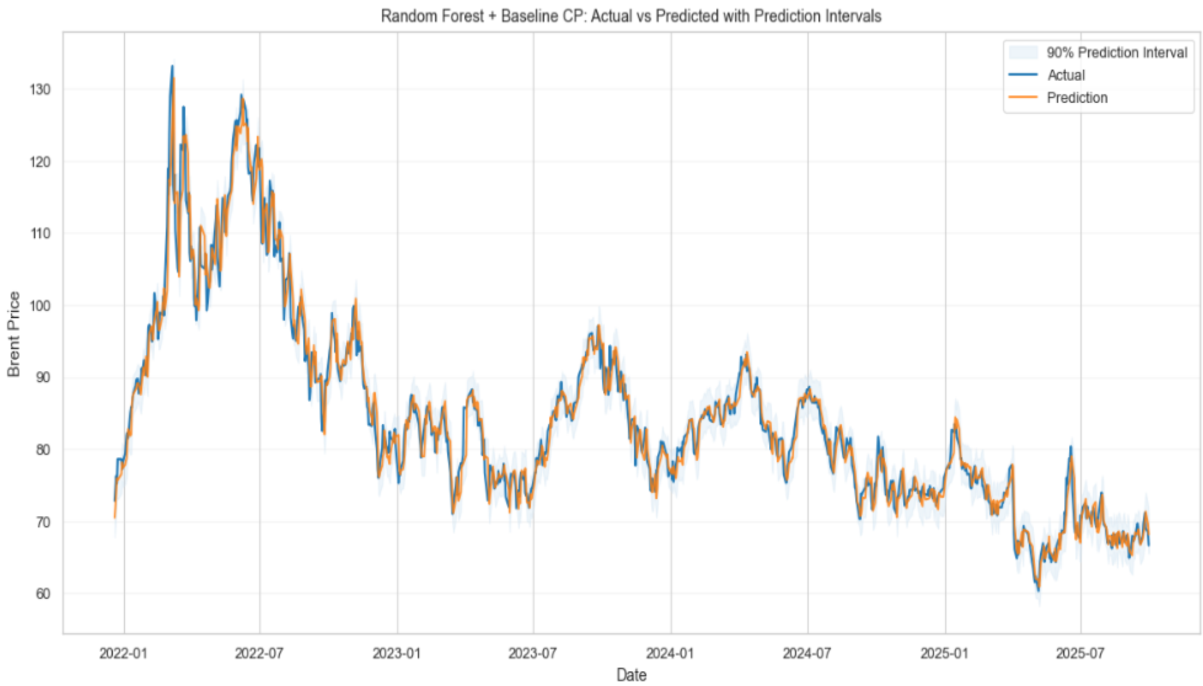

Figure 26: Test set Random Forest + Baseline CP

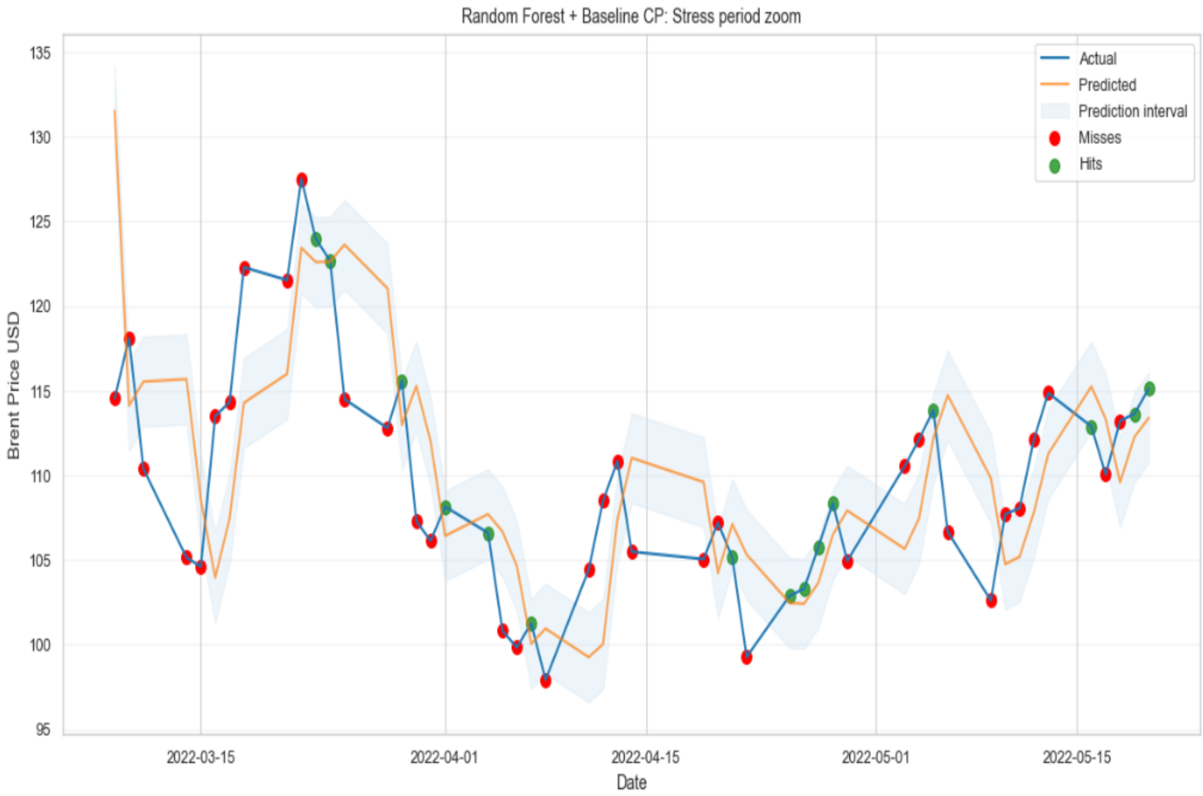

Figure 27: Random Forest + Baseline CP under Stress

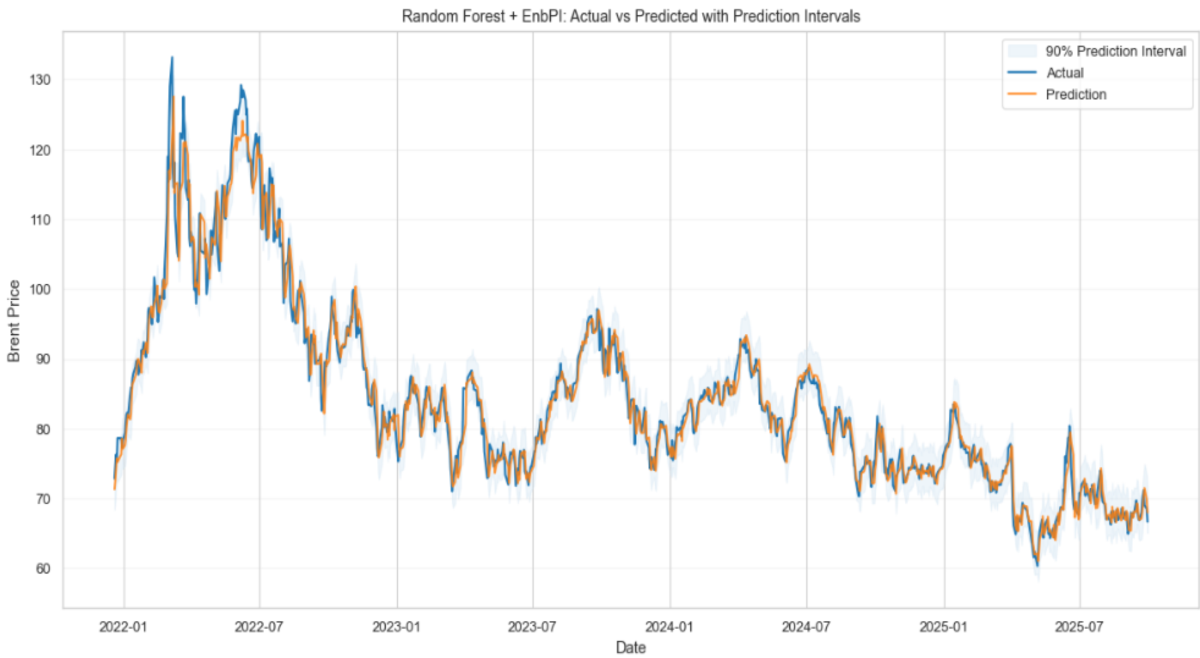

Figure 28: Test set Random Forest + EnbPI

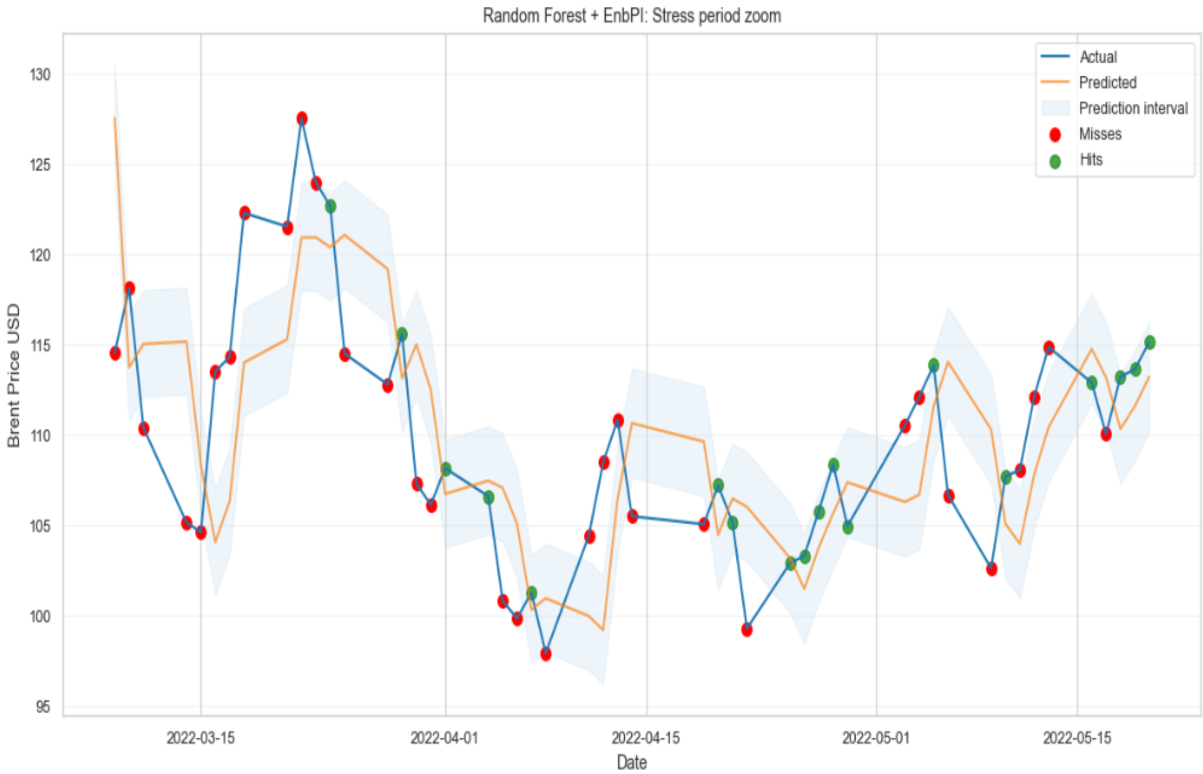

Figure 29: Random Forest + EnbPI under Stress

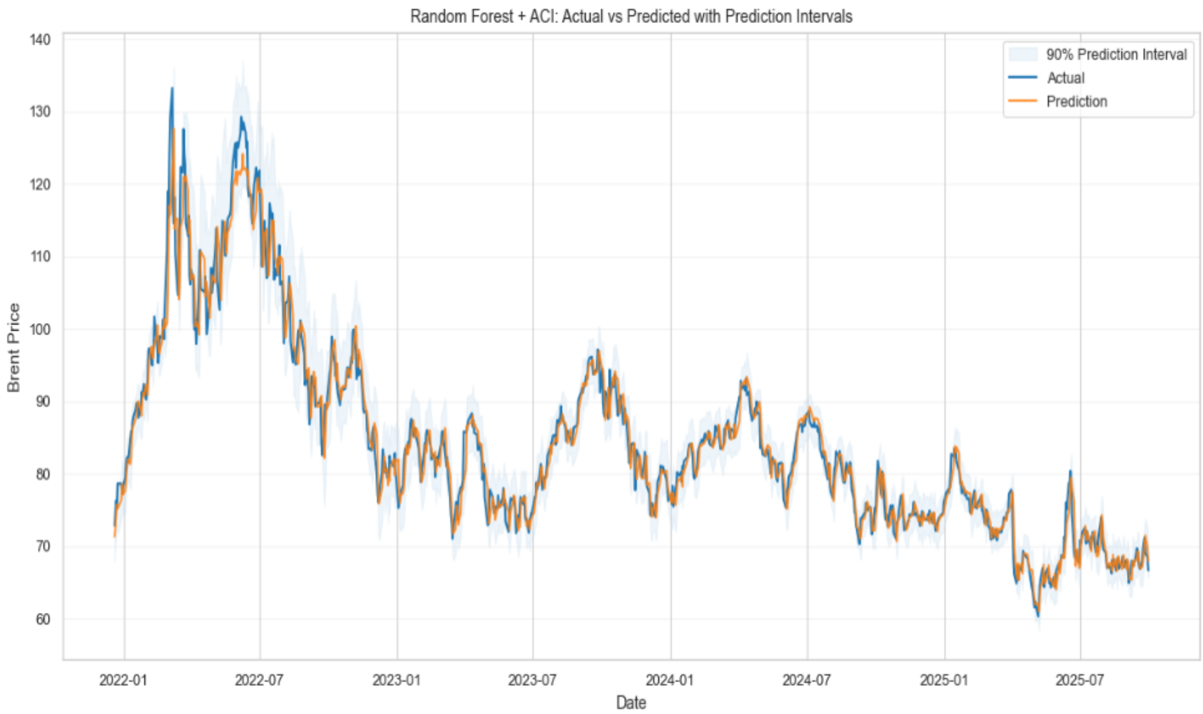

Figure 30: Test set Random Forest + ACI

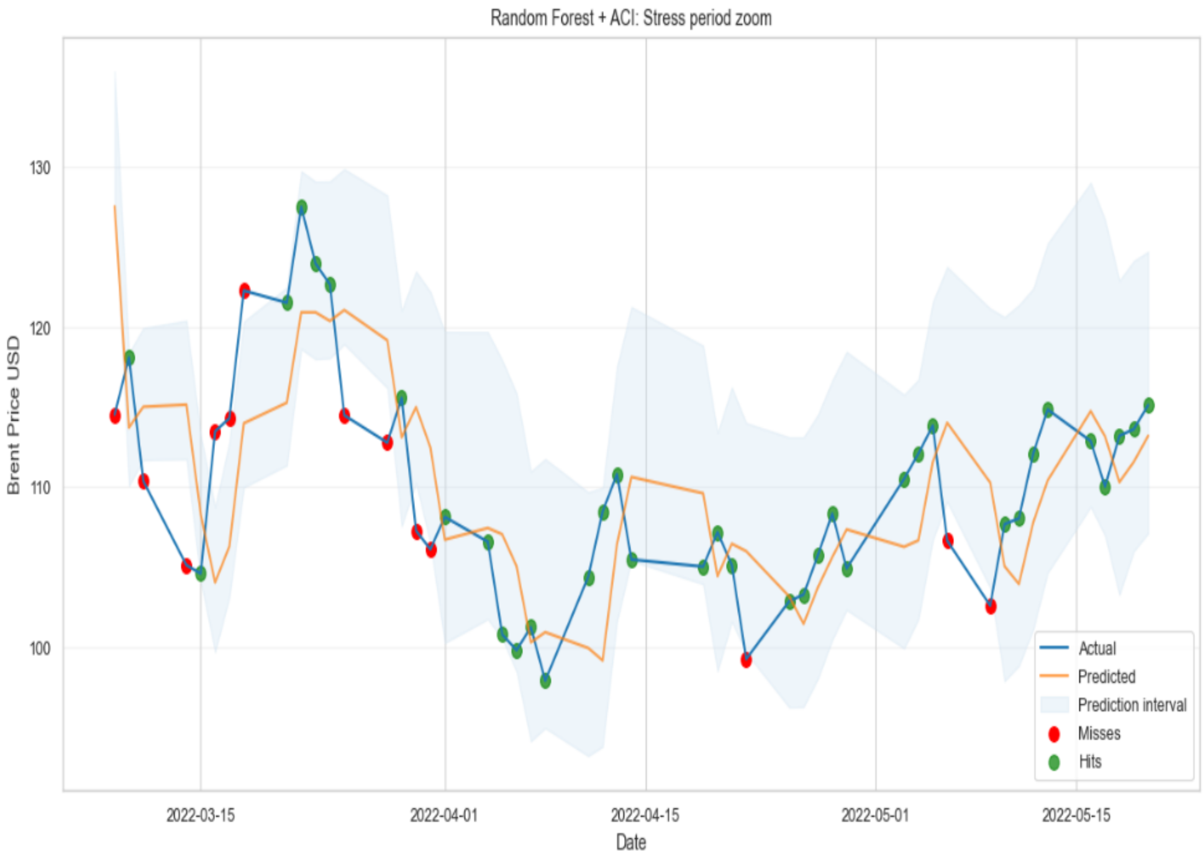

Figure 31: Random Forest + ACI under Stress

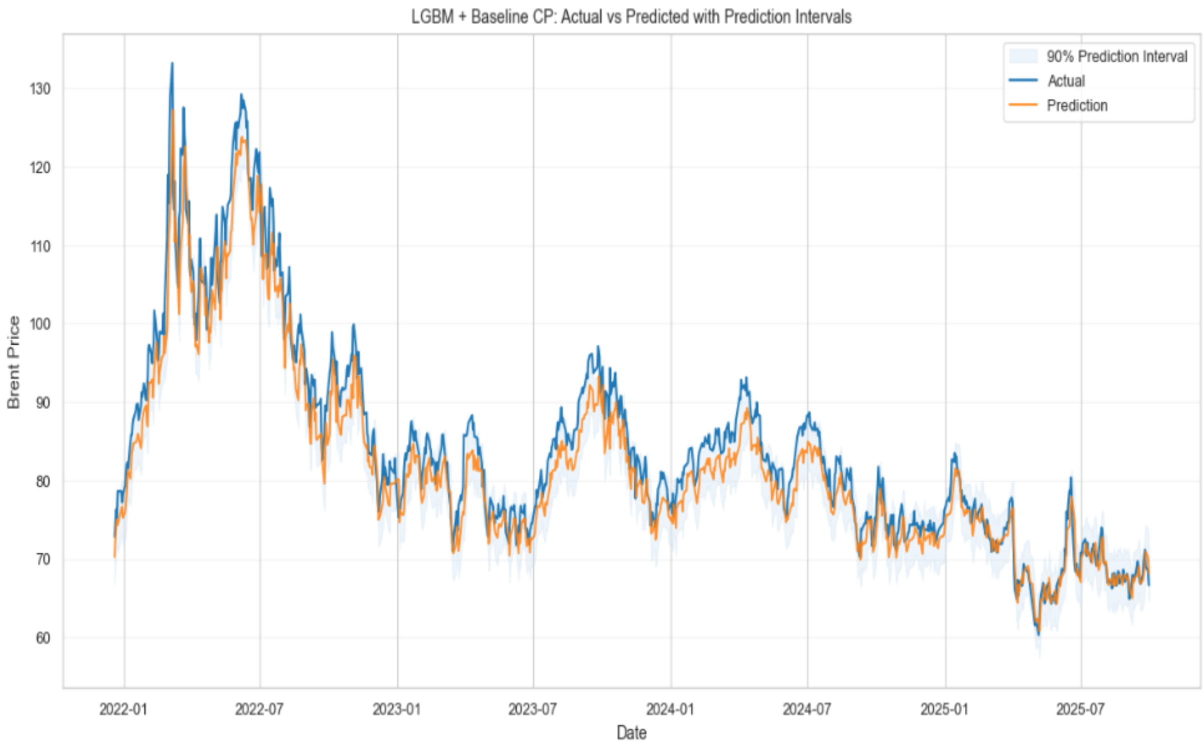

Figure 32: Test set LGBM + Baseline CP

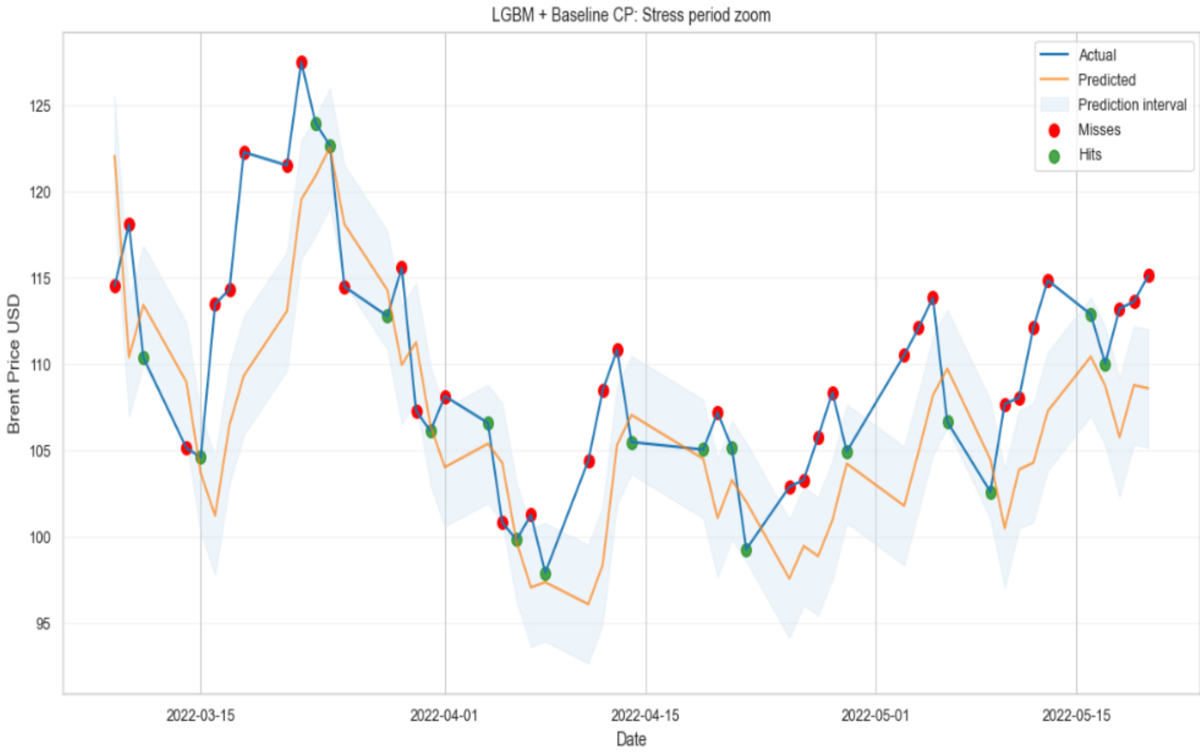

Figure 33: LGBM + Baseline CP under Stress

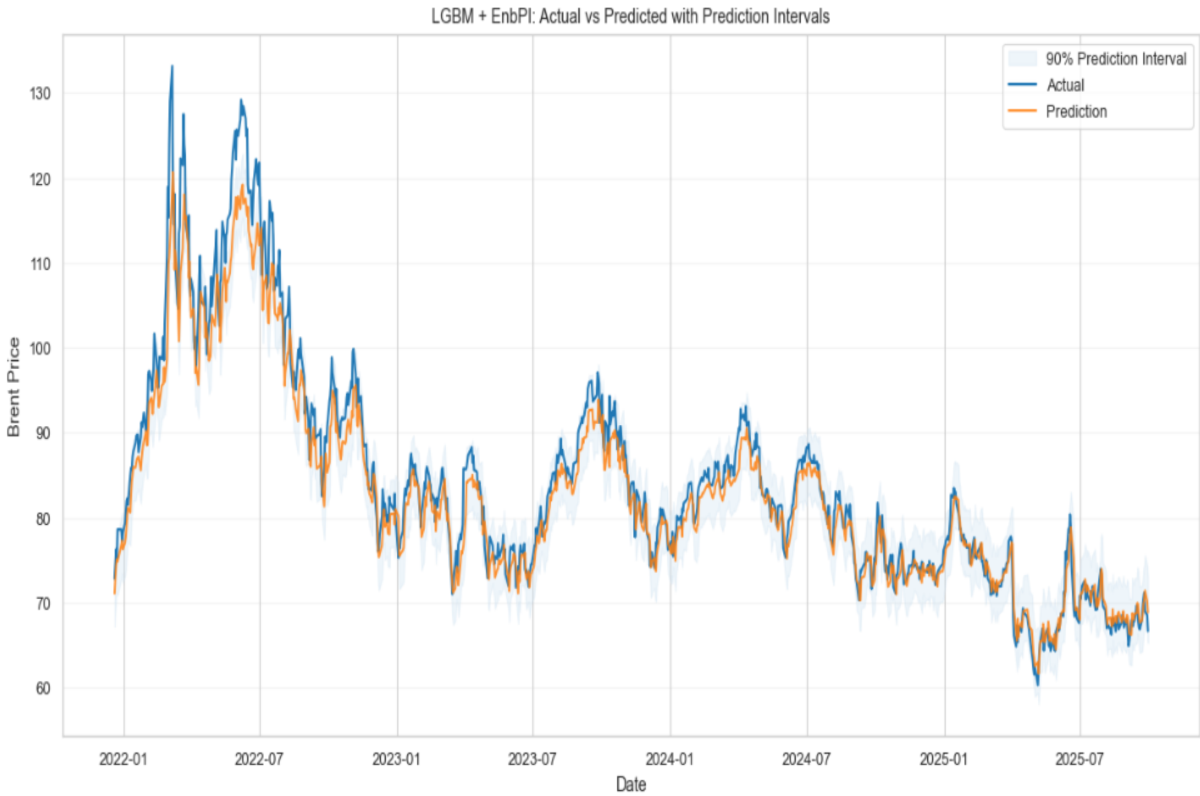

Figure 34: Test set LGBM + EnbPI

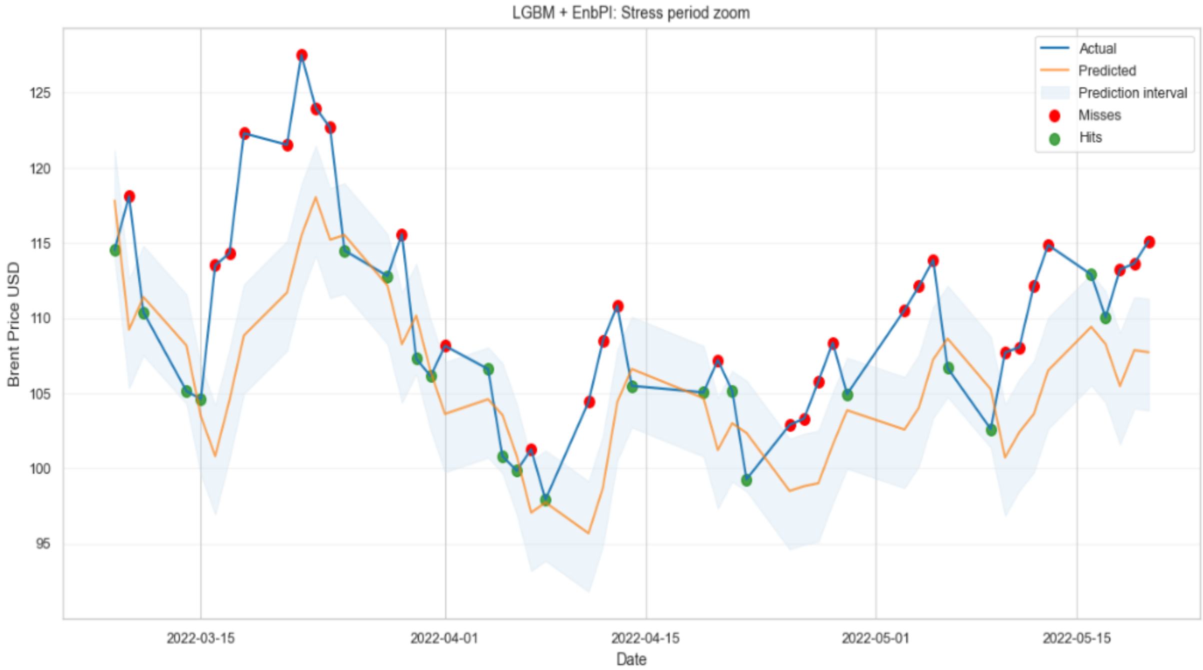

Figure 35: LGBM + EnbPI under Stress

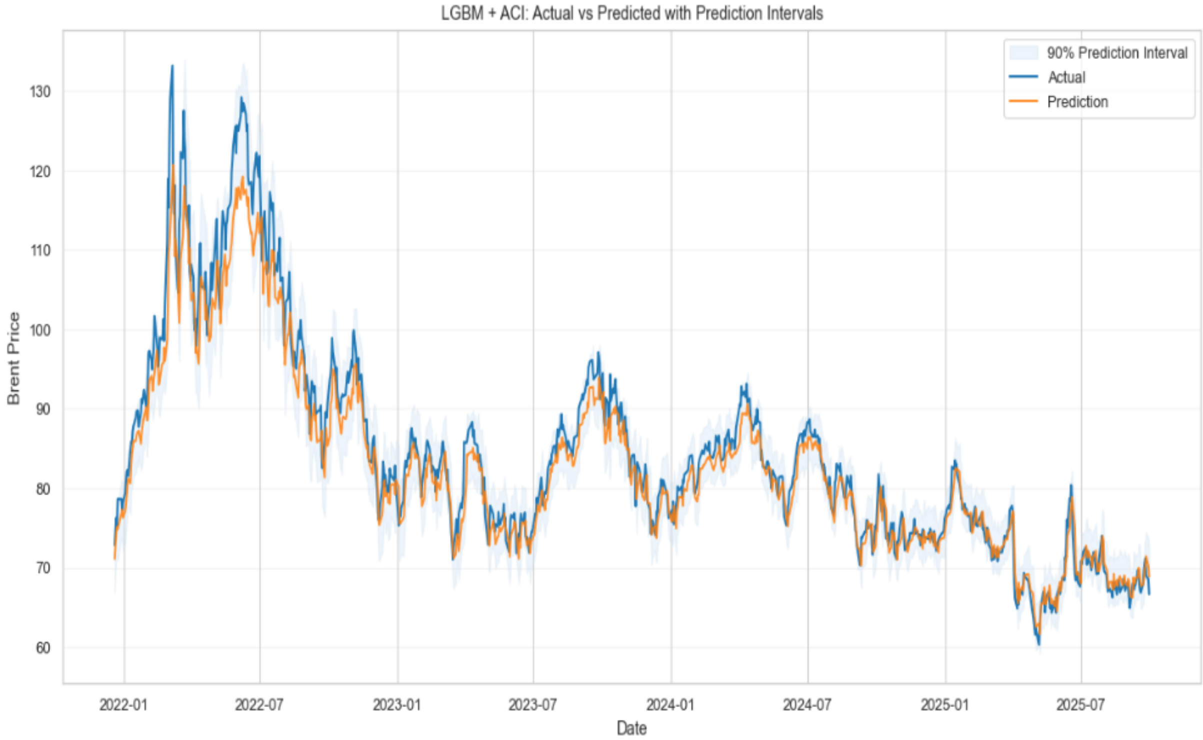

Figure 36: Test set LGBM + ACI

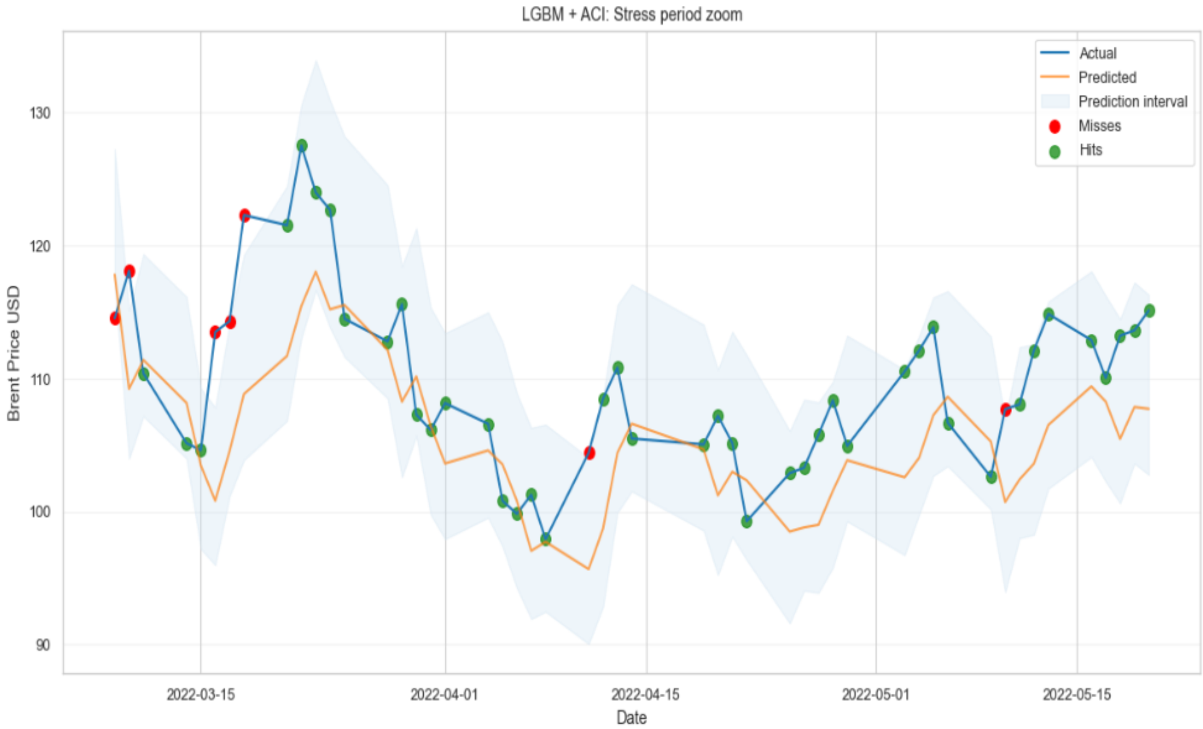

Figure 37: LGBM + ACI under Stress

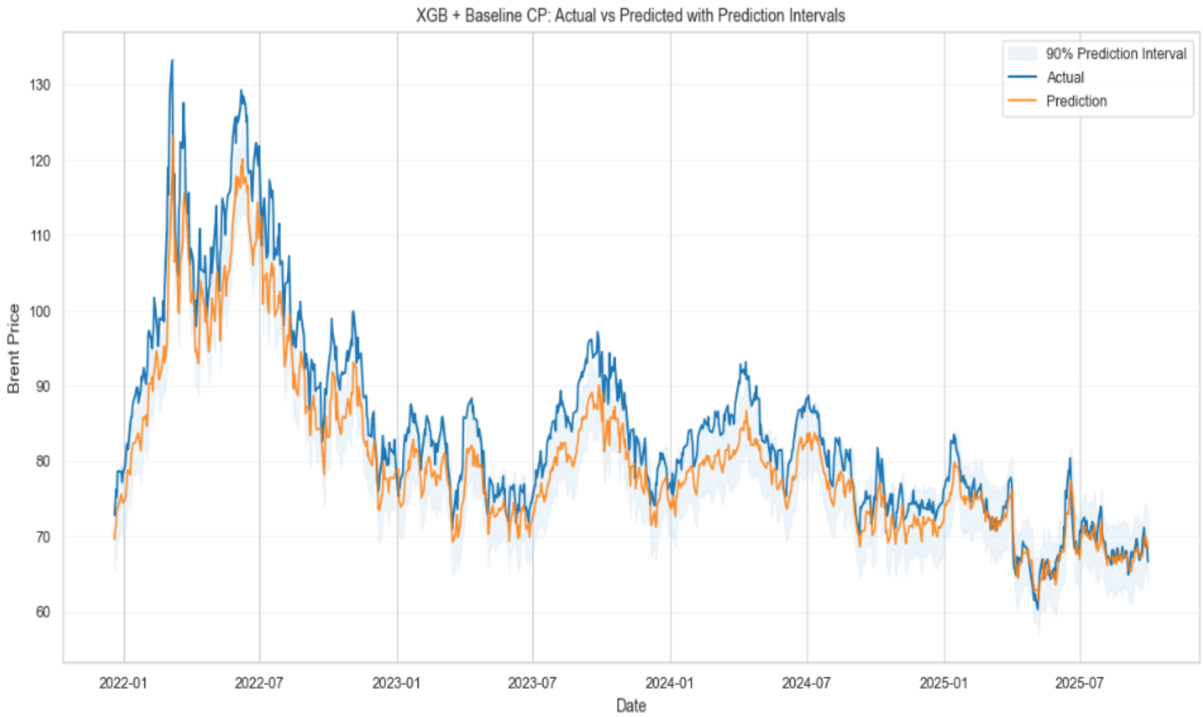

Figure 38: Test Set XGB + Baseline CP

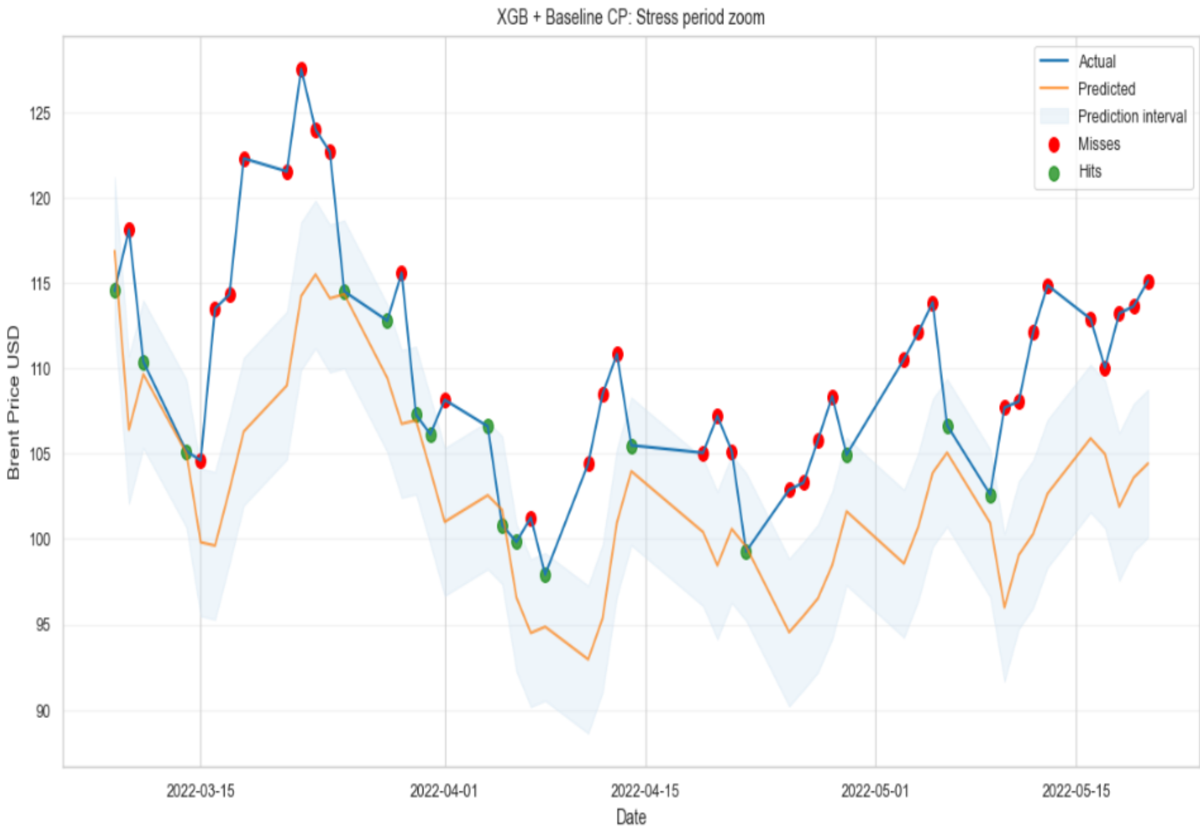

Figure 39: XGB + Baseline CP under Stress

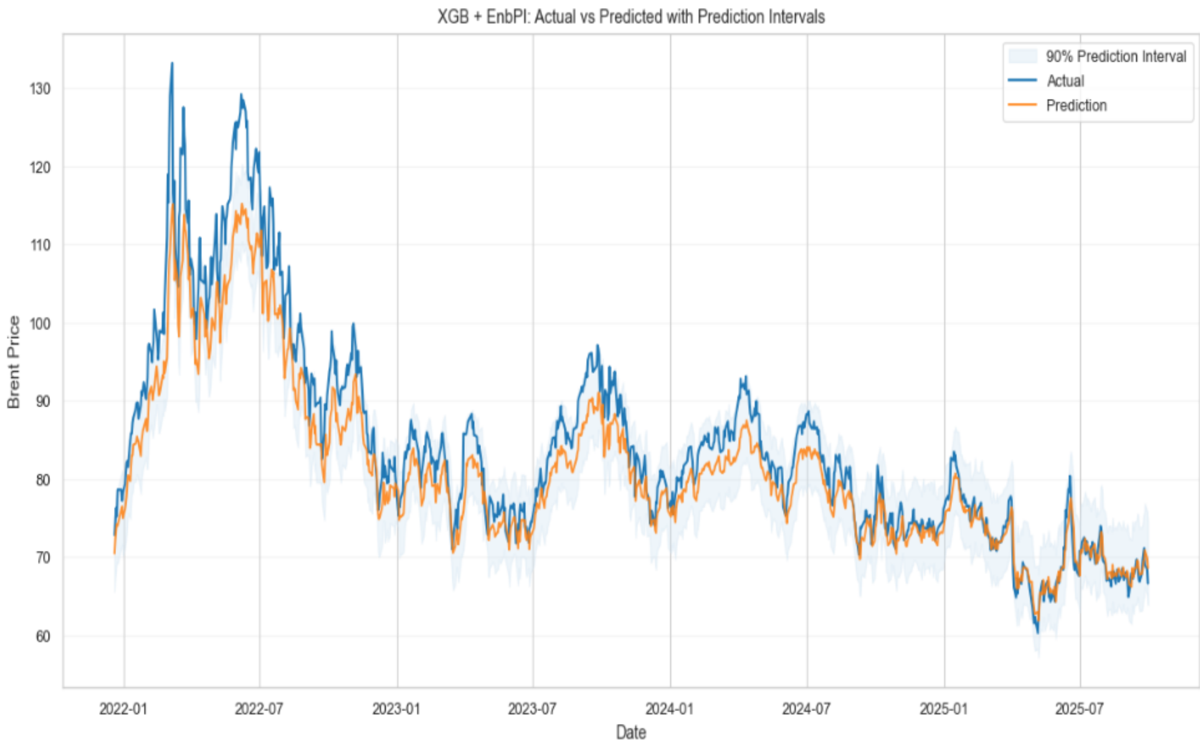

Figure 40: Test Set XGB + EnbPI

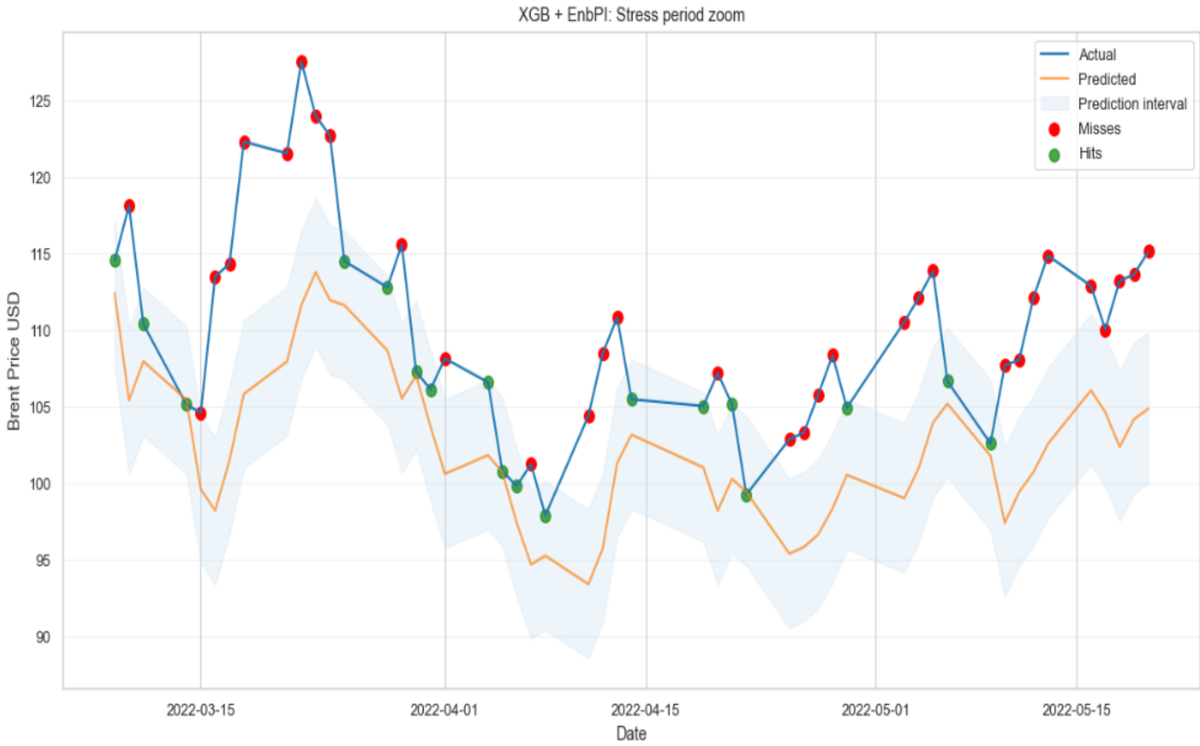

Figure 41: XGB + EnbPI under Stress

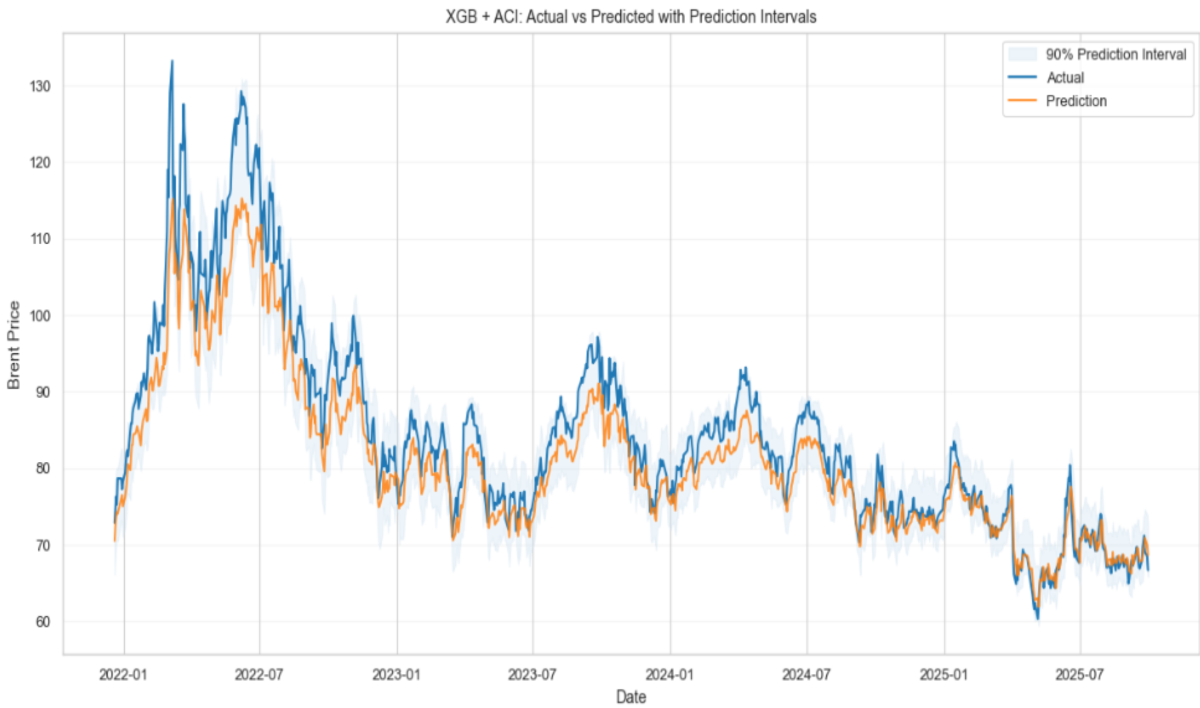

Figure 42: Test set XGB + ACI

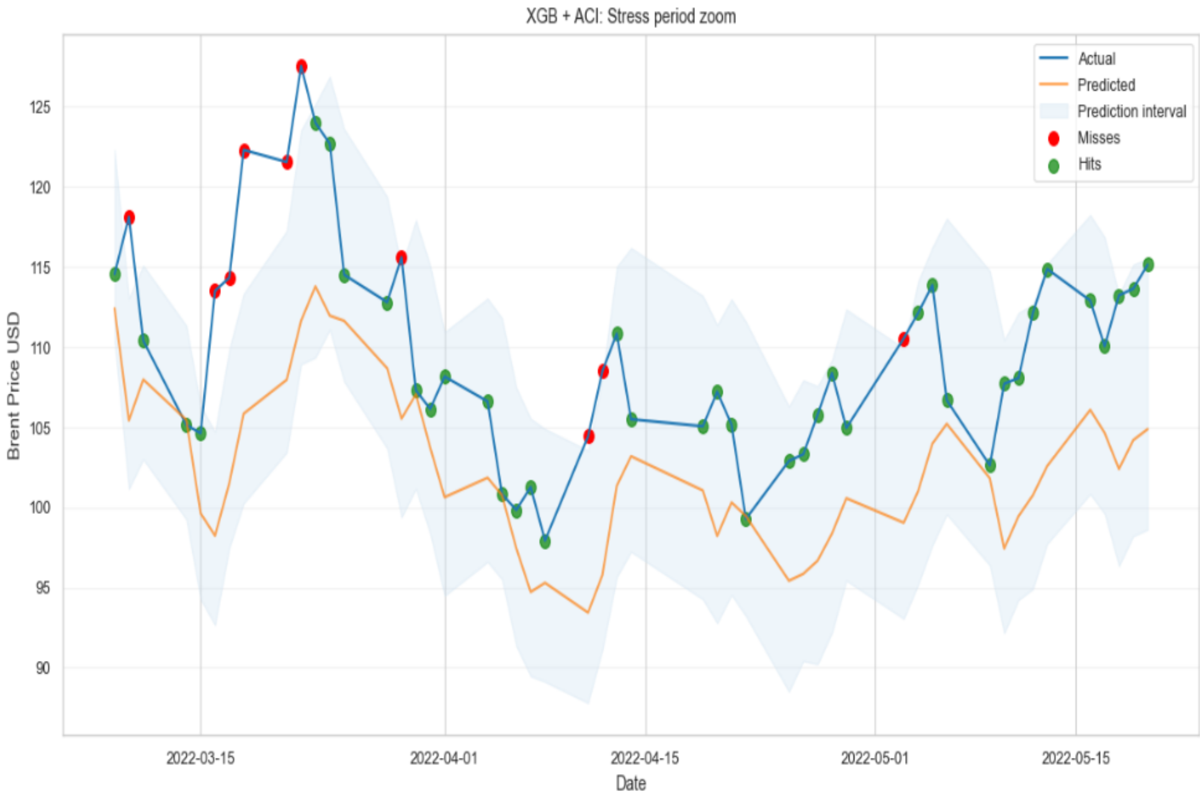

Figure 43: XGB + ACI under Stress